Exchange Rate Disconnect Revisited*

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Abstract

We find that variation in expected US productivity explains around half of US/G7 real exchange rate fluctuations. These fluctuations reflect both imperfectly anticipated changes in productivity and noise that affects beliefs without affecting realized productivity. This "noisy news" is primarily related to TFP growth at medium-to-long horizons. Moreover, noisy news gives rise to several well-known exchange rate puzzles, including the consumption-real exchange rate anomaly, uncovered interest parity violations, and excess volatility. These empirical findings pose challenges for leading classes of open-economy models.

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The real exchange rate—the relative price of consumption across countries—plays a crucial role in clearing markets for goods and financial assets in open-economy models. As a result, these models typically imply that exchange rates are tightly linked to cross-country differentials in macro aggregates, real interest rates, and other asset prices in the economy. However, in the data, real exchange rates appear largely "disconnected" from these macro fundamentals: they exhibit virtually no correlation with current and past macro aggregates (e.g. Meese and Rogoff, 1983) or interest rates (e.g. Fama, 1984), and are also surprisingly volatile (e.g. Rogoff, 1996). Consequently, a large literature has proposed structural models to explore the shocks and mechanisms capable of generating these empirical features. Yet, direct empirical evidence on the origins of these features remains limited.

This paper aims to uncover the main empirical drivers of exchange rate fluctuations without relying on a structural model. Our key finding is that noisy news about future TFP accounts for half or more of the overall variation in *both* real exchange rates and many macro variables. Thus, our results indicate that the exchange rate is indeed connected to macro fundamentals: the apparent disconnect documented in previous literature arises because the effects of news manifest at *different horizons* for exchange rates and macro aggregates. Specifically, we find that the response of the exchange rate leads that of many macro variables by two years or more, resulting in low contemporaneous correlations between the exchange rate and macro aggregates, even though both sets of variables ultimately react to the same underlying source of fluctuations. Moreover, the conditional responses to noisy news give rise to several patterns typically associated with well-known exchange rate puzzles, suggesting that these regularities have a common origin.

We begin by documenting the main comovement patterns associated with exchange rate fluctuations using a reduced-form approach. In particular, following Uhlig (2003), we employ a VAR framework to isolate the dynamic covariances associated with the bulk of real exchange rate movements. This approach reveals that US/G7 real exchange rate appreciations are typically associated with future improvements in macro variables, such as US consumption and productivity, while their contemporaneous comovement with these macro variables is weak. The resulting lead-lag structure unfolds over three to five years. These patterns suggest that the exchange rate responds to news about future macro fundamentals.

We then seek to identify structural disturbances to expectations of future US TFP and estimate their effects on the real exchange rate and other macro variables. To do so, we follow Chahrour and Jurado (2022) to separately identify two orthogonal disturbances: (i) changes in productivity, which may be imperfectly anticipated, and (ii) expectational "noise" disturbances that move expectations of productivity, *but never materialize in realized productivity.*

We find that anticipated future TFP changes and noise in TFP expectations both play an important role in driving exchange rates in the data. Noise disturbances give rise to volatile but relatively short-lived fluctuations in the exchange rate, while imperfectly anticipated TFP changes drive longer-run changes in the exchange rate. As a result, expectational noise appears more relevant at higher frequencies, while lower-frequency movements in exchange rates predominantly reflect technological disturbances. Taken together, the two disturbances account for half or more of the variation in the level of the real exchange rate, while explaining a comparable fraction of the variation in real macro aggregates. Echoing the reduced-form evidence mentioned above, the effects on macro aggregates are delayed relative to the response of exchange rates. We also study the impulse responses of a number of other variables, such as the trade balance, equity prices, and long-term bond yields. We find that, in anticipation of higher future US productivity, the US trade balance deteriorates, US and foreign equity prices rise, and US and foreign long-term bond yields fall. Thus, noisy news offers a potential explanation for the elusive connection between real exchange rates, international business cycles, and other asset prices.

Noisy news also generates empirical patterns typical of well-known "exchange rate puzzles," suggesting that these patterns originate from the same fundamental disturbances. First, these disturbances cause significant fluctuations in expected excess currency returns, generating violations of uncovered interest parity (UIP) that are consistent with both the Forward Premium Puzzle (Fama, 1984) and the reversal at longer horizons (Engel, 2016; Valchev, 2020). Our results therefore complement the evidence of time-series predictability of excess currency returns in Kremens and Martin (2019). In that vein, we also document that trading strategies that exploit the predictable component of excess currency returns associated with our identified disturbances are profitable out of sample. Second, the conditional responses of exchange rates and cross-country consumption differentials align with the consumption-real exchange rate anomaly (Backus and Smith, 1993): when US consumption is relatively high, the US dollar is, on average, appreciated. Third, noisy news leads the exchange rate to be highly persistent and more volatile than other macro quantities, features that the literature calls the "PPP Puzzle" and excess volatility, respectively (e.g. Rogoff, 1996).

Two observations help explain why other studies that have examined the forward-looking nature of exchange rates have generally not found a robust relationship between exchange rates and future macro aggregates. First, our estimates suggest that the information driving the exchange rate is mainly *medium-to-long horizon* news—specifically news about TFP changes around three to five years in advance. Most existing studies have focused on much shorter horizons, seeking lead-lag relationships at horizons of one to two years (Engel and

West, 2005). Indeed, we find that extending the forecast horizon of Engel and West's regressions reveals stronger evidence of Granger causality. Second, unlike previous literature, we separately identify and account for expectational noise. Because expectational noise weakens the correlation between expectations and *realized* fundamentals, it reduces the statistical power of empirical approaches that focus on the lead-lag relationship between exchange rates and *realized* future macro variables.

We conclude the paper with a discussion of the implications of our results for several popular types of open-economy models. We find that leading models with complete asset markets cannot replicate our key finding that higher expected home TFP is associated with relatively high current home consumption and an appreciated exchange rate. For example, in a benchmark endowment complete-markets model, consumption and the exchange rate do not respond to news of future TFP, and move only after the actual TFP improvement. Complete-markets models with recursive preferences do generate expectational effects (e.g. Colacito and Croce, 2013). However, in these models, complete risk sharing implies that high expected home productivity causes a *fall* in current home consumption and a *depreciation* of the real exchange rate, contrary to our evidence.

We then examine whether a canonical incomplete-markets model can reproduce the empirical effects of expected productivity changes.[1] We show that the performance of this model is better in some respects, but it remains at odds with some of our evidence. In the model, positive news about future TFP raises expected household income, prompting a higher consumption differential, higher interest rate differentials, and an exchange rate appreciation—all qualitatively consistent with the patterns we estimate in anticipation of a TFP improvement. Nevertheless, this incomplete-markets model produces a key counterfactual implication: the exchange rate depreciates once the TFP improvement occurs, whereas in the data it remains appreciated for several years following the productivity increase.

Related literature This paper is related to several strands of the international finance and macro literatures. First, we speak to the literature on exchange rate determination, which documents low empirical correlations between current or past macro fundamentals and current exchange rates (Meese and Rogoff, 1983; Cheung et al., 2005; Rogoff and Stavrakeva, 2008; Rossi, 2013).[2] A related observation is that the exchange rate is highly persistent, and significantly more volatile than macroeconomic fundamentals (Obstfeld and Rogoff, 2000;

[1] Incomplete-markets models are central to several recently proposed explanations of exchange rate dynamics. Our focus, however, is on how they propagate productivity changes, rather than financial shocks (Itskhoki and Mukhin, 2021) or discount-rate shocks (Kekre and Lenel, 2024).

[2] A related literature documents a weak relationship between currency returns and other asset returns (Burnside, 2012; Chernov and Creal, 2023; Chernov et al., 2023).

Chari et al., 2002; Sarno, 2005; Corsetti et al., 2008a; Steinsson, 2008).[3]

Contrary to this strand of literature, we find a strong connection between exchange rates and macro fundamentals, albeit one that relates current exchange rates to future fundamentals. Our evidence is consistent with the argument of Engel and West (2005) that exchange rates are forward-looking and therefore should lead, rather than lag behind, macro variables. We contribute to this discussion in several ways. First, we show that the link between current exchange rates and future fundamentals runs specifically through *imperfect* and *noisy* anticipation of future productivity. Second, we show that the expectations reflected in exchange rates primarily concern TFP at medium-to-long horizons (three to five years in the future). Third, our analysis explicitly accounts for expectational noise, i.e. fluctuations in expectations that are not associated with actual subsequent changes in fundamentals. We argue that ignoring this component of expectations leads one to understate the forward-looking nature of exchange rates.[4]

This paper also contributes by showing that many well-known exchange rate puzzles have a common origin in noisy news about future TFP. These empirical moments have received extensive theoretical attention, although few mechanisms have been linked to news about future TFP.[5] The evidence that fundamental disturbances account for a sizable share of fluctuations in both macro aggregates and exchange rates contrasts with the view that exchange rate movements are primarily driven by financial shocks originating in foreign exchange markets with limited effects on macro quantities (Itskhoki and Mukhin, 2021). More broadly, the conditional moments that we identify provide discipline for candidate models of exchange rate determination. Against these moments, we evaluate how leading classes of complete- and incomplete-markets open-economy models perform.

A related literature examines the relationship between cross-country productivity differences and international business cycle comovement or exchange rate dynamics (Corsetti

[3]There is some evidence of contemporaneous relationships between exchange rates and specific variables, including commodity prices (Chen et al., 2010; Ayres et al., 2020), external imbalances (Gourinchas and Rey, 2007), order flow and microstructure dynamics (Evans and Lyons, 2002, 2008), as well as proxies for global risk (Lilley et al., 2022) and monetary factors (Engel and Wu, 2024) in recent decades. Hassan et al. (2016) further link exchange rate stochastic properties to cross-country differences in capital-output ratios.

[4]A related literature uses survey expectations to measure surprises in macro announcements and examines their effects on exchange rates (Andersen et al., 2003; Faust et al., 2007; Engel et al., 2008; Stavrakeva and Tang, 2026). While there are important conceptual differences between our analysis and this literature—including in the definition of "news"—we find that revisions in expectations of future US TFP are partly linked to information revealed in macro announcements (see Section 4.3).

[5]The deep body of theoretical work on exchange rate puzzles includes, among others, Devereux and Engel (2002); Jeanne and Rose (2002); Gourinchas and Tornell (2004); Kollmann (2005); Bacchetta and van Wincoop (2006); Corsetti et al. (2008b); Alvarez et al. (2009); Bacchetta and Van Wincoop (2010); Verdelhan (2010); Burnside et al. (2011); Ilut (2012); Bansal and Shaliastovich (2012); Colacito and Croce (2013); Hassan (2013); Karabarbounis (2014); Farhi and Gabaix (2016); Gabaix and Maggiori (2015); Engel (2016); Valchev (2020); Itskhoki and Mukhin (2021); Kekre and Lenel (2024); Candian and De Leo (2025).

et al., 2014; Beaudry and Portier, 2014; Siena, 2015; Levchenko and Pandalai-Nayar, 2020; Huo et al., 2023, 2025; Gornemann et al., 2025). Most closely related to our analysis is Nam and Wang (2015), who adopt the Barsky and Sims (2011) approach to identify TFP news shocks in an open-economy context. In contrast to that paper, we depart from the conventional "news shock" framework to separately identify the effects of imperfectly anticipated TFP improvements and noise disturbances. This distinction offers both practical advantages in terms of identification assumptions and conceptual advantages for interpretation.[6] Our approach helps explain why previous studies found only a weak link between exchange rates and future realized fundamentals, clarifies the distinct roles of technological and noise disturbances in giving rise to various exchange rate puzzles, and provides a natural framework for confronting model predictions with the data.

1 Data and basic empirical framework

Our empirical analysis centers on a VAR

$$Y_t = B(L)Y_{t-1} + u_t, \tag{1}$$

where the column vector Y_t contains data on the US and a trade-weighted aggregate for the other G7 economies. Going forward, we refer to the US and the other G7 economies as the "home" and "foreign" economies, respectively, and use the * notation to denote foreign variables. For our baseline analyses, the vector Y_t contains eight variables: the nominal exchange rate S_t expressed in US dollars per unit of foreign currency, Fernald's (2012) series on utilization-adjusted US TFP, US real consumption and investment, foreign real consumption and investment, the nominal interest rate differential, and the CPI price level differential between the US and a trade-weighted aggregate for the other G7 economies. In summary,

$$Y_t \equiv \left[\log\left(S_t\right), \log\left(TFP_t\right), \log\left(C_t\right), \log\left(C_t^\star\right), \log\left(I_t\right), \log\left(I_t^\star\right), \log\left(\frac{1+i_t}{1+i_t^\star}\right), \log\left(\frac{CPI_t}{CPI_t^\star}\right)\right]'. \tag{2}$$

We follow Engel (2016) in constructing the trade-weighted aggregates of the other G7 economies. The weight for each of the six other G7 economies is the sum of the economy's

[6] Regarding identification, the news shock literature (Barsky and Sims, 2011) relies on joint assumptions about the invertibility of the data-generating process and the persistence of news shocks relative to surprises. By contrast, our approach relies on the weaker recoverability condition of Chahrour and Jurado (2022) and imposes no assumptions about persistence.

world trade divided by the sum of world trade for all six economies combined.[7]

For our benchmark results, we use quarterly data for 1978:Q4–2008:Q2. The sample stops in 2008:Q2 to guard against a possible structural break in the aftermath of the financial crisis, which is a potential pitfall as argued by Baillie and Cho (2014) and Du et al. (2018). In Online Appendix C.1, we reproduce all of our main results in an extended sample covering 1978:Q4–2020:Q1.

We describe the data and their sources in detail in Online Appendix A. As a brief overview, the exchange rate is the average of the daily exchange rates within a quarter, obtained from *Datastream*. The interest rate differential is the average of daily Eurodollar rates within a quarter, obtained from *Datastream*. The CPI indices and the consumption and investment series are from the *OECD* database. Lastly, the US TFP is from John Fernald's website.

We do not have a comparable utilization-adjusted quarterly TFP series for countries other than the US. Some recent papers have constructed utilization-adjusted TFP for foreign economies (e.g. Huo et al., 2023; Comin et al., 2025), but we find that these measures are not suitable for our purposes. Most such measures of adjusted foreign TFP can only be constructed at annual frequencies (Huo et al., 2023) or over a very short sample period (Comin et al., 2025), which limits the scope of the analysis. As we discuss below, we also consider an alternative specification that includes the foreign Solow residual as part of Y_t, and it does not materially affect the conclusions of this paper.

We estimate the VAR in (1) using four lags and Bayesian methods with a Minnesota prior. This commonly used prior assumes all series in Y_t are mutually independent random walks. This choice of prior is conservative, since we are looking for relationships between the exchange rate and other macro variables. Throughout, figures report posterior medians, and the shaded areas are 16th–84th percentile bands of the posterior distribution.

Following the established convention (e.g. Sims et al., 1990; Eichenbaum and Evans, 1995), we estimate the VAR in levels without imposing any cointegration assumptions ex ante. Nevertheless, Online Appendix C shows that our results are qualitatively similar in a Vector Error Correction Model (VECM) with a rich cointegration structure.

We use our VAR to perform two different identification exercises. The first is descriptive. The goal of this exercise is to isolate the comovement of macro variables and the real exchange rate, conditional on a reduced-form innovation that explains most of the unconditional variance in the real exchange rate. The second exercise is structural, and is intended to quantify the role of productivity expectations in driving macro and real exchange rate fluc-

[7] We use the average of the 1978 and 2008 weights. The weights are: Germany: 29.45%; Japan: 18.43%; France: 15.99%; United Kingdom: 13.99%; Italy: 12.30%; Canada: 9.84%.

tuations. This approach allows us to separately identify the disturbances that drive actual changes in TFP and the noise that influences expectations, but never affects TFP.

2 The main driver of real exchange rate fluctuations

We begin with an agnostic empirical approach that isolates the main source of exchange rate fluctuations in the data while imposing minimal assumptions on the nature of the underlying structural disturbances. To do so, we follow the max-share approach of Faust (1998) and Uhlig (2003) to extract the reduced-form innovation that explains the largest share of the variation in the real exchange rate. This approach has been applied by Angeletos et al. (2020) to extract a so-called "main business cycle shock." We apply it to exchange rates and, in parallel to Angeletos et al.'s terminology, we refer to the innovation that we extract as the "main exchange rate shock."[8]

To apply the max-share procedure, write the moving average (MA) representation of the VAR in (1):

$$Y_t = \sum_{\ell=0}^{\infty} C_\ell A_0 \varepsilon_{t-\ell}, \tag{3}$$

where the $\{C_\ell\}$ correspond to the coefficients in the MA(∞) representation of $C(L) \equiv (I - LB(L))^{-1}$ and A_0 is the impact matrix relating reduced-form residuals and the orthogonalized innovations, such that $u_t = A_0 \varepsilon_t$. The (log) real exchange rate is the difference between the log nominal exchange rate and the log of relative CPIs, $q_t \equiv s_t + p_t^\star - p_t$, so that a fall in q_t represents an appreciation of the home currency (US dollar). Using the VAR variables reported in (2), we can therefore construct $q_t = e_q' Y_t$, where $e_q = [1, 0, 0, 0, 0, 0, 0, -1]'$.

The error in forecasting the real exchange rate at $t+m$ with information at $t-1$ is

$$q_{t+m} - \mathbb{E}_{t-1}[q_{t+m}] = e_q' \left[\sum_{\ell=0}^{m} C_\ell A_0 \varepsilon_{t+m-\ell} \right],$$

where $m \geq 0$. The contribution of the first element $\varepsilon_{1,t}$ to the forecast error variance of the real exchange rate is then

$$\text{var}\left(q_{t+m} - \mathbb{E}_{t-1}[q_{t+m}] \,|\, \varepsilon_{s,t+j} = 0 \;\forall\, s \neq 1,\; j = 0, \ldots, m\right) = e_q' \left[\sum_{\ell=0}^{m} C_\ell A_0 e_1 e_1' A_0' C_\ell' \right] e_q, \tag{4}$$

where e_1 is the selection vector so that $A_0 e_1$ selects the first column of A_0, and $\varepsilon_{s,t}$ is the sth

[8] Kurmann and Otrok (2013), Basu et al. (2021), Cormun and De Leo (2026), Chahrour et al. (2023), and Boer et al. (2026) also contain applications of the max-share approach.

Table 1: Share of forecast error variance explained by the main exchange rate shock (ε_1)

	Forecast Horizon (Quarters)					
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.03	0.05	0.18	0.35	0.45	0.44
Home Consumption	0.03	0.04	0.17	0.43	0.49	0.41
Foreign Consumption	0.01	0.02	0.06	0.21	0.34	0.31
Home Investment	0.17	0.21	0.30	0.46	0.49	0.44
Foreign Investment	0.08	0.10	0.18	0.24	0.35	0.34
Interest Rate Differential	0.29	0.32	0.28	0.30	0.34	0.37
Real Exchange Rate	0.46	0.62	0.78	0.70	0.67	0.65
Expected Excess Returns	0.38	0.30	0.31	0.40	0.41	0.43

Notes: The table reports the variance shares accounted for by the main exchange rate shock.

element in the vector ε_t.

We choose the impact matrix A_0 by maximizing (4) subject to $A_0 A_0' = \mathbb{E}[u_t u_t']$. This requires us to specify a horizon m at which the forecast error variance in (4) is computed, and for that we choose $m = 100$ quarters, which effectively gives us the unconditional variance of the real exchange rate q_t. This procedure yields a partially identified system, in that the above maximization problem uniquely determines only the first column of A_0 and thus the first element of the vector ε_t, our object of interest. The resulting series $\varepsilon_{1,t}$ represents the reduced-form innovation that accounts for the largest share of fluctuations in q_t. It should not be interpreted as a structural shock; indeed, $\varepsilon_{1,t}$ likely reflects a linear combination of underlying structural disturbances (Dieppe et al., 2026; Dou et al., 2025). We address the question of its structural origin in the following sections.

Table 1 reports the share of the forecast error variance of each macro variable that is explained by the main exchange rate shock, across horizons from 1 to 100 quarters. We find that this innovation is indeed a dominant driver of exchange rate fluctuations—it explains roughly 65% of the variance of the real exchange rate.

To examine the comovement patterns associated with this variation, Figure 1 presents the impulse response functions of the exchange rate and other macroeconomic variables following a one-standard-deviation increase in the main exchange rate shock. The responses reveal a clear comovement: high interest rates along with an appreciation of the real exchange rate, which exhibits persistent, hump-shaped dynamics—continuing to appreciate for about four quarters after the initial impact before gradually returning to its long-run mean.

These dynamics are reflected in the non-monotonic behavior of expected excess returns of holding foreign bonds, defined following Engel (2014) as

$$\mathbb{E}_t[\lambda_{t+1}] \equiv \mathbb{E}_t[\Delta q_{t+1}] - (r_t - r_t^\star), \tag{5}$$

where $r_t - r_t^\star$ is the three-month real interest rate differential.[9] These expected excess foreign-currency returns, which are computed using VAR-implied expectations, are initially negative and remain so for around five quarters before turning significantly positive and staying elevated for several years thereafter. While our empirical results measure time variation in the *predictable component* of excess currency returns, they do not allow us to determine whether this predictability arises from classical risk premia (i.e. compensation for bearing risk) or from systematic expectational errors. Henceforth, however, we follow standard terminology and refer to these predictable excess currency returns as a time-varying "currency risk premium." Also, in Section 5 we only consider models under the assumption of rational expectations.

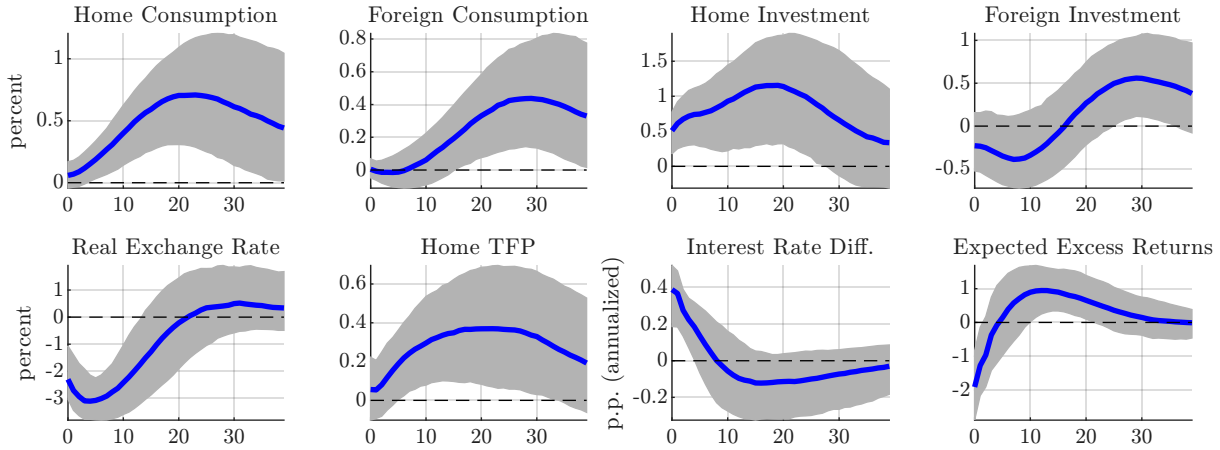
The joint behavior of these variables displays the defining features of exchange rate dynamics. The hump-shaped, persistent appreciation of the real exchange rate with a half-life of about 2.5 years echoes the estimates of Steinsson (2008) based on a univariate reduced-form innovation to the exchange rate. The observed comovement with the interest rate differential reflects the classic Forward Premium Puzzle: the high-interest-rate currency (the US dollar) initially earns high returns (Fama, 1984), but over the medium run, the direction of the UIP violation reverses, with the US dollar subsequently earning low returns for an extended period (Engel, 2016; Valchev, 2020).

Table 1 also shows that the main exchange rate shock explains a substantial share of the overall variation in several key macro aggregates. At long horizons, it accounts for roughly 40% of the forecast error variance of home consumption and TFP, while contributing only modestly to their short-run fluctuations. Indeed, Figure 1 shows that the responses of other macro aggregates systematically lag those of the exchange rate. For instance, home and foreign consumption peak only around 20–30 quarters after the initial response of the exchange rate. (The increase in the home consumption differential, along with the appreciated exchange rate, echoes the well-known empirical violation of the Backus-Smith condition.) Likewise, the main exchange rate shock has little immediate effect on US TFP, which rises significantly only subsequently, peaking at an increase of about 0.4% roughly 20 quarters after the initial impulse. Thus, both consumption and TFP exhibit pronounced medium-to-long horizon responses, but respond relatively little in the short run. This difference in the timing of responses implies that exchange rates are only weakly (conditionally) correlated with current and lagged macro aggregates, consistent with the empirical regularities highlighted in the literature on the "exchange rate disconnect" (e.g. Meese and Rogoff, 1983).

The main exchange rate shock at different frequencies Some authors using the max-share approach have targeted specific frequencies instead. To understand the implication of

[9] The real interest rates are computed using the VAR-implied expected inflation.

Figure 1: Impulse responses to the main exchange rate shock (ε_1)



Notes: The figure reports the impulse responses to the main exchange rate shock, along with the 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

focusing on different frequency bands, Online Appendix C.2 reports variance decomposition values when the innovation is selected to explain most of the variation in the real exchange rate at periodicities of 2–1000 and 6–32 quarters, respectively.

The first of these, reported in Online Appendix Table C.4, essentially targets the unconditional variances in the data and delivers results extremely similar to those in Table 1. By contrast, the contribution of the main exchange rate shock to other macro aggregates, reported in Online Appendix Table C.5, is about half as large when the maximization targets business-cycle periodicities of 6–32 quarters. This observation is consistent with Miyamoto et al.'s (2025) finding that the relationship between exchange rates and fundamentals appears weak when the max-share approach targets business-cycle frequencies. As we explain below, our main empirical results rationalize these contrasting results. We find that noise in the expectation of future TFP drives a lot of the high-frequency variation in the exchange rate, and this feature obscures the connection between exchange rates and *realized* fundamentals.

Taking stock Taken together, this evidence sheds new light on the "exchange rate disconnect." Our results confirm that most real exchange rate variation is essentially unrelated *contemporaneously* to aggregate consumption or TFP, two key macro variables often examined in this context. Instead, we find that the exchange rate *leads* these aggregates, suggesting that the observed "disconnect" reflects differences in the timing of responses rather than a true separation between exchange rates and fundamentals. This evidence challenges the emerging view that exchange rate fluctuations are primarily driven by currency-market-specific financial or risk shocks that are unrelated to macro fundamentals.

While this empirical approach provides a compact statistical summary of the data, it does not identify the structural origins of the reduced-form innovation $\varepsilon_{1,t}$. We argue that the observed lead-lag relationship between exchange rates and macroeconomic aggregates suggests the possibility that these variables are responding to news about future TFP. The basic idea is that when information about future productivity arrives, forward-looking asset prices—such as the exchange rate—adjust immediately, whereas it takes time for the news, and the subsequent realization of productivity improvements, to be reflected in macroeconomic quantities. Consistent with this news interpretation, we find that the main exchange rate shock explains approximately 25% of the 100-quarter forecast error variance of both home and foreign real equity prices, and 31% and 39% of that of home and foreign real bond yields, respectively.[10] The next section introduces structural assumptions that allow us to identify changes in expectations about future TFP and to assess their role in driving exchange rate dynamics.

3 Expectations of TFP and exchange rates

We follow the approach of Chahrour and Jurado (2022) to identify two distinct types of disturbances to expected TFP. The first type of disturbance captures all realized changes to TFP, and may be imperfectly anticipated at an arbitrary horizon. The second disturbance captures the noise in expectations that leads agents' forecasts about future TFP to fluctuate even when realized TFP subsequently does not change.

Our identification strategy can be applied under general information and time-series structures. We assume that the process for home TFP, $a_t \equiv \log TFP_t$, admits an MA(∞) representation,

$$a_t = \sum_{\ell=0}^{\infty} \delta_\ell \varepsilon^a_{t-\ell}, \tag{6}$$

where ε^a_t are the innovations in TFP's univariate Wold representation; we impose no additional assumptions on the time-series dynamics of a_t.

One key goal of our analysis is to identify the degree to which the future innovations to TFP, ε^a_{t+j}, are forecastable, i.e. whether $\mathbb{E}_t[\varepsilon^a_{t+j}] \neq 0$. There are different ways to model the time-t information set underlying this conditional expectation, and we seek an empirical approach that is both general and flexible. To motivate our approach to identification, we describe a concrete information structure that embodies our key identifying assumptions. As we discuss below, this structure encompasses a wide range of processes and information

[10] To obtain these variance shares, we project equity prices and bond yields on the VAR's state (Online Appendix B.3).

assumptions used elsewhere in the literature.

Specifically, assume that each period agents perfectly observe current and past productivity, along with an additional signal about future productivity,

$$\eta_t = \sum_{j=1}^{\infty} \zeta_j a_{t+j} + v_t. \tag{7}$$

Again, we put no ex ante restrictions on the weights ζ_j in the signal. The process $v_t \perp a_{t+j}, \forall j$ represents informational noise. It also follows an arbitrary MA(∞) process,

$$v_t = \sum_{\ell=0}^{\infty} \nu_\ell \varepsilon^v_{t-\ell}. \tag{8}$$

Evidently, the information structure in (6)–(8) allows for a wide range of signal structures with noisy information about future TFP. But this structure also encompasses a range of processes in which *there is no explicit expectational noise*, including the well-known "long-run risk" process often considered in international asset pricing models (Colacito and Croce, 2011, 2013). Intuitively, Beveridge-Nelson, long-run risk, and other processes that decompose a scalar fundamental into separate components observed by agents all provide those agents with information above and beyond the observation of the fundamental history, and they can all be represented as including noisy observations of future realizations (Chahrour and Jurado, 2018). In Online Appendices D.2 and D.3, we provide an explicit mapping to (6)–(8) for the information structures of Blanchard et al. (2013) and Colacito and Croce (2013). Hence, we work with a very general representation, while imposing minimal structural assumptions.

We separately identify the technological disturbances ε^a_t and the noise disturbances ε^v_t using the approach of Chahrour and Jurado (2022). The economic assumptions behind this identification approach can be easily stated: (i) the technological disturbances ε^a_t are orthogonal to other structural disturbances (as is standard) and (ii) the noise disturbances ε^v_t are orthogonal to ε^a_t (intuitively, the procedure captures rational expectations errors, unrelated to actual TFP by construction).[11] Online Appendix B.1 provides full details of our procedure.[12]

To get some intuition for how the procedure works, consider the illustrative case where productivity a_t and the signal η_t are both directly observed by the econometrician. As in

[11] We also follow Chahrour and Jurado's (2022) choice to refer to structural "disturbances" in this context, rather than "shocks," because the ε^a_t and ε^v_t do not necessarily enter people's information sets at the same time as they are realized.

[12] Forni et al. (2017) propose an alternative approach to identifying noisy news under more restrictive assumptions.

standard VAR analysis, the joint process for $[a_t, \eta_t]$ has many potential representations that deliver the same autocovariances; identification requires some theoretical restrictions. The minimal restrictions implied by (6)–(8) can be represented by imposing zeros in the two-sided MA representation of $[a_t, \eta_t]$ in the following way:

$$\begin{bmatrix} a_t \\ \eta_t \end{bmatrix} = \cdots + \begin{bmatrix} 0 & 0 \\ * & 0 \end{bmatrix} \begin{bmatrix} \varepsilon^a_{t+1} \\ \varepsilon^v_{t+1} \end{bmatrix} + \begin{bmatrix} * & 0 \\ * & * \end{bmatrix} \begin{bmatrix} \varepsilon^a_t \\ \varepsilon^v_t \end{bmatrix} + \begin{bmatrix} * & 0 \\ * & * \end{bmatrix} \begin{bmatrix} \varepsilon^a_{t-1} \\ \varepsilon^v_{t-1} \end{bmatrix} + \cdots . \quad (9)$$

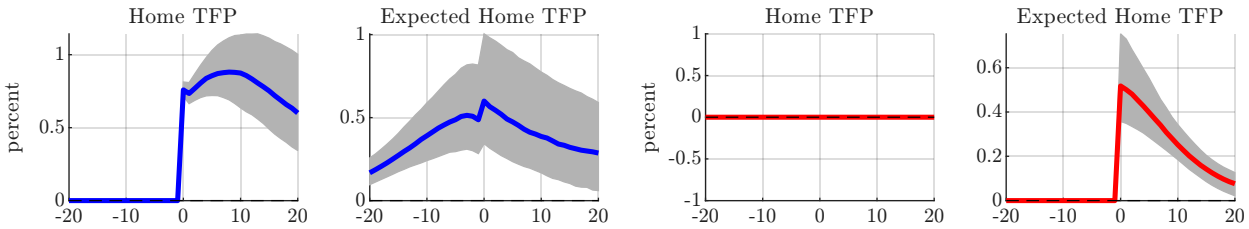
The identification assumption that the noise disturbances ε^v_t are not related to TFP at any lead or lag implies that the upper-right elements of all lead and lag coefficient matrices are zero. In addition, since ε^a_t is the Wold innovation to TFP, the upper-left corner of all *lead* matrices is zero, i.e. technological disturbances naturally move TFP only after they realize. Chahrour and Jurado (2022) show that this gives enough zero restrictions to identify the system—intuitively, the number of remaining unrestricted coefficients is equal to the number of moments one can estimate from data. With estimates for the unrestricted $*$ coefficients in (9), one can subsequently recover time-series estimates of the disturbances ε^a_t and ε^v_t.

Recovering the technological disturbances therefore requires only the history of TFP. Identifying the noise disturbances, by contrast, requires the signal η_t, and the illustrative example above implausibly assumes that the econometrician observes it directly. However, because expectations reflect the information contained in the signals, the same identification can be achieved by replacing η_t in the observation vector with the forecasts of future TFP, $\mathbb{E}_t[a_{t+h}]$. When we implement this approach in practice, we use the forecast of future TFP implied by our estimated VAR in (1). Formally, the key assumption needed for replacing η_t with the VAR-implied $\mathbb{E}_t[a_{t+h}]$ is that the forward-looking variables in the VAR, e.g. exchange rates and interest rates, reflect the forward information about TFP that agents receive through the unobserved signals η_t. Moreover, if the data in our VAR fail to capture some of agents' forward information, this will only bias us *against* finding anticipation effects in our estimation.

To implement this procedure, we must specify a target "horizon" h for the VAR-expectation $\mathbb{E}_t[a_{t+h}]$. We set $h = 20$, in line with the observation from Section 2 that TFP appears most forecastable at medium-to-long horizons. However, in principle, if agents only observe one signal η_t about future TFP, the choice of horizon h is irrelevant, as any choice will yield identical estimation results (Chahrour and Jurado, 2022). In practice, our estimation yields similar results for a wide range of h's (see Online Appendix Figure C.9).

Last, when the VAR has more than two variables, as in our baseline application, the procedure imposes only a subset of the restrictions implied by the signal structure (6)–

Figure 2: Dynamic responses of TFP and TFP expectations



(a) Technological disturbances (ε^a) (b) Noise disturbances (ε^v)

Notes: The figure displays responses to a one-standard-deviation impulse in the technological disturbance (panel (a)) and noise disturbance (panel (b)) at time $t = 0$. Expected Home TFP refers to the fixed-horizon expectation, $\mathbb{E}_t[a_{t+20}]$. The shaded areas are 16th–84th percentile bands. Each period is a quarter.

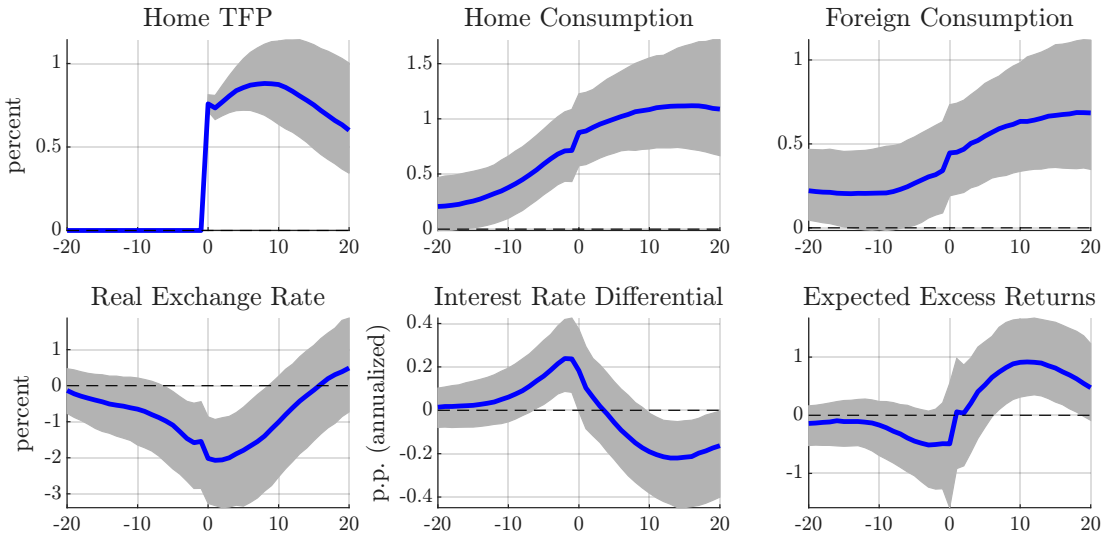
(8). In particular, it does not impose the restriction that other variables besides the target expectation, $\mathbb{E}_t[a_{t+h}]$, have a zero response prior to the occurrence of the noise disturbance ε_t^v. Because we do not impose these additional over-identifying restrictions *ex ante*, one can use these additional restrictions as an *ex post* test of the information assumptions that motivate our identification approach. In our main application, we generally find small responses of other variables before the occurrence of a noise disturbance, suggesting that these additional zero restrictions are empirically plausible.

3.1 The dynamic effects of technological and noise disturbances

Having identified the technological and noise disturbances, we next compute the impulse responses to those disturbances. Figure 2 plots the impulse response of TFP itself, a_t, as well as the impulse response of the 20-quarter-ahead expectation of TFP, $\mathbb{E}_t[a_{t+20}]$. Since our approach allows for anticipation of technological disturbances, we plot each response beginning 20 quarters *before* the actual change in productivity, which occurs at $t = 0$, and then trace the evolution of the responses up to 20 quarters after the realization of the disturbance.

The figure shows strong evidence of anticipation of technological disturbances. In response to a one-standard-deviation technological disturbance, ε_t^a, expected TFP rises by about 0.2% 20 quarters before the actual increase at $t = 0$ of roughly 0.75%, implying that around one-quarter of the improvement is foreseen five years in advance (see Figure 2a). The forecast continues to rise as the TFP improvement approaches, consistent with greater short-term predictability. That said, the further upward revision in expectations at $t = 0$ indicates that the realization of the technological disturbance still comes as a partial surprise. After the TFP increase materializes, both TFP and TFP expectations remain persistently elevated.

Figure 3: Impulse responses to technological disturbances (ε^a)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the technological disturbance at time $t = 0$. The shaded areas are 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

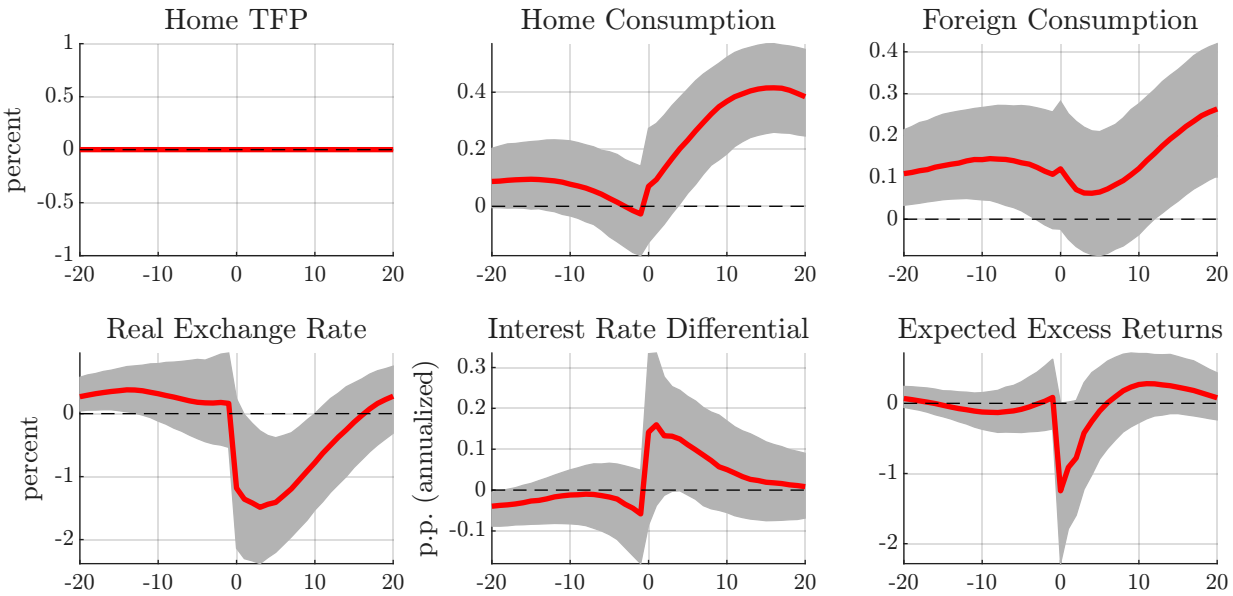
We also find evidence of expectational noise in expectations of TFP. While noise disturbances, ε_t^v, have no effect on actual TFP at any lead or lag by assumption, they move expected TFP substantially: a one-standard-deviation increase in ε_t^v, an "optimistic" revision about future TFP, raises $\mathbb{E}_t[a_{t+20}]$ by roughly 0.5% (see Figure 2b). These expectational effects are mean-reverting, as people gradually recognize that their initial optimism was unwarranted and expectations return to trend.

Overall, the results in Figure 2 support the hypothesis of a noisy-information paradigm, where future movements in TFP are partially anticipated, but expectations are noisy and sometimes move even though there is no actual change in productivity ex post.

We now turn to the effects of these two disturbances on the rest of the endogenous variables in the VAR. Figure 3 plots the responses of selected variables to a technological improvement (an increase in ε_t^a at $t = 0$). Here, we focus on describing the comovement patterns induced by our disturbances, and defer a more theoretical interpretation of them until Section 5.

Technological disturbances give rise to significant responses in exchange rates, interest rates, and consumption differentials. The real exchange rate responds in a pronounced hump-shaped fashion. In anticipation of the TFP improvement, the real exchange rate steadily appreciates and only gradually depreciates back to its long-run mean after the TFP increase materializes. The real interest rate differential also rises in anticipation of the TFP improvement, while it declines steadily and remains significantly below its long-run mean

Figure 4: Impulse responses to noise disturbances (ε^v)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the noise disturbance at time $t = 0$. The shaded areas are 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

for around 10–20 quarters following the actual TFP increase. We also observe a significant increase in US consumption preceding the TFP improvement. Although foreign consumption also rises, its response is considerably weaker, leading to a large and positive consumption differential (not shown). Notably, the exchange rate appears to lead consumption, as its peak response occurs earlier than the peak in consumption. Technological disturbances, which are partly anticipated, are indeed strongly reflected in the real exchange rate, and, with different timing, in macro aggregates.

The timing differences are even clearer in the impulse responses to the noise disturbance, ε_t^v, shown in Figure 4. Following an improvement in expectations (period 0 on the x-axis), the real exchange rate appreciates immediately. This directly speaks to anticipation effects on the exchange rate, as the noise disturbance has no impact on TFP at any horizon. The exchange rate response is also persistent, returning to its long-run mean only after about 12 quarters. The interest rate differential—another asset price—also jumps on impact following the upward revision in expectations. By contrast, the consumption response is gradual and delayed without a significant change at time 0. Delayed consumption effects relative to the immediate exchange rate response are consistent with an information structure in which agents receive news about medium-to-long horizon TFP prospects; as a forward-looking asset price, the exchange rate reacts immediately, whereas home (and, to a smaller extent, foreign)

Table 2: Variance decomposition

	Periodicities of 2–100 Quarters			Periodicities of 6–32 Quarters		
	Both	Tech.	Noise	Both	Tech.	Noise
Home TFP	1.00	1.00	0.00	1.00	1.00	0.00
Home Consumption	0.69	0.55	0.14	0.28	0.10	0.18
Foreign Consumption	0.62	0.51	0.12	0.27	0.13	0.14
Home Investment	0.71	0.56	0.14	0.40	0.25	0.15
Foreign Investment	0.66	0.43	0.23	0.42	0.12	0.30
Interest Rate Differential	0.53	0.43	0.10	0.30	0.19	0.11
Real Exchange Rate	0.62	0.45	0.17	0.31	0.12	0.20
Expected Excess Returns	0.47	0.33	0.14	0.32	0.16	0.16

Notes: The table reports the estimated variance shares (at periodicities of 2–100 quarters and of 6–32 quarters) explained by technological disturbances ("Tech."), noise disturbances ("Noise"), and both combined ("Both").

consumption responds only as the expected TFP improvement approaches.[13]

The US trade balance also responds significantly to technological and noise disturbances, as shown in Online Appendix Figure C.5. We find that the trade balance (as a percentage of US GDP) deteriorates in anticipation of future TFP gains. These dynamics are generally consistent with the "intertemporal approach to the current account," as well as with the observation that survey-based expectations of relative US GDP growth are significantly correlated with the US current account (Engel and Rogers, 2006; Hoffmann et al., 2019).[14]

Variance decomposition To quantify the role of technological and noise disturbances, Table 2 reports the contribution of these disturbances to various endogenous variables. We report these variance decompositions both at periodicities of 2–100 quarters and at the business-cycle periodicities of 6–32 quarters that many models target. As per our identification restrictions, technological disturbances account for 100% of the variation in TFP, while noise disturbances are orthogonal to TFP.

We estimate that the two disturbances explain around 60% of the variation at periodicities of 2–100 quarters in the levels of the real exchange rate, as well as US and foreign real consumption and investment. Noisy news about future TFP therefore represents a key driver of both exchange rates and real quantities, highlighting a fundamental link between exchange rates and the broader macroeconomy. At business-cycle frequencies, the two disturbances remain important, accounting for 27–42% of the variation in real aggregates and 31% of the

[13]Recall that our procedure does not impose zero restrictions before the arrival of noise. We interpret the small pre-$t = 0$ responses in Figure 4 as evidence that these over-identifying restrictions are plausible.

[14]For recent analyses of the joint properties of the trade balance and the exchange rate, see Mac Mullen and Woo (2025) and Bodenstein et al. (2024).

variation in the exchange rate.[15]

The relative importance of the technological disturbance ε_t^a and the noise disturbance ε_t^v varies across frequencies. At lower frequencies, technological disturbances dominate, explaining more than two-thirds of the exchange rate variation captured by the two disturbances combined (45% versus 62% in total). By contrast, at the business-cycle frequencies, expectational noise is the main driver, accounting for roughly two-thirds of the variation generated by both disturbances (20% versus 31% in total). This pattern mirrors the impulse responses discussed above: noise disturbances generate short-lived, volatile exchange rate movements, while imperfectly anticipated TFP changes yield more persistent effects. Similar frequency patterns appear in the variance decompositions of other real variables.

This sizable role of fundamental disturbances stands in contrast to the view of Itskhoki and Mukhin (2021) that exchange rate movements are primarily driven by financial shocks originating in foreign exchange markets. If these shocks were the dominant source of exchange rate variation, the disturbances emphasized in our analysis would account for only a small fraction of the exchange rate variation. Our variance decomposition results instead suggest that noisy news plays an important role across the spectrum of exchange rate fluctuations.

We emphasize that the variance contribution of technological disturbances reflects both movements in anticipation of and reactions to realized TFP changes (i.e. before and after time 0 in Figure 3). To isolate the pure anticipation effect, we compute the variance contribution of the leading terms in the impulse responses, which occur before TFP is realized. This anticipation channel alone accounts for one-third of the 45% of exchange rate variation explained by technological disturbances, highlighting the substantial contribution of anticipated TFP changes to exchange rate fluctuations, beyond the effects accounted for by expectational noise. Thus, a large fraction of TFP changes are anticipated, and the anticipated component of TFP accounts for a substantial share of exchange rate variance.

4 Extensions and empirical implications

In this section, we present additional empirical results and relate them to previous findings on the link between exchange rates and fundamentals. Section 4.1 shows that technological and noise disturbances generate exchange rate fluctuations exhibiting many well-known puzzles. Section 4.2 examines the effects of these disturbances on other asset prices, while Section 4.3 explores why prior studies have struggled to identify robust correlations between

[15]The evidence that noisy news about US TFP explains a large share of the variation in the real exchange rate and in foreign real variables is consistent with the notion that the US is a dominant source of technological progress and international spillovers, a view supported by the fact that US inventors account for the vast majority of foreign patents in all major advanced economies (e.g. Eaton and Kortum, 1999).

exchange rates and macro fundamentals. Finally, Section 4.4 reports robustness tests for the identification of our disturbances.

4.1 Common origin in exchange rate puzzles

The international macro literature has traditionally been organized around a series of "puzzles," that is, empirical exchange rate patterns that are at odds with standard international macro models. Given the prominent role the two identified disturbances play in driving exchange rate dynamics, it is natural to ask whether they reproduce some of these classic puzzles. The answer is yes, which leads us to conclude that many exchange rate puzzles may share a common origin in noisy news about TFP.

The Forward Premium Puzzle Our results indicate that noisy news generates significant deviations from uncovered interest parity and reproduces the dynamics characteristic of the Forward Premium Puzzle (e.g. Fama, 1984; Hassan and Mano, 2019). Indeed, Figures 3 and 4 show a negative time-series comovement between interest rate differentials and expected excess foreign-currency returns: when US interest rates are high relative to foreign rates, the dollar is not only strong but also tends to deliver high subsequent returns.

Formally, consider the following version of the Fama regression:

$$\lambda_{t+1} = \alpha + \beta_\lambda (r_t - r_t^\star) + \epsilon_{t+1}^F. \tag{10}$$

Because we employ trade-weighted variables, equation (10) estimates a specific version of what Hassan and Mano (2019) denote as Dollar β (as the US dollar is the base currency). Unlike Hassan and Mano (2019), who use simple averages of foreign variables, we use trade-weighted averages—an approach that we consider more appropriate for our context.[16] Nevertheless, using simple averages yields very similar results for β_λ and other related estimates.

Panel A of Table 3 summarizes our estimates of (10). Unconditionally, we obtain a significantly negative estimate of $\beta_\lambda = -2.29$, consistent with earlier evidence (e.g. Engel, 2014). We then compute the β_λ implied by a counterfactual in which only the two disturbances we identify, ε_t^a and ε_t^v, are active, obtained from our estimated VAR by setting the variances of all other disturbances to zero. In this counterfactual, $\beta_\lambda = -2.07$, indicating that the combination of the two disturbances qualitatively and quantitatively reproduces the Forward Premium Puzzle. We also examine how much of the unconditional $\text{cov}(\lambda_{t+1}, r_t - r_t^\star)$—the

[16] This approach is consistent with standard practice in the empirical international macro literature, where foreign countries are typically weighted by their economic size (GDP) or their trade relevance with respect to the US (e.g. Engel, 2016; Miyamoto et al., 2025; Corsetti et al., 2014).

numerator in the β_λ expression—our identified disturbances can explain. As shown in the table, the two disturbances together account for roughly two-thirds of the total covariance: the unconditional covariance equals -1.29, whereas in the counterfactual data driven only by our two disturbances, it is -0.79. Hence, the empirical tendency for the dollar to deliver high subsequent returns when relative US interest rates are high can be largely attributed to noisy news.

In addition to the "classic" Forward Premium Puzzle, the responses to noisy news replicate the Engel (2016) and Valchev (2020) finding that the puzzle reverses at longer horizons. That is, when US interest rates are high relative to foreign rates, the dollar tends to deliver high subsequent returns in the short run, but low returns in the medium run. To summarize this relationship, we follow Engel (2016) and examine the coefficient β_Λ from the regression:

$$\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}] = \alpha_0 + \beta_\Lambda(r_t - r_t^\star) + \epsilon_t^E.$$

Although the 16th–84th percentile band is wide, the median value of $\beta_\Lambda = 4.55$ in Table 3 indicates that the overall sum $\sum_{j=0}^{\infty} \text{cov}(\lambda_{t+j+1}, r_t - r_t^\star)$ is positive, even though the first term in the sum, i.e. the coefficient in regression (10), is negative. Moreover, noisy news alone implies a $\beta_\Lambda = 8.40$ and accounts for almost all of the total $\text{cov}\left(\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}], r_t - r_t^\star\right)$, although the credible bands of both are wide. The impulse responses in Figure 3 show how this reversal is connected to productivity: periods of high expected excess foreign-currency returns following TFP improvements are preceded by high interest rates several years in advance.

As for the estimates of the Engel β_Λ, we acknowledge that in small samples there is a downward bias in the estimated autocorrelation of the interest rate differential. This implies that our VAR may attribute too much of the variation in the exchange rate to fluctuations in expected excess currency returns at distant horizons.

In addition, we note that our two disturbances not only replicate empirically relevant regression coefficients (β's), but also generate interest rate differential dynamics that are comparable to their unconditional empirical counterparts. This can be seen from the moments $\sigma(r_t - r_t^\star)/\sigma(\Delta q_t)$ and $\rho(r_t - r_t^\star)$ reported in Table 3.

Our results speak to the time variation in expected excess currency returns, or the currency risk premium, but they reflect in-sample predictability relationships. In Online Appendix C.3, instead, we construct out-of-sample trading strategies that exploit the predictable component of excess currency returns associated with the identified disturbances. In our simple strategy, a US investor takes a position in foreign currency proportional to the VAR-implied expectation of excess foreign-currency returns. Our preferred specification

Table 3: Exchange rate related puzzles and TFP expectations

Panel A: Forward Premium Puzzle Moments

	Technology	Noise	Both	Unconditional
Fama β_λ	-1.95	-2.72	-2.07	-2.29
	$[-3.24, -1.23]$	$[-4.20, -1.53]$	$[-3.30, -1.29]$	$[-3.29, -1.56]$
$\text{cov}(\lambda_{t+1}, r_t - r_t^\star)$	-0.65	-0.12	-0.79	-1.29
	$[-2.28, -0.21]$	$[-0.26, -0.05]$	$[-2.57, -0.31]$	$[-3.30, -0.70]$
Engel β_Λ	8.33	9.27	8.40	4.55
	$[-4.05, 21.13]$	$[-12.13, 27.02]$	$[-3.00, 20.64]$	$[-2.40, 12.26]$
$\text{cov}\left(\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}], r_t - r_t^\star\right)$	0.68	0.22	0.96	1.05
	$[-0.34, 3.57]$	$[-0.24, 0.84]$	$[-0.35, 4.33]$	$[-0.51, 4.48]$
$\sigma(r_t - r_t^\star)/\sigma(\Delta q_t)$	0.38	0.14	0.27	0.18
	$[0.24, 0.69]$	$[0.08, 0.24]$	$[0.17, 0.47]$	$[0.13, 0.29]$
$\rho(r_t - r_t^\star)$	0.99	0.95	0.99	0.95
	$[0.98, 1.00]$	$[0.88, 0.98]$	$[0.97, 1.00]$	$[0.91, 0.98]$

Panel B: Backus-Smith Puzzle Moments

	Technology	Noise	Both	Unconditional
$\text{corr}(\Delta q_t, \Delta(c_t - c_t^\star))$	-0.36	-0.50	-0.42	-0.26
	$[-0.68, 0.03]$	$[-0.81, 0.08]$	$[-0.70, -0.05]$	$[-0.37, -0.16]$
$\text{cov}(\Delta q_t, \Delta(c_t - c_t^\star))$	-0.09	-0.13	-0.28	-0.65
	$[-0.34, 0.00]$	$[-0.38, 0.02]$	$[-0.66, -0.03]$	$[-1.05, -0.38]$

Panel C: Excess Volatility and Persistence

	Technology	Noise	Both	Unconditional
$\rho(q_t)$	0.99	0.96	0.98	0.96
	$[0.98, 0.99]$	$[0.93, 0.98]$	$[0.96, 0.99]$	$[0.93, 0.98]$
$\sigma(\Delta q_t)/\sigma(\Delta c_t)$	3.94	7.55	5.21	6.05
	$[2.47, 5.87]$	$[4.58, 11.89]$	$[3.51, 7.14]$	$[4.97, 6.89]$

Notes: The table reports the estimated moments conditional on technological disturbances ("Technology"), noise disturbances ("Noise"), and both combined ("Both"), along with the unconditional moments ("Unconditional"). The moments including Engel β_Λ, $\text{cov}\left(\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}], r_t - r_t^\star\right)$, and $\rho(q_t)$ are computed on the data filtered to periodicities of 2–100 quarters. The values in brackets represent the 16th–84th percentile bands. The moments in the table are defined in the text.

yields a Sharpe ratio comparable to those obtained from alternative currency trading strategies that exploit time variation in return predictability (Lustig et al., 2014; Burnside et al., 2011; Menkhoff et al., 2012; Moskowitz et al., 2012; Hassan and Mano, 2019). This evidence complements the time-series predictability documented by Kremens and Martin (2019).

The Backus-Smith Puzzle The classic international risk-sharing condition of Backus and Smith (1993) predicts that relative consumption across countries should be strongly positively correlated with real exchange rates. That is, it predicts a weak US dollar when US consumption is relatively high. In the data, however, this condition is largely violated: if anything, the US dollar is typically strong when US consumption is relatively high (Kollmann, 1995; Corsetti et al., 2008b).

To quantify this relationship, we consider the moment $\text{corr}(\Delta q_t, \Delta c_t - \Delta c_t^\star)$, where $c_t \equiv \log C_t$. The results, reported in Panel B of Table 3, confirm the empirical pattern: the correlation in the data is mildly negative at -0.26, while in the counterfactual driven solely by the two identified disturbances, it equals -0.42. In addition, examining the underlying covariance, $\text{cov}(\Delta q_t, \Delta c_t - \Delta c_t^\star)$, we find that the two disturbances together account for roughly 43% of this moment in the data. Thus, noisy news plays a substantial role in giving rise to the empirical tendency for the US dollar to be strong when US consumption is relatively high, a pattern typically associated with the Backus-Smith Puzzle.

Persistence and excess volatility of the real exchange rate Two additional puzzling features of exchange rate behavior concern its persistence and volatility (e.g. Chari et al., 2002). We next examine whether the responses to noisy news display these patterns.

Panel C of Table 3 reports additional moments. Two results stand out. First, the real exchange rate exhibits pronounced persistence when conditioned on the identified disturbances: its autocorrelation is approximately 0.98, compared with 0.96 in the unconditional data. Second, fluctuations in the real exchange rate remain highly volatile relative to macroeconomic aggregates. The standard deviation of quarterly exchange rate growth is about 5.21 times that of consumption growth under the two disturbances, close to the ratio of 6.05 observed unconditionally. Notably, noise disturbances alone generate substantially higher "excess volatility" in exchange rates than technological disturbances: the ratio of the standard deviation of quarterly exchange rate growth to that of consumption growth is 7.55 under noise disturbances, compared with 3.94 under technological disturbances. This feature is intuitive: expectational noise alters beliefs about future productivity, and thus asset prices, without corresponding movements in realized fundamentals, generating large exchange rate fluctuations but comparatively muted fluctuations in consumption and output.

4.2 Technology, noise, and other asset prices

Do other asset prices respond to news about future productivity? Figure 5 plots the responses of home and foreign equity prices and long-term bond yields to technological and noise

disturbances. The patterns exhibited in these asset price series are generally consistent with our interpretations of the disturbances as noisy news about future productivity.

The first row of panel (a) in Figure 5 plots the responses of home and foreign real equity price indices to technological disturbances.[17] In anticipation of the productivity increase, both equity prices rise, with the home–foreign equity price differential widening. Following the realization of the productivity improvement, both equity prices and their differential gradually revert toward their original levels. The first row of panel (b) shows that noise disturbances that raise expected productivity also increase equity prices in both economies, with a more immediate and relatively larger response at home.[18] To the extent that positive news about future productivity raises expected future home profits, with gradual spillovers to the foreign economy, the responses of equity prices and their differential are broadly consistent with our interpretation of these disturbances as noisy news about future productivity.[19] Moreover, the significant response of equity prices to noisy news echoes the evidence in Beaudry and Portier (2006) that equity prices incorporate news about future economic fundamentals. Indeed, our results show that noisy news accounts for approximately 44% and 43% of the variation in home and foreign equity prices, respectively, at periodicities of 2–100 quarters.

The second row of panel (a) in Figure 5 plots the responses of real long-term bond yields.[20] In anticipation of a future home productivity improvement, the home–foreign bond yield differential increases, as does the short-term interest rate differential. Underlying this response, both home and foreign 10-year bond yields decline in anticipation of the productivity improvement, with foreign yields falling by more. Following the realization of the productivity improvement, the differential gradually converges back to steady state, while both home and foreign bond yields remain below their steady-state levels for an extended period. Again, panel (b) shows that the anticipation patterns associated with noise disturbances are broadly similar to those associated with genuine changes in fundamentals, although the 16th–84th percentile band is wide. The responses of bond yields and their differential are consistent with an interpretation of these disturbances as noisy news about future productivity: an anticipated productivity improvement lowers expected future short-term interest rates, while stronger home demand keeps current home short-term interest

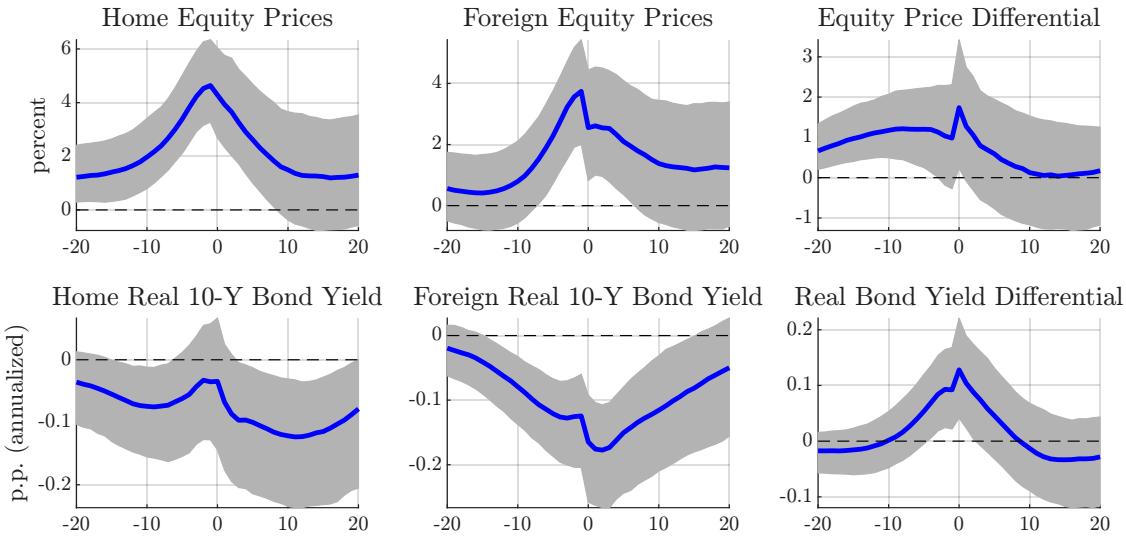
[17] Because these series are not in the VAR, we obtain their responses by projecting each series on the VAR's state using the Bayesian procedure described in Online Appendix B.3.

[18] Following noise disturbances, home equity prices begin to revert monotonically toward zero just outside the window plotted, with a zero effect entering the 16th–84th percentile band around period 45.

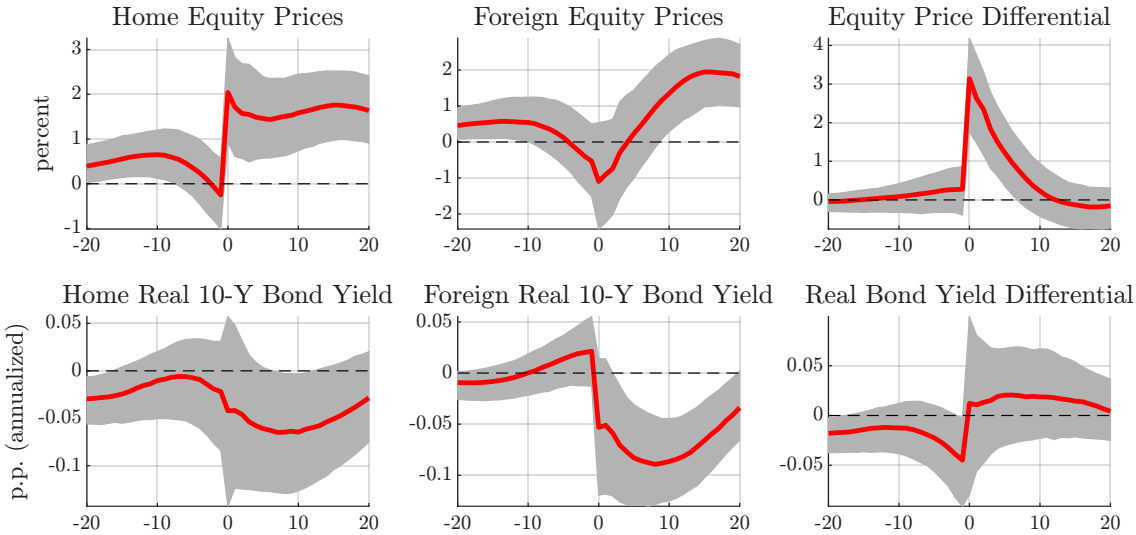
[19] This sentence describes a cash-flow growth channel. The valuation channel, through the responses of interest rates and risk premia, could either reinforce or work against the cash-flow growth channel.

[20] We construct 10-year inflation expectations by projecting respective price indices on the VAR's state. All of our statements here hold for nominal yields as well.

Figure 5: Technology, noise, and equity prices and bond yields



(a) Impulse responses to technological disturbances (ε^a)



(b) Impulse responses to noise disturbances (ε^v)

Notes: The figure displays responses to a one-standard-deviation impulse in the technological disturbance (panel (a)) and noise disturbance (panel (b)) at time $t = 0$. Equity prices are deflated by each economy's CPI. Bond yields are real annualized percentage points, deflated by the VAR-implied expectation of inflation. The shaded areas are 16th–84th percentile bands. Each period is a quarter.

rates relatively elevated.[21] Overall, noisy news explains approximately 33% and 53% of the variation in home and foreign long-term bond yields, respectively, at periodicities of 2–100

[21]The empirical responses of home and foreign real short-term interest rates (not shown) are consistent with this interpretation. This interpretation works through expected future short-term interest rates, whereas the effects of noisy news through the term premium channel are generally ambiguous.

quarters.

Finally, Online Appendix Figure C.6 reports the responses of risk premia, measured as VAR-implied expected excess returns on equities and long-term bonds. Both respond to our identified disturbances, though the relation is somewhat less robust than we find for asset price levels. These responses are consistent with the evidence of comovement between excess equity and currency returns (e.g. Verdelhan, 2010; Hau and Rey, 2006) as well as between excess bond and currency returns (e.g. Lustig et al., 2019; Greenwood et al., 2023; Gourinchas et al., 2025; Lloyd and Marin, 2020; Kekre and Lenel, 2024).

4.3 Exchange rates and future fundamentals

Our empirical results reveal a connection between exchange rate fluctuations and imperfectly forecasted changes in future TFP. In this section we discuss possible explanations for why this relationship has largely been overlooked in the empirical literature. Along the way, we provide further intuition on the key empirical features that drive our results.

Our initial observation is that many previous studies start with the premise that technological disturbances arrive as pure surprises, as in typical macroeconomic theories. From that point of view, one would only look for a relationship between the exchange rate and current and past TFP (or current and past macro aggregates more generally). Regressions of real exchange rate changes on past variables, however, typically reveal a very weak relationship.

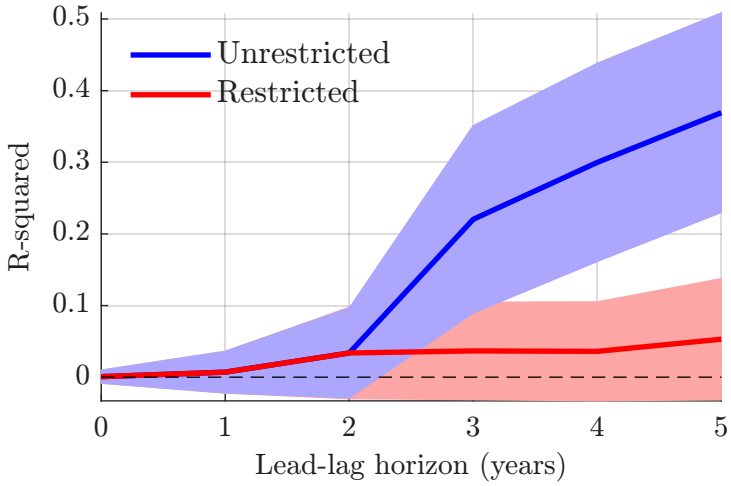
According to our empirical estimates, however, the real exchange rate *leads* TFP changes, rather than lagging them. Do simpler empirical approaches also capture the same relationship? To explore this question, we consider the following exercise, in which we regress the annual change in the real exchange rate on leads and lags of the annual change in TFP:

$$\Delta q_t = \alpha + \beta_0 \Delta TFP_t + \sum_{\ell=1}^{H} \beta_{-\ell}^{lag} (\Delta TFP_{t-\ell}) + \sum_{j=1}^{H} \beta_j^{lead} (\Delta TFP_{t+j}) + \epsilon_t. \tag{11}$$

If we estimate (11) with only a constant and ΔTFP_t on the right-hand side, the regression captures the standard relationship between changes in the exchange rate and contemporaneous TFP. If we then add the first summation term on the right-hand side to the regression, it captures the additional explanatory power of lagged changes in TFP of up to H years in the past. When we include the second summation term, it captures a potential correlation between the current exchange rate and *future* TFP changes, up to H years ahead.

Figure 6 reports the R^2 from two versions of regression (11): a "Restricted" backward-looking version including only current and lagged TFP growth (red line), and an "Unrestricted" version including all right-hand-side terms (blue line).

Figure 6: Real exchange rate growth and leads and lags of TFP growth



Notes: The figure reports the R^2 of regression (11). The right-hand side includes either present and past TFP ("Restricted"), or present, past, and future TFP ("Unrestricted"). The sample period is 1976:Q1–2008:Q2.

The R^2 of the purely backward-looking regression is statistically indistinguishable from zero regardless of the number of lags, reflecting the typical "disconnect" result. The picture changes once terms capturing *future* TFP growth are included. The relationship between exchange rate changes and TFP growth remains insignificant if only TFP growth up to two years ahead is included, but becomes increasingly significant when including growth three to five years out. Hence, a simple regression reveals that exchange rates do lead TFP, consistent with the predictive patterns underlying our VAR results.[22]

These results further highlight that exchange rates contain substantial information about future TFP growth in the *medium-to-long horizon*. The long-horizon nature of the noisy news we identify is particularly relevant in relation to the existing literature. While the existing literature recognizes that, as a forward-looking asset price, the exchange rate should lead macroeconomic fundamentals, it has also generally struggled to find robust correlations between exchange rates and future fundamentals.

Engel and West (2005) find a weak predictive relationship between exchange rates and future "macroeconomic fundamentals" using Granger causality regressions. Their regressions focus on quarterly exchange rate changes and forecast horizons of up to one year for variables such as GDP and interest rate differentials.[23] We offer three observations with regard to

[22] Online Appendix Table C.8 reports p-values from a Wald test of Granger causality based on (11), for both the aggregate of the other G7 economies and individual countries across annual horizons. We reject the null in favor of exchange rates Granger-causing TFP for most countries at horizons beyond two years.

[23] Sarno and Schmeling (2014) adopt a more non-parametric approach, but still limit their null hypothesis to testing predictive relationships to a maximum of two years ahead.

this evidence. First, given that our estimated news informs about low-frequency dynamics, we observe that *short-horizon* regressions tend to yield limited predictive power in finite samples. Second, we note that Granger causality should not be expected to manifest for *all* macroeconomic fundamentals considered by Engel and West (2005), even when TFP expectations are important drivers of exchange rates. Finally, we observe that the *noise* disturbances we estimate play the role of measurement error in Granger causality regressions, reducing their ability to identify the link between exchange rates and future productivity.

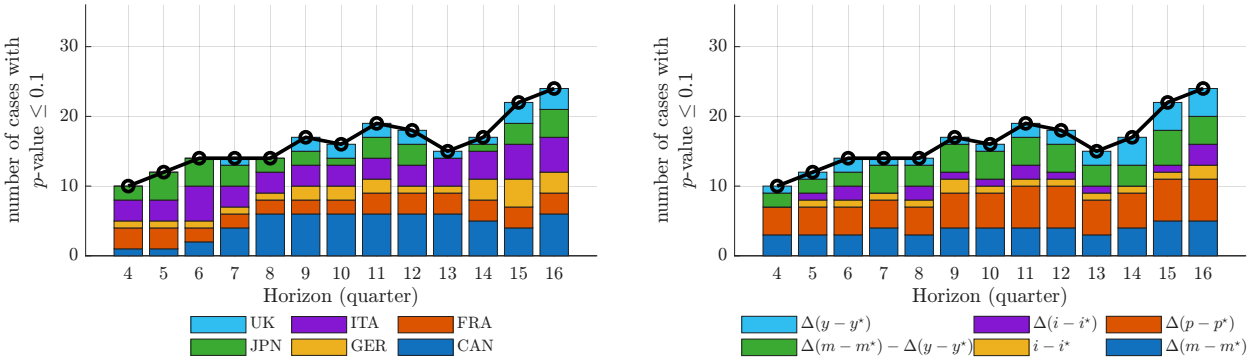
To the first observation, Figure 7 summarizes the significance of Granger causality tests for a range of variables and countries, using the 1974:Q1–2001:Q3 sample and variables originally considered by Engel and West. All regressions are estimated with Newey and West (1987) standard errors to account for serial correlation in errors. Consistent with our hypothesis that lower-frequency relationships are key, the figure shows that the predictive relationships studied in Engel and West strengthen at longer horizons, which they did not consider. A similar pattern emerges when we reproduce these tests using an updated version of their dataset, spanning 1976:Q1–2020:Q1 (see Online Appendix Figure C.14). This pattern is intuitive and consistent with a noisy-news view: long-horizon exchange rate changes reflect accumulated information about future macro movements, when the influence of transitory noise tends to average out.

To the second observation, Figure 7 shows that Granger causality results are not uniformly significant: the exchange rate–fundamentals relationship is stronger for output, money, and price differentials, but weaker for interest rate differentials. This pattern makes sense, however, because different macro variables respond to news at different horizons. For example, if both exchange rates and interest rate differentials respond to news about future TFP at the same time, the current exchange rate would *not* contain information about future interest differentials (independent of current interest differentials) and exchange rate changes would *not* Granger-cause interest differentials. In general, we should expect stronger evidence of the exchange rate Granger-causing slower-moving fundamentals, such as TFP or consumption differentials, and this is what our analyses show.

Finally, we observe that expectational noise (i.e. ε_t^v) generates changes in TFP forecasts that do not materialize. Since exchange rates respond to expectations regardless of their subsequent materialization, simple correlations between realized fundamentals and exchange rates are likely to understate the role of expected TFP fluctuations. In Granger causality regressions, noise disturbances play the role of measurement error, and may bias these regressions toward zero. This is especially true at high frequencies, where our results indicate the impact of the noise disturbances on the exchange rate is the strongest.

This final observation also helps explain why studies using survey-based macro expec-

Figure 7: Bivariate Granger causality tests *à la* Engel and West (2005) (1974:Q1–2001:Q3)



(a) Granger causality tests, by country

(b) Granger causality tests, by variable

Notes: For each horizon, we report the number of cases in which the p-value from a Wald test of Granger causality is below 0.1. There are 36 cases per horizon: six other G7 economies (each compared to the US) and six macroeconomic fundamentals, following the data and methodology of Engel and West (2005). Each test is based on a bivariate VAR estimated using the quarterly change in the nominal exchange rate, Δs, and one fundamental variable, f. The horizon on the x-axis indicates the number of lags included in the VAR. The reported p-value corresponds to a Wald test of the joint significance of the lags of Δs in predicting f. The variables are defined as follows: Δm is the percentage change in M1; Δy is the percentage change in real GDP; Δp is the percentage change in consumer prices; and i is the three-month eurocurrency deposit rate. The six fundamentals are the home-minus-foreign differences of these series: $\Delta(m-m^\star)$, $\Delta(y-y^\star)$, $\Delta(p-p^\star)$, $i-i^\star$, $\Delta(i-i^\star)$, and $\Delta(m-m^\star)-\Delta(y-y^\star)$. All series are from the Engel and West (2005) replication data. Following Engel and West (2005), exchange rates and interest rates are measured at the end of the quarter, rather than as the quarterly averages used in the VAR. Newey-West standard errors are computed using a data-driven bandwidth, a Bartlett kernel, and prewhitening. Panel (a) highlights the cases in which the p-value from the Wald test of Granger causality is below 0.1, *by country*, while panel (b) highlights the corresponding cases *by variable*.

tations, including those based on macro announcements, often find a stronger connection between exchange rates and fundamentals (e.g. Engel et al., 2008; Stavrakeva and Tang, 2026; Cormun and Ristolainen, 2026). To connect our findings to this literature, we compare the quarterly trade-weighted exchange rate changes implied by our technological and noise disturbances with the "exchange rate news index" of Stavrakeva and Tang (2026), which aggregates daily announcement effects to a quarterly frequency. Over the overlapping 2002:Q2–2020:Q1 period, using our extended-sample estimates, the two series have a correlation of approximately 0.32, indicating that TFP news partly overlaps with macro announcements but also contains distinct information; similarly, macro announcements reflect revisions in future TFP expectations, but not exclusively.

4.4 Additional observations on the identification of news

Proxies of technological innovation In our baseline analysis, our measure of total factor productivity is Fernald's utilization-adjusted TFP. The literature has often considered

various proxies of technological innovation such as the number of patent applications, the value of granted patents, the adoption of new ICT standards, or R&D expenditure. In Online Appendix Figure C.15 we plot the responses of these proxies of technological innovation to our identified disturbances. We make two observations. First, these measures rise in response to both technological and noise disturbances. This result is natural: since not every innovation has the same impact on eventual productivity, these measures are best interpreted as a noisy signal of future productivity and should be responsive to both technological *and* noise disturbances. Since they contain informational noise, however, it would be incorrect to use these series as direct proxies for *realized* TFP in our identification strategy. Second, Online Appendix Figure C.15 shows that these variables react relatively little *in anticipation* of realized productivity. This pattern suggests that these variables capture a relatively small share of anticipated TFP movements. Using these series as direct proxies for *expected* TFP changes would also likely give misleading results.[24]

Estimation of the VAR without the exchange rate The reader may wonder whether it is conceptually appropriate to include the real exchange rate in our structural VAR procedure, since any identified disturbance is allowed to load on the reduced-form residuals from all variables in the VAR, including those from the exchange rate equation. We argue that it is indeed conceptually correct to include the exchange rate in the VAR. The reason is that we want to test the hypothesis that the exchange rate responds to the structural disturbances of interest. If the exchange rate does respond to the disturbances, then its reduced-form residuals contain information about those disturbances, as well as any others that influence the economy. It is then the role of the identification procedure to extract the information from the set of reduced-form residuals of all variables and map this into structural disturbances.

A distinct, but related, question is how precisely the exchange rate contributes to our estimates of technological and noise disturbances. To answer this question, we use the baseline VAR estimates to compute the coefficients of a restricted VAR system that excludes the exchange rate. The details of this procedure are described in Online Appendix B.2. Since there is no exchange rate equation in the restricted VAR, no reduced-form exchange rate residual will contribute to the estimated disturbance. At the same time, because the exchange rate is included in the estimation of the unrestricted system, it is possible to compute the exchange rate response to the disturbance without doing any additional estimation step, and the results are directly comparable to our baseline.

[24] To ensure that our procedure does not inadvertently capture the endogenous productivity effects of non-technological disturbances, we verify that the correlations between our identified technological and noise disturbances and US monetary policy shocks—measured using the "high-frequency approach" of Gertler and Karadi (2015)—are close to zero and statistically insignificant.

Online Appendix Figure C.12 compares the resulting structural disturbances to the baseline, while Online Appendix Figure C.13 reports the implied exchange rate impulse responses. The correlation for the technological disturbance is high (above 0.99) and the response to the technological disturbance is essentially unchanged relative to the baseline. This extremely close match for the technological disturbance is natural because its recoverability requirement is less stringent: one only needs the restricted VAR to accurately reflect the (unrestricted) autocovariances of TFP. By contrast, for the noise series, the extracted series correlation of 0.82 is high-but-imperfect and the impulse responses are somewhat different. The reason is that recovering the noise disturbances requires the VAR to accurately reflect both autocovariances *and* expected productivity. Evidently, removing the exchange rate from the information set changes the forecast of TFP in a non-trivial, albeit modest, way. Still, the other endogenous variables in the VAR (e.g. investment, interest rates) also contain information about future TFP, hence the restricted VAR still has statistical power to (partially) identify the underlying noisy news.

Another indication that the exchange rate embeds additional information about future TFP is the fact that the exchange rate in Online Appendix Figure C.13 moves prior to the realization of a noise disturbance. This means that endogenous variables respond before the measured expectation has changed, indicating that agents in the economy hold more information than is reflected in the restricted VAR.

Alternative specifications of the baseline VAR Online Appendix Figures C.7 and C.8 report impulse responses to technological and noise disturbances under alternative VAR specifications. One specification considers a VECM with rank-four cointegration, following the Bayesian VECM approach of Villani (2005). Panel A of Online Appendix Tables C.6 and C.7 reports the corresponding variance shares explained by the main exchange rate shock and by the technological and noise disturbances, respectively. The other specifications augment the baseline VAR by adding one set of variables at a time: the trade balance, equity prices, long-term bond yields, or the foreign Solow residual. Online Appendix Figures C.7 and C.8 show that the impulse responses implied by these additional specifications are very similar to those obtained from the baseline VAR.

We also estimate six bilateral VARs between the US and each of the other G7 economies. Online Appendix Figures C.10 and C.11 show that the conclusions emerging from these bilateral VARs are similar, and Panel B of Online Appendix Tables C.6 and C.7 reports the variance shares, as medians across the six bilateral VARs.

5 Implications for open-economy models

Our empirical analysis shows that noisy news about future TFP plays an important role in driving exchange rate fluctuations. Anticipation of future home TFP improvements leads to a relative increase in home consumption and an appreciation of the exchange rate. When the (imperfectly) anticipated TFP change materializes, home consumption remains elevated and the dollar remains appreciated, with a gradual depreciation thereafter. In this section, we explore whether leading open-economy models can match these patterns.

We consider two classes of two-country endowment economies. The first class assumes complete international financial markets, and encompasses the time-separable constant relative risk aversion (CRRA) preferences of Backus et al. (1994) and the recursive Epstein-Zin preferences of Colacito and Croce (2011, 2013). The second class maintains time-separable CRRA preferences, but assumes incomplete international asset markets and a debt-elastic interest rate premium. These features introduce endogenous deviations from uncovered interest parity, a feature shared with Itskhoki and Mukhin (2021) and Kekre and Lenel (2024).

All of the models we explore face qualitative challenges in reproducing our main empirical findings. We characterize these challenges with two propositions about linearized versions of the economies. Our propositions are stated in terms of the expectations response function defined by Chahrour and Jurado (2025). This approach isolates the theoretical effects of changes in expectations without making particular assumptions about the information set of agents or the dynamics of the productivity process. We conclude with a quantitative demonstration of those challenges using realistically calibrated versions of each model.

The models share a common physical environment. It consists of two countries, with exogenous endowments given by A_t and $A_t^\star$, respectively. The representative household of each country consumes a CES bundle of the two world goods,

$$C_t = \left(\varsigma^{\frac{1}{\sigma}} X_t^{\frac{\sigma-1}{\sigma}} + (1-\varsigma)^{\frac{1}{\sigma}} Y_t^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \quad C_t^\star = \left((1-\varsigma)^{\frac{1}{\sigma}} (X_t^\star)^{\frac{\sigma-1}{\sigma}} + \varsigma^{\frac{1}{\sigma}} (Y_t^\star)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \quad (12)$$

where $\{X_t, Y_t\}$ and $\{X_t^\star, Y_t^\star\}$ denote the time-t consumption of goods X and Y in the home and foreign countries, respectively, $\sigma > 0$ is the cross-good price elasticity and home bias is parameterized by $\varsigma > 0.5$.[25] The real exchange rate is given by $Q_t = P_t^\star / P_t$, where P_t and $P_t^\star$ are the price indices of the home and foreign baskets, respectively. Households maximize the present value of lifetime utility and the aggregate resource constraint implies

$$A_t \geq X_t + X_t^\star, \qquad A_t^\star \geq Y_t + Y_t^\star. \quad (13)$$

[25] We assume throughout that $\sigma > 1 - \frac{1}{2\varsigma}$, which is necessary and sufficient for the economy to have a unique non-stochastic steady state (see Corsetti et al., 2008b; Bodenstein, 2011).

The steady state is symmetric. We outline the environment in more detail in Online Appendix D.1.

The real exchange rate under complete markets Let M_t and $M_t^\star$ denote the home and foreign stochastic discount factors (SDFs), each expressed in units of the corresponding country's consumption basket. Complete markets imply the risk-sharing condition:

$$\frac{Q_t}{Q_{t-1}} = \frac{M_t^\star}{M_t}. \tag{14}$$

Under Epstein-Zin preferences, which nest the CRRA utility as a special case, the home stochastic discount factor is

$$M_t = \beta \left(\frac{C_t}{C_{t-1}}\right)^{-\frac{1}{\psi}} \left(\frac{U_t}{\left(\mathbb{E}_{t-1}[U_t^{1-\gamma}]\right)^{\frac{1}{1-\gamma}}}\right)^{\frac{1}{\psi}-\gamma}, \tag{15}$$

where γ denotes the coefficient of relative risk aversion, while ψ denotes the elasticity of intertemporal substitution. The foreign discount factor is analogous.

In equation (15), U_t is lifetime utility and $\left(\mathbb{E}_{t-1}[U_t^{1-\gamma}]\right)^{\frac{1}{1-\gamma}}$ its certainty equivalent. The presence of these terms distinguishes recursive preferences from time-separable preferences. Under CRRA utility, $\gamma = 1/\psi$, these terms disappear and the SDF depends only on realized consumption growth. By contrast, when $\gamma \neq 1/\psi$, revisions in expected future consumption growth affect *current* marginal utility. In particular, most of the asset-pricing literature, including Colacito and Croce (2011), assumes that $\gamma > 1/\psi$, which implies that agents prefer an early resolution of uncertainty. In this case, positive news about future consumption raises U_t above its certainty equivalent and results in lower M_t.

Defining $z_t \equiv \log Z_t - \log Z$ as the log deviation of any variable Z_t from its steady state Z, equations (14) and (15) together imply that the real exchange rate satisfies:[26]

$$\begin{aligned} \Delta q_t &= \frac{1}{\psi}\Delta\left(c_t - c_t^\star\right) + \left(\gamma - \frac{1}{\psi}\right)\mathcal{N}_t, \\ \mathcal{N}_t &\equiv \left[\log U_t - \log\left(\mathbb{E}_{t-1}[U_t^{1-\gamma}]\right)^{\frac{1}{1-\gamma}}\right] - \left[\log U_t^\star - \log\left(\mathbb{E}_{t-1}[(U_t^\star)^{1-\gamma}]\right)^{\frac{1}{1-\gamma}}\right]. \end{aligned} \tag{16}$$

Equation (16) highlights that under recursive utility ($\gamma \neq 1/\psi$) the relative surprise in lifetime utility $\mathcal{N}_t$ can change equilibrium comovements between the real exchange rate and relative consumption.

For intuition, note that, to a first-order approximation, the innovation to the relative

[26] Derivations of all first-order relationships are presented in Online Appendix D.1.

lifetime utility term, $\mathcal{N}_t$, is equal to the revision in the present discounted value of future consumption differential growth:

$$\mathcal{N}_t = \sum_{j=0}^{\infty} \beta^j \left(\mathbb{E}_t - \mathbb{E}_{t-1}\right) \left[\Delta c_{t+j} - \Delta c^\star_{t+j}\right]. \tag{17}$$

Furthermore, goods-market clearing and goods' demand functions imply

$$c_t - c^\star_t = \frac{1}{2\varsigma - 1} \left(a_t - a^\star_t\right) - \frac{4\varsigma\sigma(1-\varsigma)}{(2\varsigma-1)^2} q_t, \tag{18}$$

where a_t and $a^\star_t$ denote the log of the home and foreign good endowments, which we refer to as productivity going forward. Combining these equations, we obtain Proposition 1.

Proposition 1. *Suppose $\gamma \geq \frac{1}{\psi}$. Then, to a first-order approximation,*

1. *An* **expected future** *increase in relative productivity causes a decrease in relative consumption and a depreciation of the home real exchange rate today. That is,*

$$\frac{\partial(c_t - c^\star_t)}{\partial \mathbb{E}_t[a_{t+j}]} \leq 0 \quad \textit{and} \quad \frac{\partial q_t}{\partial \mathbb{E}_t[a_{t+j}]} \geq 0, \quad \forall j > 0.$$

   Moreover, both derivatives are zero if and only if $\gamma = 1/\psi$ (the CRRA case).

2. *A* **current** *increase in relative productivity causes an increase in relative consumption and a depreciation of the real exchange rate. That is,*

$$\frac{\partial(c_t - c^\star_t)}{\partial a_t} > 0 \quad \textit{and} \quad \frac{\partial q_t}{\partial a_t} > 0.$$

Proposition 1 shows that our empirical estimates pose two important *qualitative* challenges for complete-markets models. First, in anticipation of future productivity improvements, current consumption either does not move (under CRRA preferences) or moves in the wrong direction, with home relative consumption falling and the real exchange rate depreciating. Second, upon an actual improvement of home productivity, the real exchange rate depreciates. Both of these patterns are inconsistent with the empirical effects we uncover in Section 3.

To understand the effects of changes in expectations about the future (Clause 1 of Proposition 1), combine (16) and (18) to get

$$\Delta\left(c_t - c^\star_t\right) = \frac{2\varsigma - 1}{\Theta} \Delta\left(a_t - a^\star_t\right) - \frac{4\varsigma\sigma(1-\varsigma)\left(\gamma - 1/\psi\right)}{\Theta} \mathcal{N}_t, \tag{19}$$

$$\Delta q_t = \frac{2\varsigma - 1}{\psi\Theta} \Delta \left(a_t - a_t^\star \right) + \frac{(2\varsigma - 1)^2 \left(\gamma - 1/\psi \right)}{\Theta} \mathcal{N}_t, \tag{20}$$

where $\Theta \equiv (2\varsigma - 1)^2 + 4\varsigma\sigma(1-\varsigma)/\psi > 0$. We can then express the lifetime utility innovation $\mathcal{N}_t$ from (17) as a revision in the expected future growth of the productivity differential:[27]

$$\mathcal{N}_t = \frac{2\varsigma - 1}{\Theta_\gamma} \sum_{j=0}^{\infty} \beta^j \left(\mathbb{E}_t - \mathbb{E}_{t-1} \right) \left[\Delta a_{t+j} - \Delta a_{t+j}^\star \right], \tag{21}$$

where $\Theta_\gamma \equiv (2\varsigma - 1)^2 + 4\varsigma\sigma(1-\varsigma)\gamma > 0$. Equation (21) says that favorable news about future home productivity makes the lifetime utility innovation positive, $\mathcal{N}_t > 0$. When $\gamma > 1/\psi$, (19) and (20) imply that this change in continuation value lowers the consumption differential and depreciates the real exchange rate, even though current endowments are unchanged.

Intuitively, this pattern emerges because the positive news itself is a resolution of uncertainty that benefits the home country and directly lowers current-period marginal utility. To compensate foreign agents for this change in relative utility, the social planner transfers current resources to them. Because of home bias in preferences, market clearing then requires a real exchange rate depreciation.

Clause 2 of Proposition 1, instead, is a well-known limitation of complete-markets models, and is the reason a number of studies look toward incomplete markets as a potential resolution of the Backus-Smith Puzzle (e.g. Corsetti et al., 2008b, and subsequent literature).

The real exchange rate under incomplete markets We next consider a model with CRRA preferences and incomplete international asset markets, in which agents trade a single international bond and countries face a debt-elastic interest rate premium. The debt-elastic interest rate premium guarantees the stationarity of debt dynamics (Schmitt-Grohé and Uribe, 2003). It also generates uncovered interest parity violations that depend on the country's external position, a relationship that is also present in the segmented-market models of Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021), and Kekre and Lenel (2024). Those papers, however, use this framework to study the propagation of different shocks—e.g. financial shocks in Itskhoki and Mukhin (2021) and discount-rate shocks in Kekre and Lenel (2024)—whereas here we ask how productivity and productivity expectations transmit in this class of models.

With incomplete asset markets, the risk-sharing condition in (14) no longer applies. By

[27] Apply $(\mathbb{E}_t - \mathbb{E}_{t-1})$ to (19), substitute into (17), and solve for $\mathcal{N}_t$, using $\mathbb{E}_t[\mathcal{N}_{t+j}] = 0$ for all $j > 0$.

combining the households' Euler equations, we obtain a condition that resembles (16),

$$q_t = \gamma\,(c_t - c_t^\star) - \sum_{j=1}^{\infty} \mathbb{E}_t[\lambda_{t+j}], \qquad \mathbb{E}_t[\lambda_{t+j}] = \phi\,\mathbb{E}_t[b^\star_{t+j}]. \tag{22}$$

Here λ_{t+j} is the $t+j$ excess foreign-currency return, defined as in Section 2. Because of the friction affecting cross-border holdings, this term is determined by net foreign assets, $b^\star_{t+j}$.[28]

Proposition 2 characterizes the effects of expected productivity in the incomplete-markets setup, in parallel to the complete-markets results in Proposition 1:

Proposition 2. *Suppose $\gamma(2\sigma\varsigma - 1) > 2\varsigma - 1$. Then, to a first-order approximation,*

1. *An* **expected future** *increase in relative productivity causes an increase in relative consumption and an appreciation of the home real exchange rate. That is,*

$$\frac{\partial(c_t - c_t^\star)}{\partial \mathbb{E}_t[a_{t+j}]} > 0 \quad \textit{and} \quad \frac{\partial q_t}{\partial \mathbb{E}_t[a_{t+j}]} < 0, \quad \forall j > 0.$$

2. *A* **current** *increase in relative productivity causes an increase in relative consumption and a depreciation of the real exchange rate. That is,*

$$\frac{\partial(c_t - c_t^\star)}{\partial a_t} > 0 \quad \textit{and} \quad \frac{\partial q_t}{\partial a_t} > 0.$$

The results in Proposition 2 show that the incomplete-markets model can address some of the qualitative challenges faced by the complete-markets model. In particular, unlike the complete-markets environment, the incomplete-markets model can exhibit a high consumption differential and an appreciated real exchange rate in *anticipation* of a productivity improvement. Yet, the incomplete-markets model still delivers counterfactual implications once the productivity improvement *materializes*.

To gain intuition for the partial success of this model, combine the modified risk-sharing condition (22) with the market clearing condition (18), to obtain the following counterparts to the complete-markets equations (19)–(20):

$$(c_t - c_t^\star) = \frac{2\varsigma - 1}{\Theta_\gamma}\,(a_t - a_t^\star) + \frac{4\varsigma\sigma(1-\varsigma)}{\Theta_\gamma}\,\phi \sum_{j=1}^{\infty} \mathbb{E}_t[b^\star_{t+j}], \tag{23}$$

$$q_t = \frac{\gamma(2\varsigma - 1)}{\Theta_\gamma}\,(a_t - a_t^\star) - \frac{(2\varsigma - 1)^2}{\Theta_\gamma}\,\phi \sum_{j=1}^{\infty} \mathbb{E}_t[b^\star_{t+j}], \tag{24}$$

[28] The term $b^\star_{t+1}$ denotes the net amount of bonds issued by the foreign agent. In equilibrium, this is the net foreign asset position of the home country.

where Θ_γ is as defined above. Equations (23)–(24) take the same form as their complete-markets counterparts (19)–(20) but carry a different wedge. The two models thus deliver different implications insofar as their respective wedges respond differently to productivity.

How does the infinite sum in (23) and (24) respond to news? Intuitively, an upward revision in expected future home productivity raises the permanent income of home households. When the elasticity of intertemporal substitution is sufficiently low (the condition in Proposition 2), the home household prefers to raise its consumption today.[29] Since current endowments are unchanged, equation (23) implies that $\sum_{j=1}^{\infty} \mathbb{E}_t[b^\star_{t+j}]$ must rise. Meanwhile, the home bias in preferences requires a real exchange rate appreciation, as per (24).

Clause 2 of Proposition 2 shows that the incomplete-markets framework still faces a major empirical challenge. In response to a realized productivity improvement, the model still predicts an immediate depreciation of the real exchange rate—that is, the direct effect of the increase in supply of home goods dominates the increase in demand from the home agents, and the real exchange rate depreciates (while the consumption differential also rises). This is contrary to our empirical evidence, which shows that the real exchange rate is appreciated when TFP is high. This failure to generate a simultaneous appreciation of the real exchange rate and a rise in home consumption following a productivity improvement is a well-known feature of these models, and one reason why the recent literature has focused on exploring different sources of exogenous variation.[30] Our empirical evidence argues for new models that not only propagate news about future TFP appropriately but also match the observation that the exchange rate remains appreciated after a productivity improvement. We leave the development of such models to future work.[31]

Numerical demonstration Figures 8a and 8b compare the implications of these model economies with our empirical responses for a specific calibration. In each of these models, we embed the same signal structure and process for productivity. In particular, for home productivity, we use the noise representation of the information structure proposed by Blanchard et al. (2013). This process is nested within the general structure of (6)–(8) and has

[29] The countervailing force is a substitution effect, since the home good is expected to be relatively inexpensive in the future. When the condition in the proposition is satisfied, the wealth effect dominates, and news about future endowments raises current consumption demand.

[30] Corsetti et al. (2008b) show that the depreciation can be overturned when the trade elasticity is very low, or when shocks are nearly permanent, so that the wealth effect of higher expected future productivity dominates. The first mechanism requires a particular equilibrium selection (Akinci and Chahrour, 2025) and is precluded by our parametric assumptions. The second mechanism is essentially the one captured by Clause 1 of Proposition 2, but our estimated information structure implies that most of the revision in expected productivity occurs before the improvement materializes.

[31] For recent extensions of incomplete-markets models that incorporate forms of idiosyncratic risk, see Huang et al. (2025), Marin and Singh (2025), and Acharya et al. (2026).

Table 4: Parameter calibration

Parameter	Description	Complete Markets		Incomp. Markets
		CRRA	EZ	CRRA
β	Subjective Discount Factor	0.99	0.99	0.99
γ	Relative Risk Aversion	5	5	5
ψ	Elas. of Intertemporal Subs.	$1/\gamma$	1.2	$1/\gamma$
ς	Home Bias	0.80	0.80	0.80
σ	Trade Elasticity	1	1	1
ϕ	Elas. of Int. Rate Premium	-	-	0.01

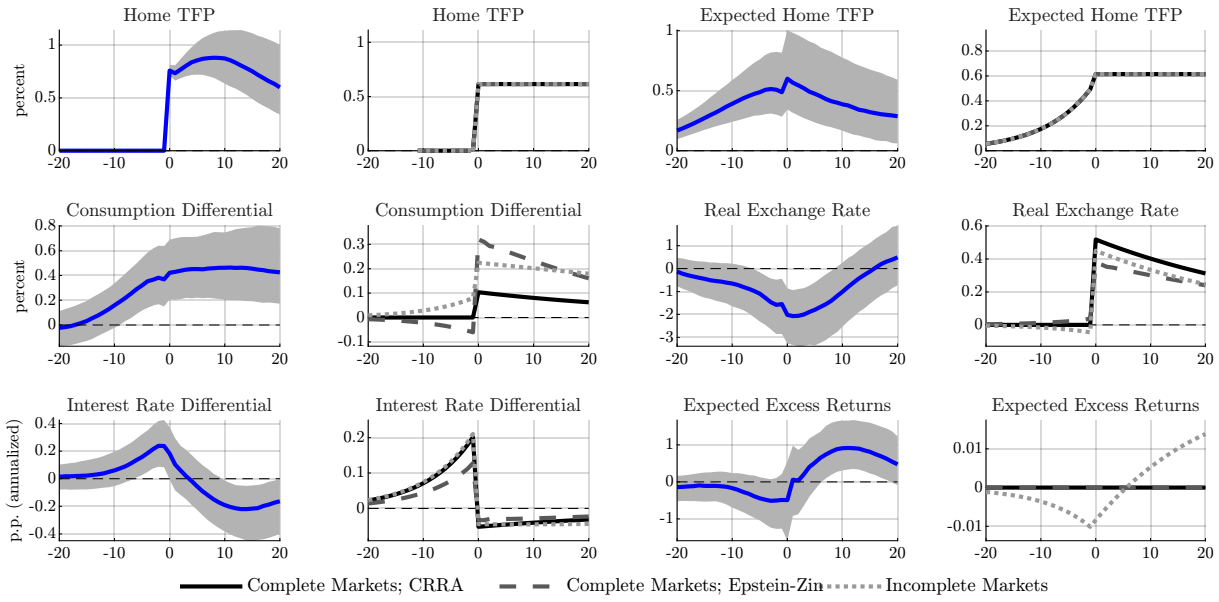
Notes: The models are calibrated at a quarterly frequency.

two useful properties: it introduces only three free parameters, and it can be calibrated to closely match the joint evolution of TFP and expectations about TFP that we observe in the data. We also assume that foreign productivity is weakly cointegrated with home productivity, converging to it at a rate of $\tau = 0.025$ per quarter. Details on the calibration of this information structure can be found in Online Appendix D.2.

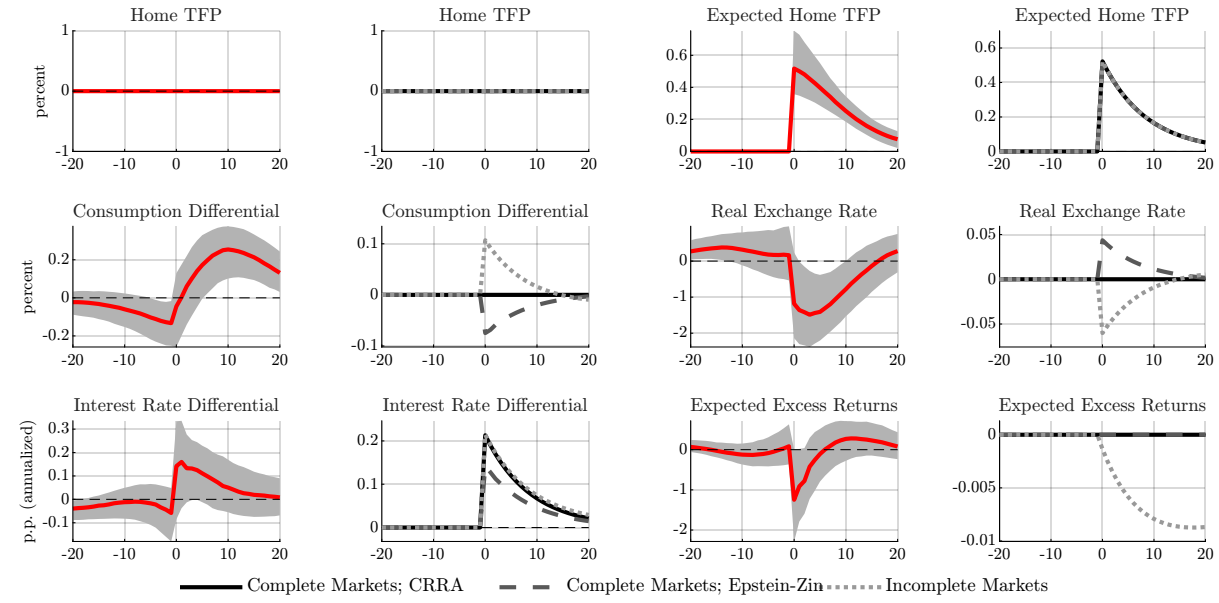
The remaining parameters are set to standard values. We set the discount factor $\beta = 0.99$ to match the quarterly frequency of our data and we set risk aversion $\gamma = 5$. We assume a trade elasticity $\sigma = 1$ and home bias in consumption preferences $\varsigma = 0.80$. With $\varsigma = 0.80$, exports plus imports amount to 40% of output in the models. Lastly, in the case of Epstein-Zin preferences, we set the elasticity of intertemporal substitution $\psi = 1.2$ as in Colacito et al. (2018). Table 4 summarizes our calibration. We solve all of the models using a first-order approximation, except for the version with Epstein-Zin preferences for which we use a third-order perturbation technique (though these additional terms turn out to be very small in our calibration).

The top row of Figures 8a and 8b compares the responses of TFP and expectations that we find in the data to those implied by our calibrated process. These figures show that the process for productivity and the news thereof match the data closely.

The figures show that during the period of anticipation of future productivity (periods -20 to -1 in Figure 8a and periods 0 to 20 in Figure 8b) the real exchange rate is unchanged in the complete-markets CRRA model, while it depreciates in the model with generalized Epstein-Zin preferences. Only the incomplete-markets model delivers, even qualitatively, an appreciation of the home currency, though in this calibration it is much smaller than we see in the data. Similarly, for the consumption differential, only the incomplete-markets model qualitatively matches the data during the anticipation phase.



(a) Technological disturbances



(b) Noise disturbances

Figure 8: Impulse responses to technological and noise disturbances: data vs models

Notes: Each variable is shown in a pair of panels: on the left, the empirical impulse response (posterior median, with the 16th–84th percentile band shaded); on the right, the impulse responses of the three models. The top row shows TFP and expected TFP, for which the three models coincide. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Details on the models can be found in Section 5 and Online Appendix D. Each period is a quarter.

While the incomplete-markets model succeeds qualitatively in the period of anticipation, Figure 8a shows that none of the three models matches the empirical patterns when improved productivity is realized (denoted by period 0). The models all imply a depreciation of the exchange rate, while in the data the exchange rate remains appreciated and only gradually depreciates after the change. Meanwhile, all three models roughly capture a similar pattern of first high, then lower, interest rate differentials.

The two benchmark classes thus display distinct limitations. Complete markets cannot rationalize the anticipation phase: with time-separable preferences nothing happens before TFP changes, and with recursive preferences relative consumption and the real exchange rate move in the wrong direction. The single-bond economy generates the anticipation phase with the right signs, but it joins the complete-markets economies in predicting a depreciation once the improvement materializes. These results thus suggest some limitations of current theory and, we hope, provide new moments for future researchers to incorporate in their work.

6 Conclusions

The empirical evidence in this paper supports the existence of a strong connection between exchange rates and noisy news about US productivity. Our finding that noisy news accounts for around half of the variation in the US/G7 real exchange rate, and a comparable share of variation in macroeconomic aggregates, contrasts with many earlier findings that the exchange rate is disconnected from macroeconomic fundamentals. Because exchange rates respond to productivity news before many other macro variables, we believe these previous findings primarily reflect timing differences rather than an absence of linkage.

Two features of this relationship between exchange rates and fundamentals have helped to obscure it in previous studies. First, the news concerns medium-to-long horizons, while the literature has focused on horizons of one to two years. When sufficient leads are included, even simple regressions of exchange rate changes on TFP growth detect a significant relationship. Second, TFP expectations are imperfect: expectational noise drives volatile, short-lived exchange rate movements that weaken the correlation between exchange rates and realized fundamentals.

The responses to noisy news also generate several well-known puzzles regarding exchange rate behavior, puzzles that leading open-economy models have been designed to explain. While these models can generate realistic unconditional moments, we show they cannot fully replicate the dynamic relationships between the exchange rate, macro variables, and productivity uncovered by our empirical procedure. None of the models we explore can jointly account for a real exchange rate appreciation in anticipation of higher domestic pro-

ductivity and for its persistence after a productivity improvement materializes. Our empirical estimates thus provide new discipline for theory, and we hope they will spur new research regarding the transmission of productivity changes in open-economy contexts.

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Online Appendix

A Data sources

- **US Total Factor Productivity.** Utilization-adjusted total factor productivity (TFP), quarterly, from Fernald (2012), 2026/06/04 vintage. Retrieved from John Fernald's website: https://www.johnfernald.net/TFP.

- **Nominal Exchange Rates.** Daily bilateral spot exchange rates for the six other G7 economies, expressed as USD per unit of foreign currency. For the transition to the euro, the data use a country's currency/euro conversion rate to construct a synthetic exchange rate for the whole period. Quarterly aggregation: period average over available daily observations. Retrieved from *LSEG Datastream*.

- **Nominal Interest Rates.** Daily three-month eurocurrency deposit rates (middle rates) for the G7 countries (`ECCAD3M`, `ECFFR3M`, `ECWGM3M`, `ECITL3M`, `ECJAP3M`, `ECUKP3M`, `ECUSD3M`). Quarterly aggregation: period average over available daily observations. Retrieved from *LSEG Datastream*. Figures 7 and C.14 instead use end-of-quarter values of both exchange rates and interest rates, following Engel and West (2005).

- **Consumer Price Indices.** Quarterly, not seasonally adjusted, all-items consumer price indices for the G7 countries. Source: OECD Consumer Price Indices database (https://data-explorer.oecd.org). Retrieved via the OECD SDMX API using MATLAB.

- **Consumption.** Real private final consumption expenditure, chain-linked volume, seasonally adjusted, quarterly. Source: *OECD Economic Outlook No. 119* (https://data-explorer.oecd.org). Retrieved via the OECD SDMX API using MATLAB.

- **Investment.** Real gross fixed capital formation, total, chain-linked volume, seasonally adjusted, quarterly. Source: *OECD Economic Outlook No. 119* (https://data-explorer.oecd.org). Retrieved via the OECD SDMX API using MATLAB.

- **Equity Prices and Returns.** Daily MSCI total-return indices in local currency for the G7 countries (`.TRXFLDUST`, `.TRXFLDDET`, `.TRXFLDJPT`, `.TRXFLDFRT`, `.TRXFLDGBT`, `.TRXFLDITT`, `.TRXFLDCAT`), deflated by each country's CPI. Quarterly aggregation: period average. Source: *MSCI*. Retrieved from *LSEG Datastream*. The equity price series is the MSCI total-return index, which cumulates returns with dividends reinvested.

- **Long-Term Bond Yields and Returns.** Monthly 10-year government bond yields (`IGUSA10D`, `IGDEU10D`, `IGJPN10D`, `IGFRA10D`, `IGGBR10D`, `IGITA10D`, `IGCAN10D`), price indices (`PRUSG10M`, `PRDEUGVM`, `PRJPNGVM`, `PRFRAGVM`, `PRGBRGVM`, `PRITAGVM`, `PRCANGVM`), and total-return indices (`TRUSG10M`, `TRDEUGVM`, `TRJPNGVM`, `TRFRAGVM`, `TRGBRGVM`, `TRITAGVM`, `TRCANGVM`) in local currency for the G7 countries. Source: *Global Financial Data* (purchased extract, 2026). Quarterly aggregation: three-month average of monthly levels, dropping quarters with a missing month.

- **US R&D.** Real gross domestic product for research and development, chained 2012 dollars, seasonally adjusted at annual rates, quarterly. Code: `Y694RX1Q020SBEA`. Source: *US Bureau of Economic Analysis*, retrieved from FRED (https://fred.stlouisfed.org/series/Y694RX1Q020SBEA).

- **US–G7 Goods Trade Balance.** Quarterly US bilateral goods trade balances with the six other G7 economies, in US dollars, summed across partners and divided by nominal US GDP. Source: *International Monetary Fund, International Trade in Goods by Partner Country (IMTS)*. Retrieved from https://data.imf.org/en/datasets/IMF.STA:IMTS.

- **Exchange Rate News Index.** Exchange rate news index for CAD, EUR, GBP, and JPY, quarterly, from Stavrakeva and Tang (2026). Provided by the authors.

- **Foreign Solow Residual.** Quarterly Solow residuals for the six other G7 economies, constructed following Monacelli et al. (2023). Source: *OECD Economic Outlook No. 119* (https://data-explorer.oecd.org). Retrieved via the OECD SDMX API using MATLAB. Each country's series starts when all of its inputs are available: Italy, Japan, and Canada in 1970:Q1, the United Kingdom in 1971:Q1, France in 1980:Q1, and Germany in 1991:Q1. The foreign aggregate is the trade-weighted average of the countries' quarterly log changes, with weights rescaled each quarter over the countries available, cumulated into a log level.

- **Proxies of Technological Innovation.**

  - **Number of US patent applications**: total patent applications filed at the US Patent and Trademark Office, from Marco et al. (2015). Log of monthly counts summed to quarterly totals, 1981:Q1–2014:Q4.
  - **Value of granted patents as a share of GDP**: the patent-based innovation index of Cascaldi-Garcia and Vukotić (2022). Log of the value of granted patents relative to GDP (their series `lxiq_pc`), 1976:Q1–2010:Q4.
  - **Adoption of new ICT standards**: number of new ICT standards released per quarter, from Baron and Schmidt (2014) (their series `count_basic`). Log of quarterly counts, 1976:Q1–2014:Q4.

- **Money (Narrow Aggregates).** Quarterly M1 for the US and the six other G7 economies, measured in the last month of the quarter, from 1976:Q1. These series are used in Online Appendix Figure C.14; Figure 7 uses the Engel and West (2005) replication data (`https://users.ssc.wisc.edu/~cengel/Data/Fundamentals/data.htm`) for all variables. Canada, Japan, and the US: OECD narrow money (M1), *Key Short-Term Economic Indicators*, throughout. France, Germany, and Italy: M1 from the Engel and West (2005) data through 1998:Q4, extended from 1999:Q1 with the growth of euro-area M1 (*OECD*, via FRED, `MANMM101EZM189S`), since national M1 is not published after the euro changeover; the series for France starts in 1978:Q1. United Kingdom: OECD narrow money (M1) from 1986:Q4, its first observation, extended back to 1976:Q1 with the growth of M4 (Bank of England, via FRED, `MSM4UKQ`).

B Additional details of empirical analyses

B.1 Identification of technological and noise disturbances

Let the n-dimensional VAR(p)

$$y_t = B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma$$

be stable. Its spectral density is

$$f_y(\lambda) = \frac{1}{2\pi}\left(I_n - B_1 e^{-\mathrm{i}\lambda} - \cdots - B_p e^{-\mathrm{i}p\lambda}\right)^{-1} \Sigma \left[\left(I_n - B_1 e^{-\mathrm{i}\lambda} - \cdots - B_p e^{-\mathrm{i}p\lambda}\right)^{-1}\right]^*. \quad \text{(B.1)}$$

Let $a_t = e_a y_t$ pick the technology series from y_t and let $b_t = \sum_{\ell=0}^{\infty} \beta_\ell \, y_{t-\ell}$ be the forecast of a_{t+h} generated by the VAR. Collecting, the spectral density of $\{a_t, b_t\}$ is

$$f(\lambda) = \begin{bmatrix} f_a(\lambda) & f_{ab}(\lambda) \\ f_{ba}(\lambda) & f_b(\lambda) \end{bmatrix} = \begin{bmatrix} e_a \\ \beta(\lambda) \end{bmatrix} f_y(\lambda) \begin{bmatrix} e_a' & \beta(\lambda)^* \end{bmatrix}, \quad \text{(B.2)}$$

where $\beta(\lambda)$ is the Fourier transform of $\{\beta_\ell\}$. The spectral density $f(\lambda)$ can be factorized

$$f(\lambda) \;=\; \frac{1}{2\pi}\,\varphi(\lambda)\,\varphi(\lambda)^*, \qquad \varphi(\lambda) = \begin{bmatrix} \varphi_{11}(\lambda) & 0 \\ \varphi_{21}(\lambda) & \varphi_{22}(\lambda) \end{bmatrix}. \quad \text{(B.3)}$$

Chahrour and Jurado (2022) show that this factorization (B.3) is unique and, because of the zero in the top-right of $\varphi(\lambda)$, satisfies the two identification assumptions stated in Section 3.

Recovering disturbances Let the vector of structural disturbances be $\varepsilon_t = (\varepsilon_t^a,\, \varepsilon_t^v)'$. The disturbances are recovered from observables via the frequency-domain filter

$$\varepsilon(\lambda) = \psi(\lambda)\, Y(\lambda), \qquad \psi(\lambda) = \varphi(\lambda)^{-1} \begin{bmatrix} e_a \\ \beta(\lambda) \end{bmatrix}. \quad \text{(B.4)}$$

Impulse responses and variance shares The impulse response of variable i at horizon j to a unit innovation in component s of ε_t is given by $\mathrm{IR}_{i,s}(j) = \mathrm{cov}(y_{i,t+j}, \varepsilon_{s,t})$. By the standard inversion from cross-spectra to cross-covariances,

$$\mathrm{IR}_{i,s}(j) \;=\; \int_{-\pi}^{\pi} e^{\mathrm{i}\lambda j}\; e_i\, f_y(\lambda)\, \psi(\lambda)^*\, e_s'\; d\lambda, \quad \text{(B.5)}$$

where e_i selects component i of y_t and e_s selects component s of ε_t. The share of the variance in the process $\{y_{i,t}\}$ due to the disturbance $\{\varepsilon_{s,t}\}$ over the frequency range $\Delta = [\lambda_1, \lambda_2]$ is given by

$$\mathrm{VS}_{i,s}(\Delta) = 2\pi \int_{\Delta} |e_i f_y(\lambda) \psi(\lambda)^* e_s'|^2 \, d\lambda \left(\int_{\Delta} f_{y,ii}(\lambda)\, d\lambda\right)^{-1}. \quad \text{(B.6)}$$

B.2 Estimation of the VAR without the exchange rate

Let $y_t \in \mathbb{R}^n$ follow a stable VAR(p)

$$y_t \;=\; \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} \;+\; u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma, \tag{B.7}$$

with MA (reduced-form) transfer function

$$C(z) \;\equiv\; \Big(I_n - B_1 z - \cdots - B_p z^p\Big)^{-1}, \qquad y(z) \;=\; C(z)\, u(z), \tag{B.8}$$

and spectral density

$$f_y(\lambda) \;=\; \frac{1}{2\pi}\, C(e^{-\mathrm{i}\lambda})\, \Sigma\, C(e^{-\mathrm{i}\lambda})^*. \tag{B.9}$$

Subset and selection matrices Partition variables into the kept subset S of size k and the excluded block Z of size $m = n - k$. Let $P_S \in \{0,1\}^{k\times n}$ select S so that $y_{S,t} = P_S y_t$, and $P_Z \in \{0,1\}^{m\times n}$ select Z. We will use the block-partitions

$$C(z) = \begin{bmatrix} C_{SS}(z) & C_{SZ}(z) \\ C_{ZS}(z) & C_{ZZ}(z) \end{bmatrix}, \qquad f_y(\lambda) = \begin{bmatrix} f_{SS} & f_{SZ} \\ f_{ZS} & f_{ZZ} \end{bmatrix}.$$

Step 1. Implied restricted VAR on the subset S

Population autocovariances of y_t from the unrestricted VAR Form the companion state $x_t = [y_t', y_{t-1}', \ldots, y_{t-p+1}']'$, with transition $x_{t+1} = F x_t + G u_{t+1}$,

$$F \;=\; \begin{bmatrix} B_1 & B_2 & \cdots & B_{p-1} & B_p \\ I_n & 0 & \cdots & 0 & 0 \\ & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & I_n & 0 & 0 \end{bmatrix}, \quad G \;=\; \begin{bmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Let $Q = G\,\Sigma\,G'$ and solve the discrete Lyapunov equation

$$\Sigma_x \;=\; F\,\Sigma_x\,F' + Q. \tag{B.10}$$

Then for $\ell \geq 0$,

$$\Gamma(\ell) \;\equiv\; \mathrm{cov}(y_t, y_{t-\ell}) \;=\; J F^\ell \Sigma_x J', \qquad J = \Big[I_n \;\; 0 \;\; \cdots \;\; 0\Big]. \tag{B.11}$$

Extract the subset autocovariances $\Gamma_S(\ell) = P_S\,\Gamma(\ell)\,P_S'$ for $\ell = 0, 1, \ldots, p$.

Block Yule–Walker projection (best linear VAR(p) for $y_{S,t}$) Let

$$\Gamma_{S,p} \;=\; \begin{bmatrix} \Gamma_S(0) & \Gamma_S(1) & \cdots & \Gamma_S(p-1) \\ \Gamma_S(1)' & \Gamma_S(0) & \cdots & \Gamma_S(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_S(p-1)' & \Gamma_S(p-2)' & \cdots & \Gamma_S(0) \end{bmatrix} \in \mathbb{R}^{kp\times kp}, \qquad G_p \;=\; \big[\,\Gamma_S(1)\;\cdots\;\Gamma_S(p)\,\big] \in \mathbb{R}^{k\times kp}.$$

Then the population OLS/Yule–Walker coefficients are

$$\big[\,\tilde{B}_1\;\cdots\;\tilde{B}_p\,\big] \;=\; G_p\,\Gamma_{S,p}^{-1} \quad\in\; \mathbb{R}^{k\times kp}, \tag{B.12}$$

and the implied restricted reduced-form covariance

$$\tilde{\Sigma} \;=\; \Gamma_S(0) \;-\; \sum_{\ell=1}^{p} \tilde{B}_\ell\,\Gamma_S(\ell)' \;=\; \Gamma_S(0) - \big[\tilde{B}_1\;\cdots\;\tilde{B}_p\big]\,[\Gamma_S(1)';\ldots;\Gamma_S(p)']. \tag{B.13}$$

Equations (B.12)–(B.13) define the *implied* restricted VAR(p) for $y_{S,t}$:

$$y_{S,t} \;=\; \sum_{\ell=1}^{p} \tilde{B}_\ell\, y_{S,t-\ell} \;+\; \tilde{u}_t, \qquad \mathbb{E}[\tilde{u}_t\tilde{u}_t'] = \tilde{\Sigma}. \tag{B.14}$$

The restricted transfer function and spectral density are

$$\tilde{C}(z) \;\equiv\; \big(I_k - \tilde{B}_1 z - \cdots - \tilde{B}_p z^p\big)^{-1}, \qquad f_S(\lambda) \;=\; \frac{1}{2\pi}\,\tilde{C}(e^{-\mathrm{i}\lambda})\,\tilde{\Sigma}\,\tilde{C}(e^{-\mathrm{i}\lambda})^*. \tag{B.15}$$

Step 2. Estimate impulse response functions

We use our identification procedure to identify the impulse responses of the subset of $k \le n$ variables, so that

$$\varepsilon(\lambda) = \psi_S(\lambda) Y_S(\lambda), \tag{B.16}$$

where $Y_S(\lambda) = P_S\, Y(\lambda)$. Thus,

$$\varepsilon(\lambda) = \psi_S(\lambda) P_S\, Y(\lambda). \tag{B.17}$$

The impulse response of all variables is the inverse Fourier transform of

$$T(\lambda) = \frac{f_{y\varepsilon}(\lambda)}{f_\varepsilon(\lambda)} = \frac{\mathbb{E}[Y(\lambda)Y(\lambda)^* P_S' \psi_S(\lambda)^*]}{f_\varepsilon(\lambda)} = \frac{f_y(\lambda) P_S' \psi_S(\lambda)^*}{f_\varepsilon(\lambda)}. \tag{B.18}$$

Note that when $k = n$, i.e. when we drop no variable from the VAR, P_S is an identity matrix and we are back to the full VAR identification.

B.3 Responses of variables not included in the VAR

Let x_t be the VAR's companion state, demeaned and standardized, which holds the data at t and its first $p-1$ lags, and let w_t be the demeaned external series. We assume the following statistical model, in which the error terms may be autoregressive:

$$w_t = x_t'\vartheta + \mu_t, \qquad \mu_t = \rho_1\mu_{t-1} + \rho_2\mu_{t-2} + \omega_t, \qquad \omega_t \sim \mathcal{N}(0, \sigma_\omega^2) \text{ i.i.d.} \tag{B.19}$$

Define $w_t^\dagger = w_t - \rho_1 w_{t-1} - \rho_2 w_{t-2}$ as the quasi-difference of w_t, and $x_t^\dagger$ in the same way. Then (B.19) becomes $w_t^\dagger = x_t^{\dagger\prime}\vartheta + \omega_t$, on $T_\dagger = T-2$ observations. Let X and w denote the stacked observations of $x_t^{\dagger\prime}$ and $w_t^\dagger$, respectively, given $\rho = (\rho_1, \rho_2)'$.

Priors. The priors are chosen to be conditionally conjugate and are characterized by two parameters: κ_ϑ and κ_ρ. Write $\vartheta = (\vartheta_0', \vartheta_1', \ldots, \vartheta_{p-1}')'$, where ϑ_ℓ multiplies the ℓ-th lag of the VAR's variables in the state. We take improper priors on the contemporaneous coefficients, $p(\vartheta_0) \propto 1$, and on the variance, $p(\sigma_\omega^2) \propto 1/\sigma_\omega^2$, and place mean-zero priors on the subsequent coefficient blocks according to

$$\vartheta_\ell \mid \sigma_\omega^2 \sim \mathcal{N}\Big(0,\ \frac{\sigma_\omega^2}{T_\dagger\,\kappa_\vartheta\,\ell^2}\,I\Big), \qquad \rho_1, \rho_2 \sim \mathcal{N}(0, \kappa_\rho^2),$$

with ρ restricted to the stationary region. We set $\kappa_\vartheta = 0.01$ and $\kappa_\rho = 0.05$, except for the two series that are led one quarter (expected excess bond and equity returns), for which $\kappa_\vartheta = \kappa_\rho = 0.05$.

Posterior draws. The sampler alternates two blocks, each of which is a standard distribution (Zellner, 1971; Chib and Greenberg, 1994).

1. $(\sigma_\omega^2, \vartheta)$ *given* ρ. With $K = \operatorname{diag}(0, \kappa_\vartheta, \ldots, \kappa_\vartheta(p-1)^2) \otimes I$, $M = X'X + T_\dagger K$ and $\bar\vartheta = M^{-1}X'w$,

   $$\sigma_\omega^2 \mid \rho, w \sim \mathcal{IG}\big(\tfrac{d}{2}, \tfrac{\mathrm{SSR}}{2}\big), \qquad \vartheta \mid \sigma_\omega^2, \rho, w \sim \mathcal{N}\big(\bar\vartheta,\ \sigma_\omega^2 M^{-1}\big),$$

   where $\mathrm{SSR} = (w - X\bar\vartheta)'(w - X\bar\vartheta) + T_\dagger\,\bar\vartheta' K \bar\vartheta$ and d is $T_\dagger$ less the number of coefficients with a flat prior.

2. ρ *given* ϑ *and* σ_ω^2. Let $\mu_t = w_t - x_t'\vartheta$, and let N hold its two lags and μ the values they explain. Then

   $$\rho \mid \vartheta, \sigma_\omega^2, w \sim \mathcal{N}(\bar\rho, V), \qquad V = \Big(\frac{N'N}{\sigma_\omega^2} + \frac{I}{\kappa_\rho^2}\Big)^{-1}, \qquad \bar\rho = V\,\frac{N'\mu}{\sigma_\omega^2},$$

   redrawn until it is stationary.

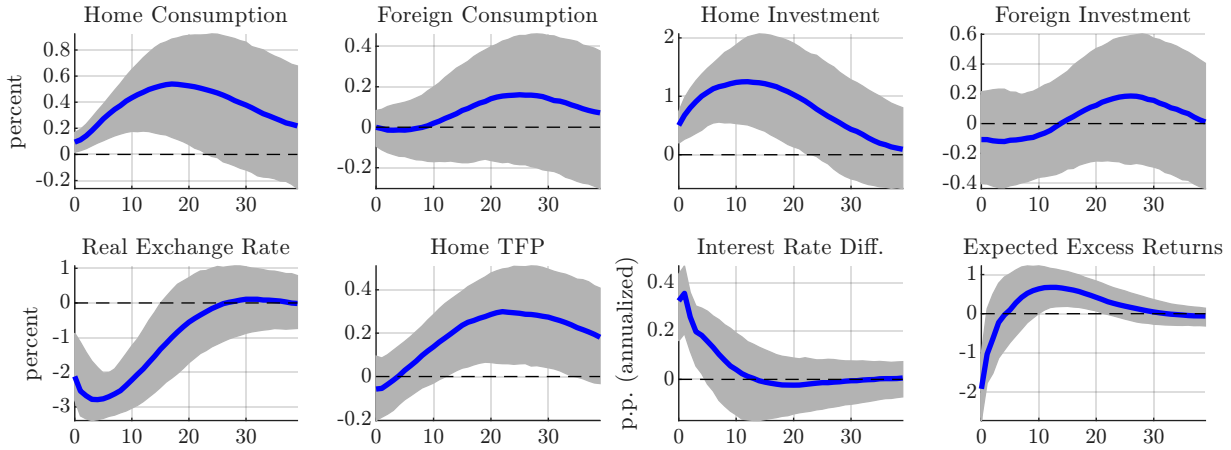
To get a consistent approximation of the posterior distribution, we start a chain from the least-squares estimate and burn in 2000 iterations before keeping every tenth draw thereafter.

C Alternative specifications

C.1 Extended sample results: 1978:Q4–2020:Q1

In the longer sample, our results remain similar. Our two estimated disturbances explain 48% of the variance of the real exchange rate at periodicities of 2–100 quarters, versus 62% in the baseline sample. The total shares for home consumption and investment are slightly higher, while those for foreign consumption and investment fall to around 40%. The two disturbances therefore remain important, and the impulse responses are qualitatively similar.

Figure C.1: Extended sample: impulse responses to the main exchange rate shock (ε_1)



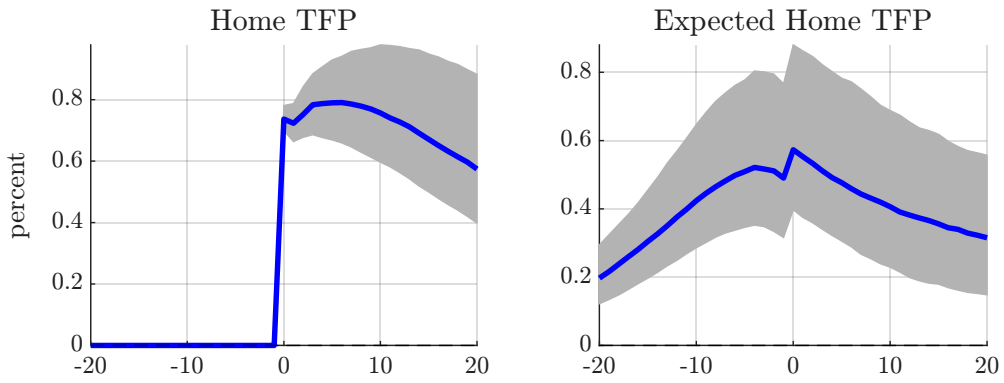
Notes: The figure reports the impulse responses to the main exchange rate shock, along with the 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Table C.1: Extended sample: share of forecast error variance explained by the main exchange rate shock (ε_1)

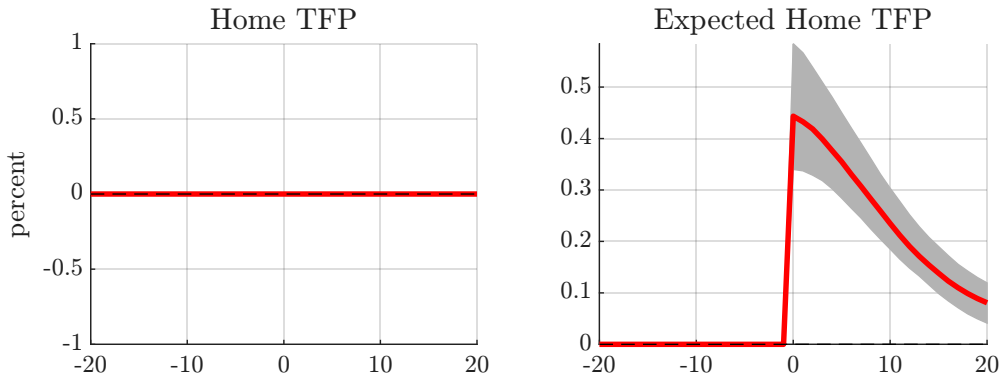
	Forecast Horizon (Quarters)					
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.02	0.03	0.08	0.23	0.34	0.31
Home Consumption	0.04	0.07	0.21	0.29	0.27	0.21
Foreign Consumption	0.02	0.02	0.04	0.07	0.09	0.13
Home Investment	0.17	0.23	0.38	0.42	0.38	0.33
Foreign Investment	0.04	0.04	0.06	0.09	0.11	0.15
Interest Rate Differential	0.31	0.35	0.31	0.30	0.31	0.33
Real Exchange Rate	0.44	0.61	0.77	0.70	0.67	0.65
Expected Excess Returns	0.52	0.30	0.28	0.33	0.34	0.34

Notes: The table reports the variance shares accounted for by the main exchange rate shock.

Figure C.2: Extended sample: dynamic responses of TFP and TFP expectations



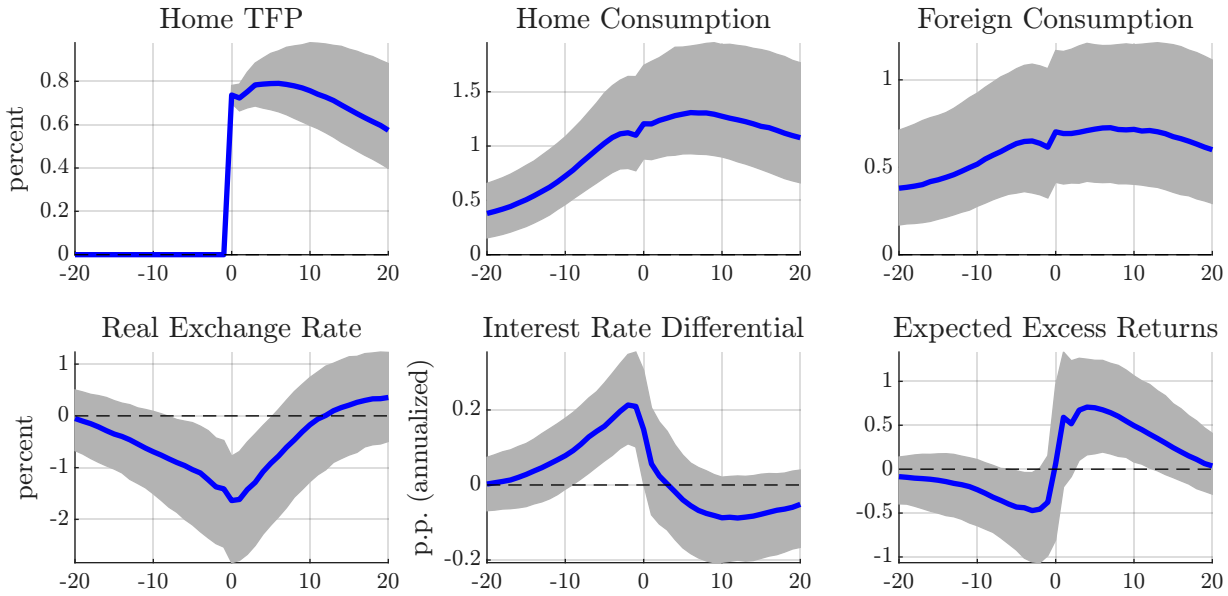
(a) Impulse responses to technological disturbances (ε^a)



(b) Impulse responses to noise disturbances (ε^v)

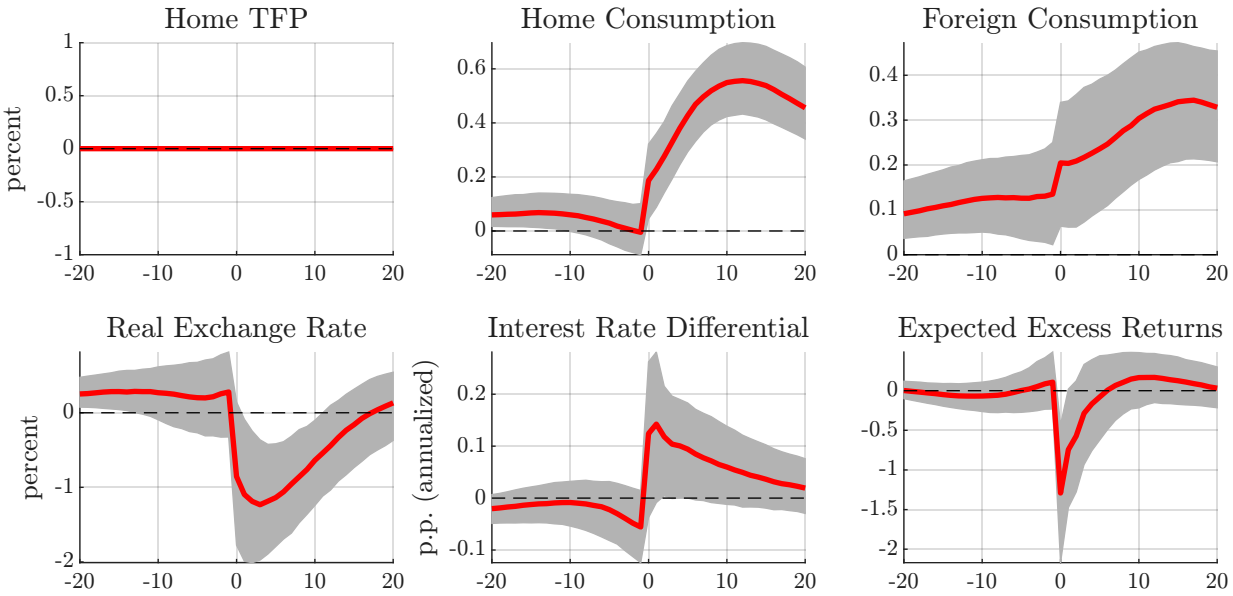
Notes: The figure displays responses to a one-standard-deviation impulse in the technological disturbance (panel (a)) and noise disturbance (panel (b)) at time $t = 0$. The shaded areas are 16th–84th percentile bands. Each period is a quarter.

Figure C.3: Extended sample: impulse responses to technological disturbances (ε^a)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the technological disturbance at time $t = 0$. The shaded areas are 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Figure C.4: Extended sample: impulse responses to noise disturbances (ε^v)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the noise disturbance at time $t = 0$. The shaded areas are 16th–84th percentile bands. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Table C.2: Extended sample: variance decomposition

	Periodicities of 2–100 Quarters			Periodicities of 6–32 Quarters		
	Both	Tech.	Noise	Both	Tech.	Noise
Home TFP	1.00	1.00	0.00	1.00	1.00	0.00
Home Consumption	0.73	0.52	0.21	0.34	0.08	0.26
Foreign Consumption	0.39	0.28	0.11	0.12	0.06	0.06
Home Investment	0.75	0.55	0.20	0.37	0.11	0.26
Foreign Investment	0.40	0.23	0.17	0.27	0.09	0.18
Interest Rate Differential	0.41	0.31	0.10	0.27	0.17	0.10
Real Exchange Rate	0.48	0.31	0.17	0.26	0.09	0.17
Expected Excess Returns	0.38	0.25	0.13	0.32	0.18	0.14

Notes: The table reports the estimated variance shares (at periodicities of 2–100 quarters and of 6–32 quarters) explained by technological disturbances ("Tech."), noise disturbances ("Noise"), and both combined ("Both").

Table C.3: Extended sample: exchange rate related puzzles and TFP expectations

Panel A: Forward Premium Puzzle Moments

	Technology	Noise	Both	Unconditional
Fama β_λ	-2.05	-2.72	-2.19	-2.53
	$[-3.31, -1.13]$	$[-4.38, -1.47]$	$[-3.37, -1.24]$	$[-3.36, -1.75]$
$\text{cov}(\lambda_{t+1}, r_t - r_t^\star)$	-0.19	-0.06	-0.26	-0.60
	$[-0.51, -0.07]$	$[-0.14, -0.02]$	$[-0.63, -0.10]$	$[-1.06, -0.39]$
Engel β_Λ	11.35	6.76	10.11	3.30
	$[-0.63, 24.15]$	$[-13.21, 23.96]$	$[-1.88, 22.05]$	$[-3.38, 10.41]$
$\text{cov}\left(\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}], r_t - r_t^\star\right)$	0.42	0.09	0.54	0.43
	$[-0.02, 1.57]$	$[-0.16, 0.41]$	$[-0.10, 1.82]$	$[-0.41, 1.79]$
$\sigma(r_t - r_t^\star)/\sigma(\Delta q_t)$	0.28	0.12	0.19	0.13
	$[0.19, 0.45]$	$[0.07, 0.21]$	$[0.13, 0.32]$	$[0.11, 0.18]$
$\rho(r_t - r_t^\star)$	0.98	0.93	0.97	0.92
	$[0.96, 0.99]$	$[0.84, 0.97]$	$[0.94, 0.99]$	$[0.88, 0.96]$

Panel B: Backus-Smith Puzzle Moments

	Technology	Noise	Both	Unconditional
$\text{corr}(\Delta q_t, \Delta(c_t - c_t^\star))$	-0.32	-0.65	-0.49	-0.18
	$[-0.58, 0.00]$	$[-0.87, -0.13]$	$[-0.72, -0.17]$	$[-0.27, -0.11]$
$\text{cov}(\Delta q_t, \Delta(c_t - c_t^\star))$	-0.06	-0.14	-0.24	-0.41
	$[-0.18, 0.00]$	$[-0.37, -0.02]$	$[-0.52, -0.06]$	$[-0.64, -0.22]$

Panel C: Excess Volatility and Persistence

	Technology	Noise	Both	Unconditional
$\rho(q_t)$	0.99	0.96	0.98	0.95
	$[0.98, 0.99]$	$[0.92, 0.98]$	$[0.96, 0.99]$	$[0.93, 0.97]$
$\sigma(\Delta q_t)/\sigma(\Delta c_t)$	3.18	5.25	4.08	5.79
	$[2.14, 4.59]$	$[3.08, 8.14]$	$[2.77, 5.82]$	$[4.93, 6.41]$

Notes: The table reports the estimated moments conditional on technological disturbances ("Technology"), noise disturbances ("Noise"), and both combined ("Both"), along with the unconditional moments ("Unconditional"). The moments including Engel β_Λ, $\text{cov}\left(\sum_{j=0}^{\infty} \mathbb{E}_t[\lambda_{t+j+1}], r_t - r_t^\star\right)$, and $\rho(q_t)$ are computed on the data filtered to periodicities of 2–100 quarters. The values in brackets represent the 16th–84th percentile bands. The moments in the table are defined in the text.

C.2 Additional figures and tables

Table C.4: Share of forecast error variance explained by the main exchange rate shock: target periodicities of 2–1000 quarters

	Forecast Horizon (Quarters)					
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.03	0.05	0.17	0.36	0.46	0.46
Home Consumption	0.03	0.05	0.20	0.46	0.51	0.44
Foreign Consumption	0.02	0.02	0.06	0.23	0.37	0.35
Home Investment	0.16	0.21	0.31	0.47	0.50	0.46
Foreign Investment	0.09	0.11	0.19	0.24	0.36	0.37
Interest Rate Differential	0.27	0.29	0.27	0.30	0.33	0.37
Real Exchange Rate	0.43	0.58	0.75	0.68	0.65	0.64
Expected Excess Returns	0.37	0.29	0.30	0.39	0.41	0.42

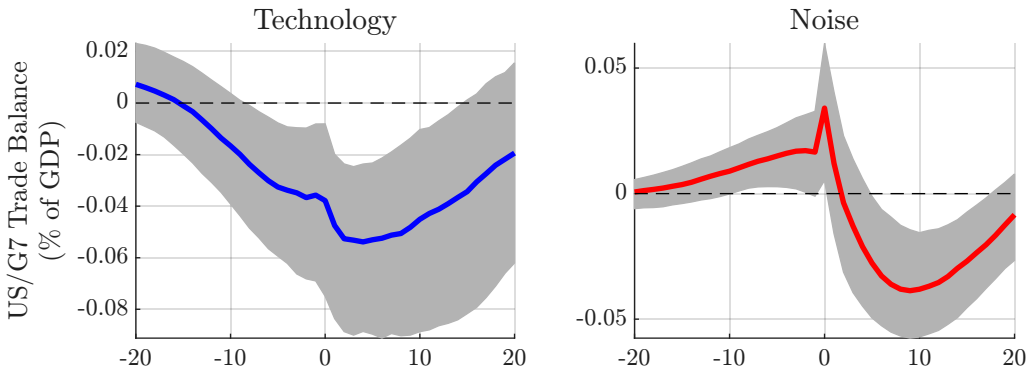
Notes: The table reports the estimated variance shares at different horizons accounted for by the main exchange rate shock that explains most of the forecast error variance of the exchange rate at periodicities of 2–1000 quarters.

Table C.5: Share of forecast error variance explained by the main exchange rate shock: target periodicities of 6–32 quarters

	Forecast Horizon (Quarters)					
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.03	0.03	0.09	0.14	0.15	0.15
Home Consumption	0.08	0.10	0.19	0.26	0.23	0.18
Foreign Consumption	0.01	0.02	0.05	0.20	0.23	0.16
Home Investment	0.07	0.10	0.13	0.18	0.17	0.17
Foreign Investment	0.01	0.02	0.04	0.12	0.18	0.16
Interest Rate Differential	0.25	0.19	0.20	0.22	0.22	0.21
Real Exchange Rate	0.91	0.92	0.67	0.52	0.48	0.43
Expected Excess Returns	0.27	0.20	0.43	0.43	0.40	0.39

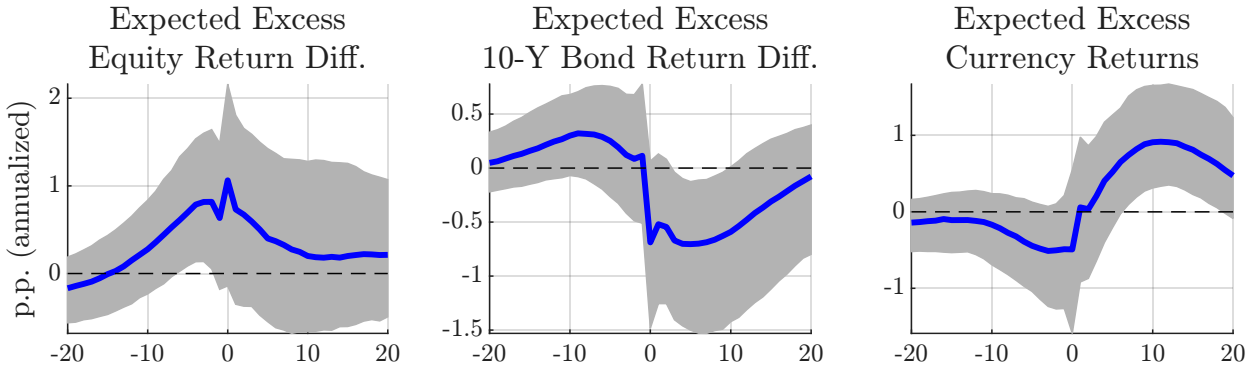
Notes: The table reports the estimated variance shares at different horizons accounted for by the main exchange rate shock that explains most of the forecast error variance of the exchange rate at periodicities of 6–32 quarters.

Figure C.5: Technology, noise, and the trade balance

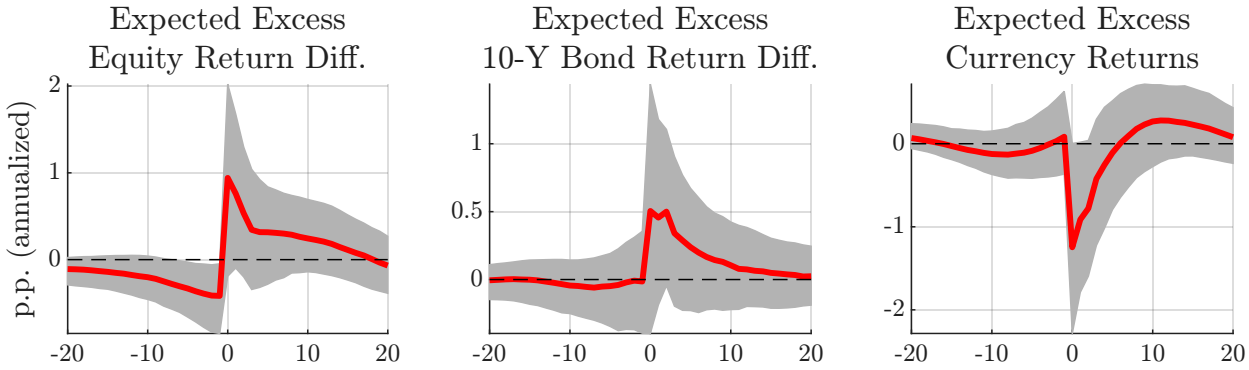


Notes: The figure displays the responses to a one-standard-deviation impulse in the technological disturbance (left column) and the noise disturbance (right column) at time $t = 0$. The shaded areas are 16th–84th percentile bands. Each period is a quarter. The trade balance data refer to the US trade balance in goods with the other G7 economies and are obtained from the International Monetary Fund, International Trade in Goods by Partner Country (IMTS), available at https://data.imf.org/en/datasets/IMF.STA:IMTS.

Figure C.6: Technology, noise, and asset returns



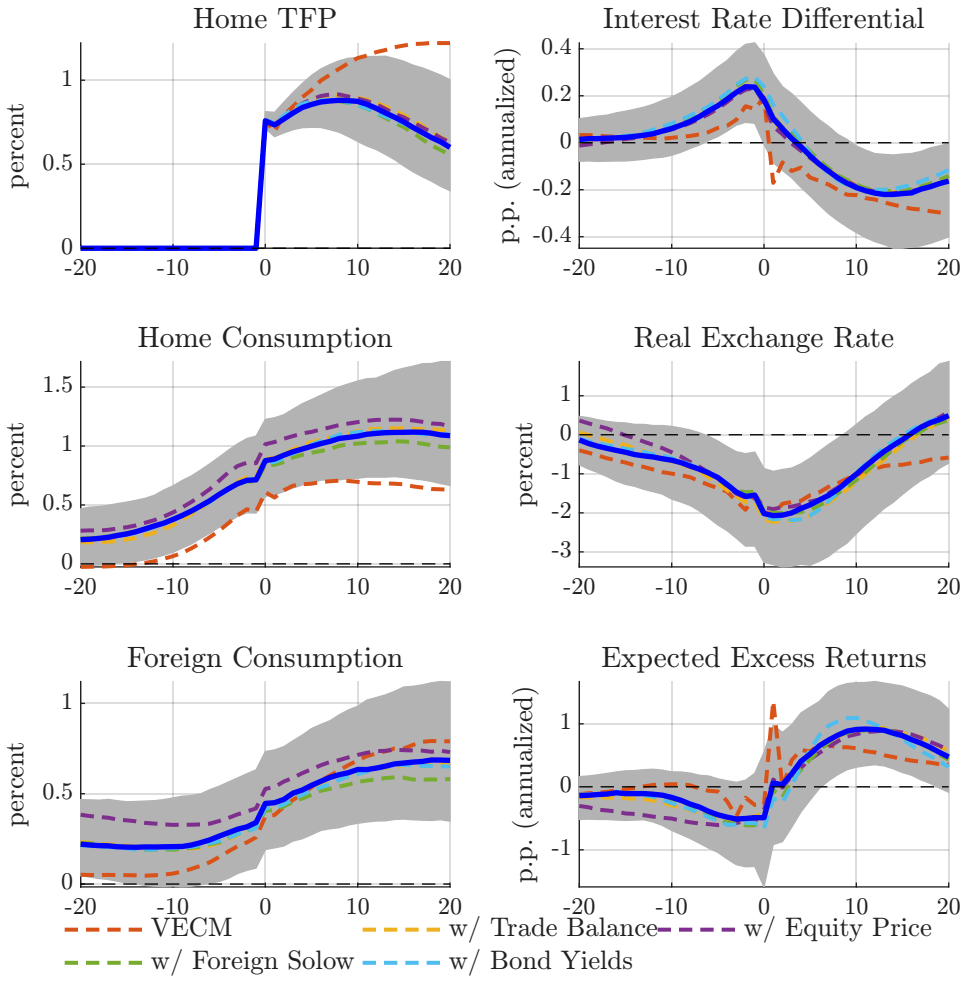
(a) Impulse responses to technological disturbances (ε^a)



(b) Impulse responses to noise disturbances (ε^v)

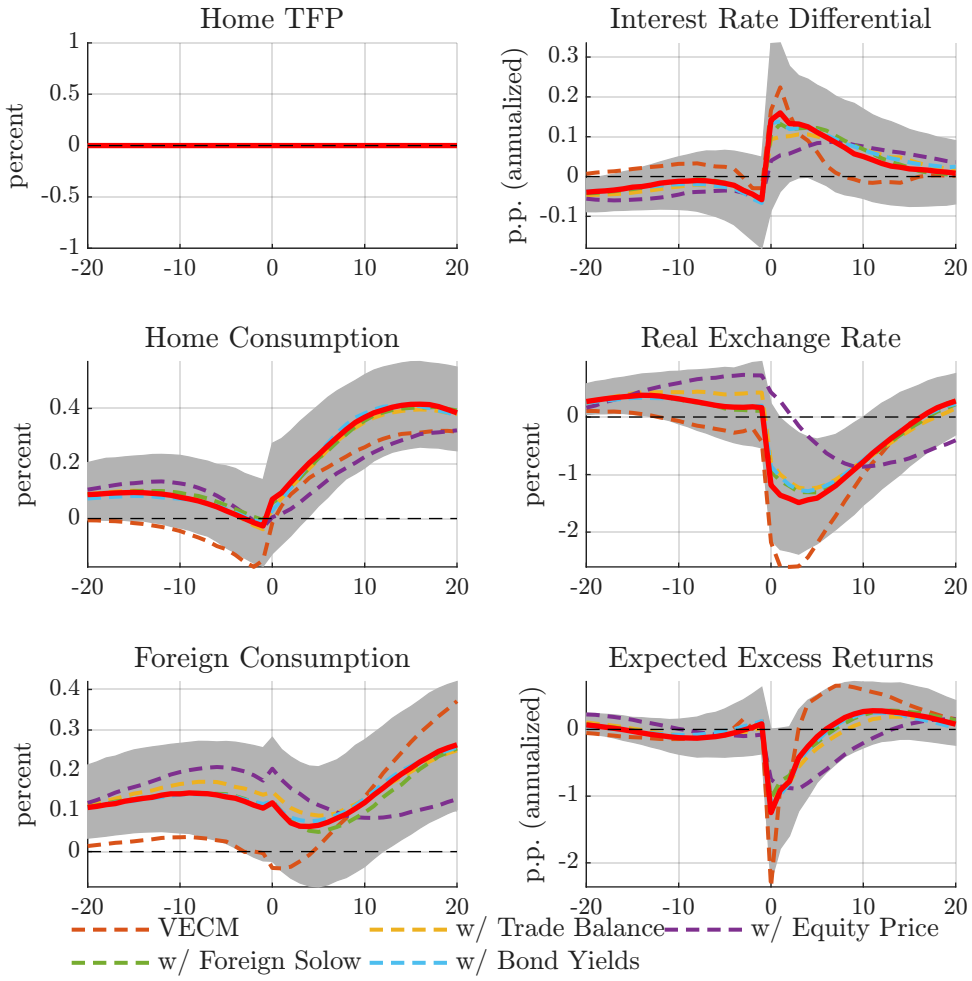
Notes: The figure displays responses to a one-standard-deviation impulse in the technological disturbance (panel (a)) and noise disturbance (panel (b)) at time $t = 0$. The return differentials shown are defined as the expected excess foreign return (in foreign currency) minus the expected excess home return (in home currency), that is, $(\mathbb{E}_t[r^{x,\star}_{t+1}] - r^\star_t) - (\mathbb{E}_t[r^x_{t+1}] - r_t)$, where r^x_{t+1} denotes the return on either equities or long-term bonds between periods t and $t+1$. The shaded areas are 16th–84th percentile bands. Each period is a quarter.

Figure C.7: Impulse responses to technological disturbances (alternative specifications)



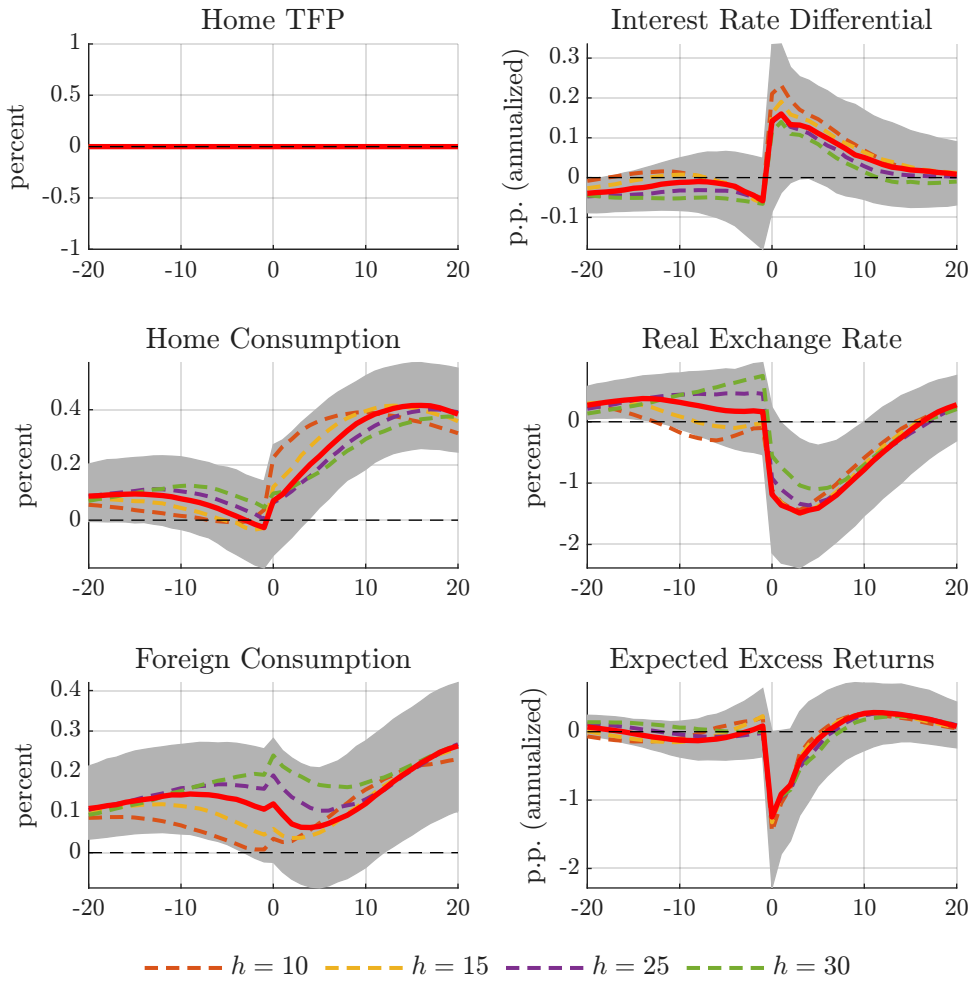
Notes: The figure displays impulse responses to a one-standard-deviation impulse in the technological disturbance at time $t = 0$ for the baseline specification (blue lines), where the shaded areas are 16th–84th percentile bands. In addition, the figure displays the posterior median responses obtained from the Bayesian VECM and from the baseline VARs with one additional variable. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Figure C.8: Impulse responses to noise disturbances (alternative specifications)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the noise disturbance at time $t = 0$ for the baseline specification (red lines), where the shaded areas are 16th–84th percentile bands. In addition, the figure displays the posterior median responses obtained from the Bayesian VECM and from the baseline VARs with one additional variable. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Figure C.9: Impulse responses to noise disturbances (alternative expectation horizons)



Notes: The figure displays impulse responses to a one-standard-deviation impulse in the noise disturbance at time $t = 0$ for the baseline specification, which targets expected TFP $h = 20$ quarters ahead (solid line), where the shaded areas are 16th–84th percentile bands. The dashed lines are the median responses obtained when the target horizon is instead $h = 10$, 15, 25, or 30 quarters. The responses to the technological disturbance do not depend on h and are not shown. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Table C.6: Share of variance explained by the main exchange rate shock (alternative specifications)

Panel A: Bayesian VECM						
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.06	0.09	0.20	0.31	0.39	0.49
Home Consumption	0.08	0.09	0.12	0.18	0.24	0.30
Foreign Consumption	0.03	0.06	0.12	0.23	0.32	0.43
Home Investment	0.22	0.25	0.25	0.28	0.35	0.44
Foreign Investment	0.10	0.11	0.17	0.27	0.37	0.47
Interest Rate Differential	0.13	0.13	0.14	0.17	0.20	0.26
Real Exchange Rate	0.29	0.40	0.53	0.63	0.71	0.84
Expected Excess Returns	0.31	0.27	0.28	0.30	0.32	0.34
Panel B: Individual countries (median)						
	$Q1$	$Q4$	$Q12$	$Q24$	$Q40$	$Q100$
Home TFP	0.04	0.06	0.16	0.32	0.45	0.47
Home Consumption	0.05	0.08	0.22	0.36	0.43	0.40
Foreign Consumption	0.04	0.05	0.11	0.23	0.36	0.39
Home Investment	0.12	0.15	0.21	0.36	0.39	0.39
Foreign Investment	0.06	0.07	0.16	0.27	0.35	0.40
Interest Rate Differential	0.15	0.21	0.23	0.23	0.26	0.31
Real Exchange Rate	0.55	0.72	0.81	0.78	0.74	0.72
Expected Excess Returns	0.37	0.25	0.29	0.33	0.38	0.41

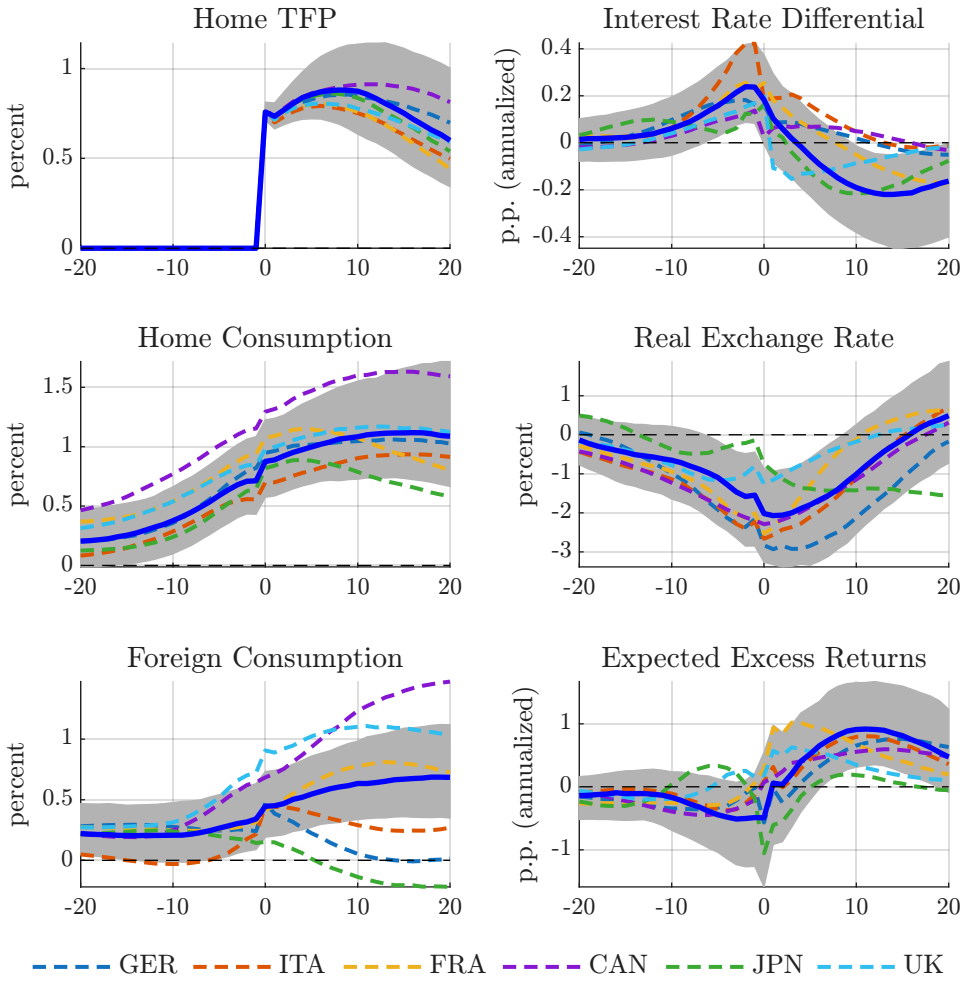
Notes: The table reports the estimated variance shares accounted for by the main exchange rate shock at different horizons.

Table C.7: Variance decomposition (alternative specifications)

Panel A: Bayesian VECM						
	Periodicities of 2–100 Quarters			Periodicities of 6–32 Quarters		
	Both	Tech.	Noise	Both	Tech.	Noise
Home TFP	1.00	1.00	0.00	1.00	1.00	0.00
Home Consumption	0.67	0.45	0.22	0.38	0.14	0.24
Foreign Consumption	0.65	0.45	0.20	0.33	0.13	0.20
Home Investment	0.68	0.49	0.18	0.50	0.23	0.27
Foreign Investment	0.53	0.31	0.23	0.29	0.09	0.19
Interest Rate Differential	0.37	0.24	0.12	0.30	0.15	0.15
Real Exchange Rate	0.62	0.28	0.34	0.48	0.09	0.39
Expected Excess Returns	0.46	0.21	0.25	0.43	0.14	0.29
Panel B: Individual countries (median)						
	Periodicities of 2–100 Quarters			Periodicities of 6–32 Quarters		
	Both	Tech.	Noise	Both	Tech.	Noise
Home TFP	1.00	1.00	0.00	1.00	1.00	0.00
Home Consumption	0.69	0.53	0.15	0.35	0.11	0.24
Foreign Consumption	0.53	0.39	0.12	0.28	0.12	0.14
Home Investment	0.71	0.57	0.14	0.41	0.20	0.22
Foreign Investment	0.54	0.41	0.15	0.37	0.11	0.23
Interest Rate Differential	0.37	0.26	0.11	0.24	0.10	0.11
Real Exchange Rate	0.60	0.39	0.18	0.36	0.10	0.25
Expected Excess Returns	0.42	0.26	0.15	0.31	0.13	0.19

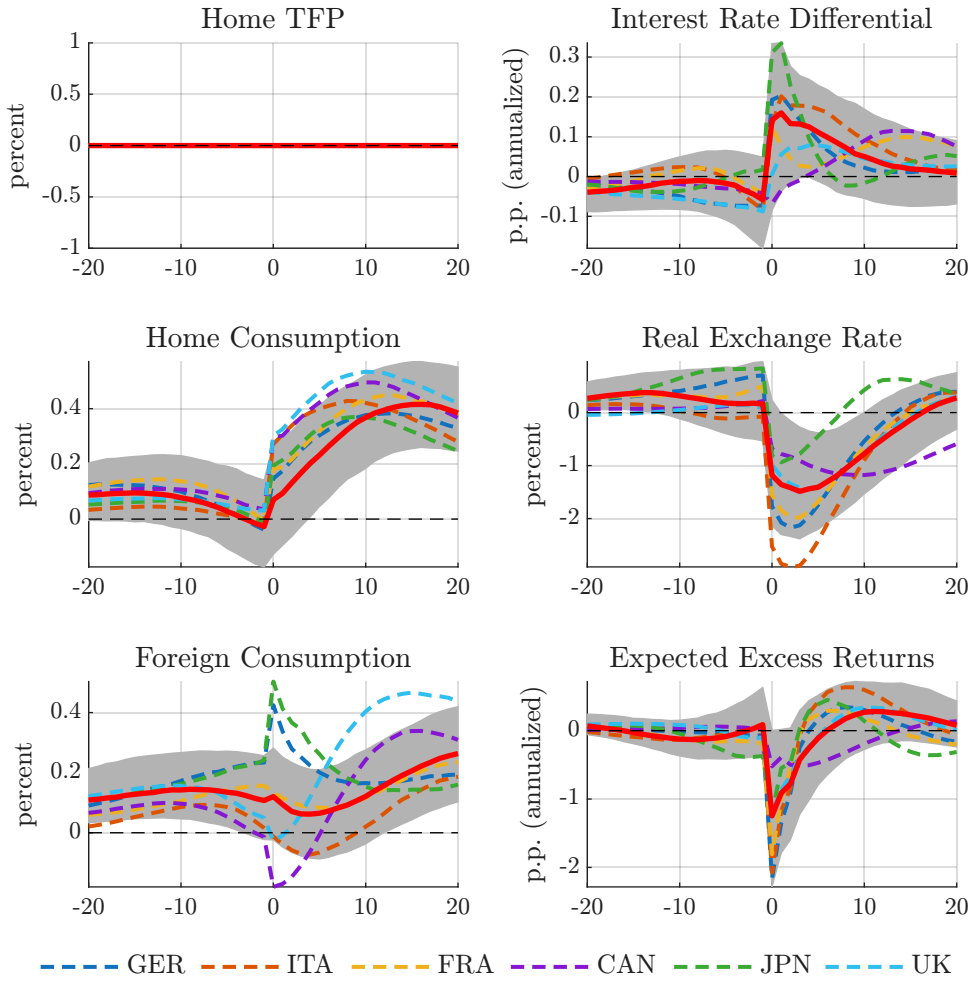
Notes: The table reports the estimated variance shares (at periodicities of 2–100 quarters and of 6–32 quarters) explained by technological disturbances ("Tech."), noise disturbances ("Noise"), and both combined ("Both").

Figure C.10: Impulse responses to technological disturbances (individual countries)



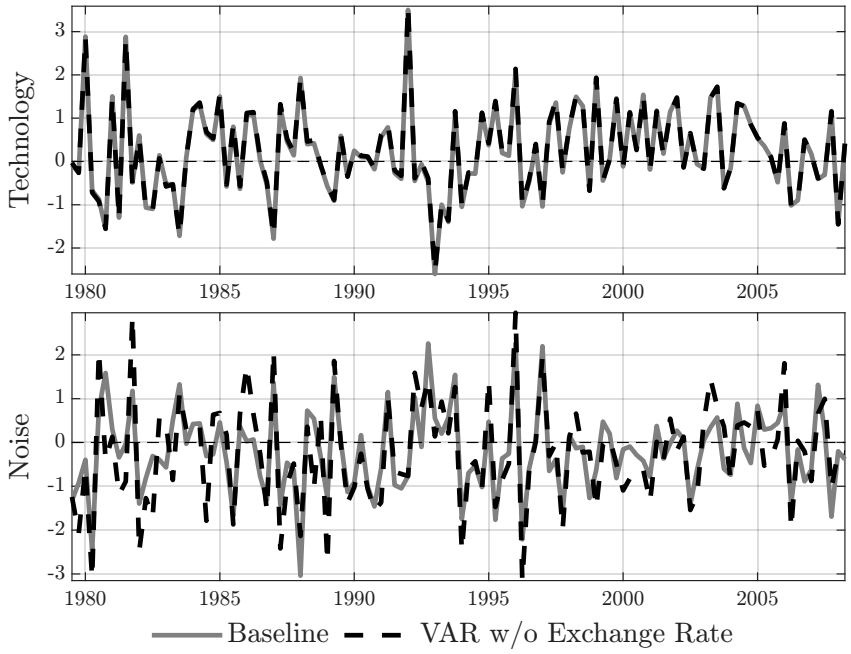
Notes: The figure displays impulse responses to a one-standard-deviation impulse in the technological disturbance at time $t = 0$ for the baseline specification (blue lines), where the shaded areas are 16th–84th percentile bands. In addition, the figure displays the posterior median responses obtained from six bilateral VARs, one for each of the other G7 economies. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Figure C.11: Impulse responses to noise disturbances (individual countries)



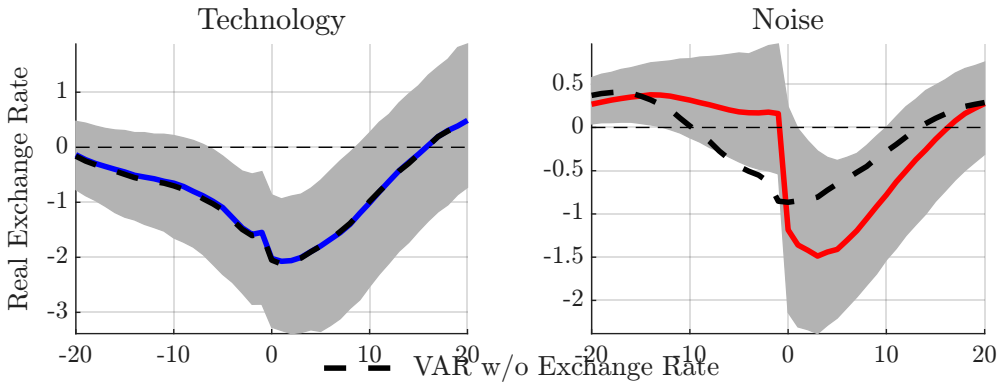
Notes: The figure displays impulse responses to a one-standard-deviation impulse in the noise disturbance at time $t = 0$ for the baseline specification (red lines), where the shaded areas are 16th–84th percentile bands. In addition, the figure displays the posterior median responses obtained from six bilateral VARs, one for each of the other G7 economies. Expected excess returns refer to those of holding foreign bonds, as defined in (5). Each period is a quarter.

Figure C.12: Series of estimated technological and noise disturbances



Notes: The figure displays the series of technological and noise disturbances estimated by the baseline VAR as well as the VAR without the exchange rate (see Online Appendix B.2).

Figure C.13: Impulse responses of the exchange rate to technological and noise disturbances (VAR without the exchange rate)



Notes: The figure displays the impulse response of the real exchange rate to a one-standard-deviation impulse in the technological disturbance (left column) and the noise disturbance (right column) at time $t = 0$. The dashed black line is the exchange rate response when the disturbances are estimated using a VAR without the exchange rate (see Online Appendix B.2). The shaded area is the 16th–84th percentile band. Each period is a quarter.

Table C.8: Granger causality: exchange rates and TFP

Panel A: p-values testing whether exchange rates Granger-cause TFP

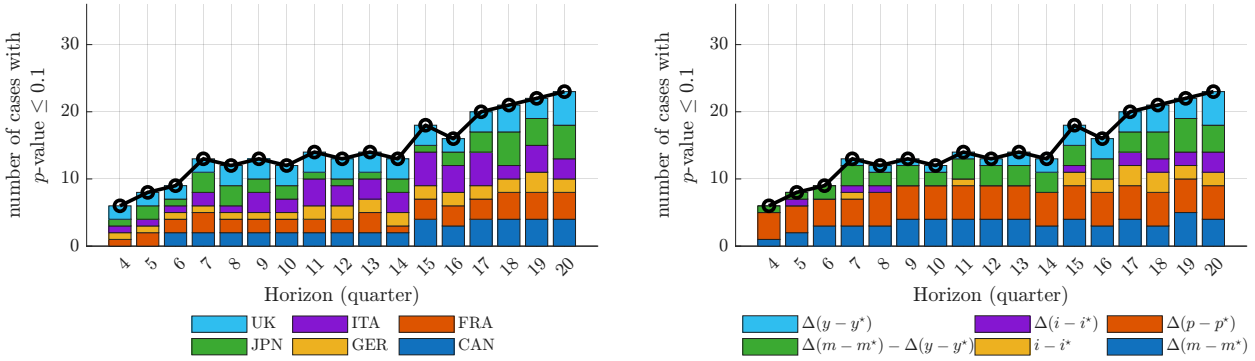
H	Canada	France	Germany	Italy	Japan	UK	Aggregate
1	0.94	0.79	0.62	0.97	0.65	0.64	0.96
2	0.06*	0.91	0.66	0.76	0.80	0.71	1.00
3	0.16	0.02**	0.07*	0.00***	0.17	0.00***	0.06*
4	0.14	0.03**	0.09*	0.02**	0.08*	0.02**	0.10*
5	0.10*	0.00***	0.00***	0.00***	0.00***	0.02**	0.00***

Panel B: p-values testing whether TFP Granger-causes exchange rates

H	Canada	France	Germany	Italy	Japan	UK	Aggregate
1	0.67	0.82	0.86	0.94	0.99	1.00	0.94
2	0.37	0.83	0.79	0.64	0.98	0.17	0.73
3	0.28	1.00	0.99	0.73	0.77	0.74	0.94
4	0.05**	0.67	0.78	0.39	0.79	0.84	0.53
5	0.00***	0.85	0.90	0.62	0.63	0.95	0.85

Notes: p-values from a Wald test of Granger causality as in Sims (1972). The test is based on the equation: $x_t = \alpha + \beta_0 y_t + \sum_{\ell=1}^{H} \beta_{-\ell}^{\text{lag}} y_{t-\ell} + \sum_{j=1}^{H} \beta_j^{\text{lead}} y_{t+j} + \epsilon_t$, where x is the quarterly observation of the annual change in the log of the real exchange rate and y is the quarterly observation of the annual change in the log of TFP in Panel A; variables are reversed in Panel B. The null hypothesis is that $\beta_j^{\text{lead}} = 0$ for all $j = 1, \ldots, H$. The sample period is 1976:Q1–2008:Q2. Newey-West standard errors are computed using a data-driven bandwidth, a Bartlett kernel, and prewhitening.

Figure C.14: Bivariate Granger causality tests *à la* Engel and West (2005) (1976:Q1–2020:Q1)

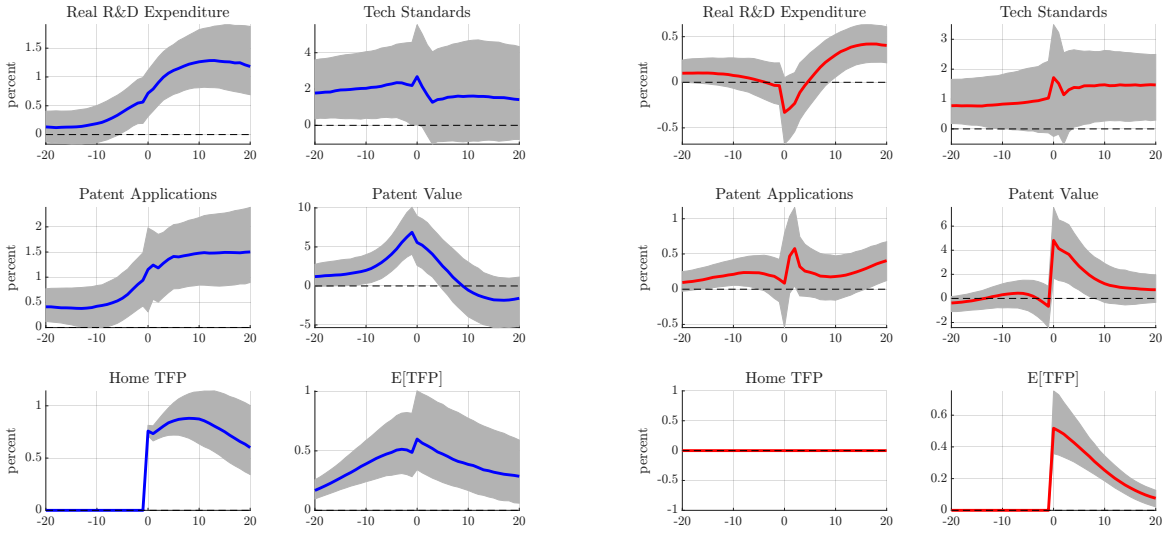


(a) Granger causality tests, by country

(b) Granger causality tests, by variable

Notes: For each horizon, we report the number of cases in which the p-value from a Wald test of Granger causality is below 0.1. There are 36 cases per horizon: six other G7 economies (each compared to the US) and six macroeconomic fundamentals, following the data and methodology of Engel and West (2005). Each test is based on a bivariate VAR estimated using the quarterly change in the nominal exchange rate, Δs, and one fundamental variable, f. The horizon on the x-axis indicates the number of lags included in the VAR. The reported p-value corresponds to a Wald test of the joint significance of the lags of Δs in predicting f. The variables are defined as follows: Δm is the percentage change in M1 (spliced for France, Germany, Italy, and the UK; see Online Appendix A); Δy is the percentage change in real GDP; Δp is the percentage change in consumer prices; and i is the three-month eurocurrency deposit rate. The six fundamentals are the home-minus-foreign differences of these series: $\Delta(m - m^\star)$, $\Delta(y - y^\star)$, $\Delta(p - p^\star)$, $i - i^\star$, $\Delta(i - i^\star)$, and $\Delta(m - m^\star) - \Delta(y - y^\star)$. Following Engel and West (2005), exchange rates and interest rates are measured at the end of the quarter in this figure, rather than as the quarterly averages used in the VAR; money is measured in the last month of the quarter. Newey-West standard errors are computed using a data-driven bandwidth, a Bartlett kernel, and prewhitening. Panel (a) highlights the cases in which the p-value from the Wald test of Granger causality is below 0.1, *by country*, while panel (b) highlights the corresponding cases *by variable*.

Figure C.15: Responses of technology proxy series to technological and noise disturbances



(a) Technology

(b) Noise

Notes: The figure displays responses to a one-standard-deviation impulse in the technological disturbance (panel (a)) and the noise disturbance (panel (b)) at time $t = 0$. The shaded areas are 16th–84th percentile bands. Each period is a quarter. The responses of these technology proxies are obtained by projecting each series on the VAR's state, as described in Online Appendix B.3.

C.3 Performance of a VAR-based trading strategy

We construct trading strategies that exploit the predictable component of excess currency returns associated with the identified technological and noise disturbances. Specifically, we consider a simple strategy in which a US investor takes a position in foreign currency proportional to the VAR-implied expectation of excess foreign-currency returns. The portfolio return from this strategy is given by the product of realized and expected excess foreign-currency returns, $\lambda_{t+1}\mathbb{E}_t[\lambda_{t+1}]$. Note that $\mathbb{E}_t[\lambda_{t+1}]$ can be negative, in which case the investor takes a long position in the dollar and a short position in the basket of foreign currencies. We estimate expected returns using a rolling window of N quarters, with an investment horizon of one quarter. After each quarter, the investor observes the new data and updates expectations accordingly.

Specifically, for each t we re-estimate the VAR on the rolling window of $t-N+1:t$, identify the disturbances, and form the prediction $\mathbb{E}_t[\lambda_{t+1}]$. Then, we evaluate the resulting trading return $\lambda_{t+1}\mathbb{E}_t[\lambda_{t+1}]$. The exercise constitutes a valid out-of-sample analysis that can be implemented by investors using variables dated until the beginning of the investment period.[32]

Table C.9 reports the annualized Sharpe ratios and corresponding portfolio returns for different estimation window lengths when conditioning on the two disturbances.

Our preferred specification uses a rolling window length of 80 quarters, a choice we justify below. In this case, the annualized Sharpe ratio is 0.53 when conditioning on the two disturbances. The magnitude of the Sharpe ratio is broadly comparable to those obtained from alternative trading strategies in the currency market that exploit related, but distinct, time variation in return predictability, such as the "Dollar trade" (Lustig et al., 2014; Hassan and Mano, 2019), and momentum strategies (Burnside et al., 2011; Menkhoff et al., 2012; Moskowitz et al., 2012).

Table C.9 also reports results for alternative window lengths (ranging from 60 to 100 quarters). With very short windows, the Sharpe ratios are small, reflecting the limited degrees of freedom adopted in the estimation of the VAR parameters. The Sharpe ratios are bigger for longer window sizes, but the number of forecast observations decreases, leading to uncertainty around the Sharpe ratio estimates. Thus, we consider a rolling window length of 80 quarters to be an appropriate choice.

[32] This procedure is equivalent to conditioning on both current and past values of TFP as well as on expected future TFP (five years ahead), all of which belong to investors' information set at time t.

Table C.9: Sharpe ratios and returns of the VAR-based trading strategy

Window (quarters)	Sharpe ratio	Portfolio return
60	0.23	1.60
64	−0.17	−1.21
68	−0.34	−2.09
72	−0.01	−0.12
76	0.51	4.25
80	0.53	5.34
84	0.63	6.18
88	0.64	5.95
92	0.79	7.26
96	0.86	7.39
100	0.63	4.66

Notes: The table reports the annualized Sharpe ratios and corresponding portfolio returns for different estimation window lengths when conditioning on the technological and noise disturbances. The Sharpe ratios are estimated from the median expected excess foreign-currency returns from 1000 VAR parameter draws in each sampled window. Since the baseline dataset contains 119 quarterly observations, this procedure yields $119 - N$ observations of portfolio returns, where N is the length of the rolling window.

D Model appendix

D.1 Environments

We study different classes of two-country endowment economies. We begin by outlining the preferences and technology, and then characterize the equilibria under complete and incomplete markets.

Utility function Let C_t denote consumption of the home country. The home-country preferences over an infinite horizon are

$$U_t = \left[(1-\beta)\cdot C_t^{1-1/\psi} + \beta \mathbb{E}_t\left[U_{t+1}^{1-\gamma}\right]^{\frac{1-1/\psi}{1-\gamma}}\right]^{\frac{1}{1-1/\psi}}. \tag{D.1}$$

With these preferences, agents are risk averse in future utility as well as future consumption. The extent of such utility risk aversion depends on the preference for early resolution of uncertainty, measured by $\gamma - 1/\psi > 0$.

When $\gamma = 1/\psi$, the agent is utility-risk neutral and preferences collapse to the standard time-additive, constant relative risk aversion (CRRA) case. When the agent prefers early resolution of uncertainty, that is, $\gamma > 1/\psi$, uncertainty about continuation utility reduces welfare and generates an incentive to trade off future expected utility for future utility risk. This trade-off drives international consumption flows—a key element of Colacito and Croce (2011; 2013).

The utility of foreign households is symmetric.

Technology Let $\{X_t, Y_t\}$ and $\{X_t^\star, Y_t^\star\}$ denote the time-t consumption of goods X and Y in the home and foreign countries, respectively. The consumption aggregates in the two countries are

$$C_t = \left((\varsigma)^{\frac{1}{\sigma}}(X_t)^{\frac{\sigma-1}{\sigma}} + (1-\varsigma)^{\frac{1}{\sigma}}(Y_t)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \quad C_t^\star = \left((1-\varsigma)^{\frac{1}{\sigma}}(X_t^\star)^{\frac{\sigma-1}{\sigma}} + (\varsigma)^{\frac{1}{\sigma}}(Y_t^\star)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}. \tag{D.2}$$

The home (foreign) country produces good X (Y) so that $\varsigma > 1/2$ introduces consumption home bias, while $\sigma > 0$ represents the cross-good price elasticity.

In this endowment economy, the following resource constraints apply:

$$A_t \geq X_t + X_t^\star, \tag{D.3}$$
$$A_t^\star \geq Y_t + Y_t^\star, \tag{D.4}$$

where A_t and $A_t^\star$ are the exogenous stochastic productivity processes that follow equation (D.48). We assume throughout that $\sigma > 1 - \frac{1}{2\varsigma}$, which is necessary and sufficient for the economy to have a unique non-stochastic steady state.

D.1.1 Complete markets

To find the solution under complete markets, we follow Colacito et al. (2018) and solve the Pareto problem. We solve the Pareto problem under the general utility function (D.1).

Pareto problem This appendix suppresses notation denoting states and histories and retains only subscripts for time. We represent the Epstein-Zin utility preferences in the following compact way:

$$U_t = W(C_t, U_{t+1}),$$

so that the dependence of current utility on j-step-ahead consumption can be denoted as

$$\frac{\partial U_t}{\partial C_{t+j}} = W_{2,t+1} \cdot W_{2,t+2} \cdots W_{2,t+j} \cdot W_{1,t+j}, \tag{D.5}$$

where $W_{2,t+j} \equiv \frac{\partial U_{t+j-1}}{\partial U_{t+j}}$ and $W_{1,t+j} \equiv \frac{\partial U_{t+j}}{\partial C_{t+j}}$.

The stochastic discount factor in C units, M_t, is

$$M_t = \beta \left(\frac{C_t}{C_{t-1}} \right)^{-\frac{1}{\psi}} \left(\frac{U_t}{\mathbb{E}_{t-1}[U_t^{1-\gamma}]^{\frac{1}{1-\gamma}}} \right)^{\frac{1}{\psi}-\gamma}. \tag{D.6}$$

The consumption aggregate combines two goods, X and Y:

$$C_t = C(X_t, Y_t).$$

The planner faces the resource constraints (D.3) and (D.4), where A_t and $A_t^\star$ are the exogenous stochastic productivity processes.

The social planner chooses sequences of $\{X_t, X_t^\star, Y_t, Y_t^\star\}_t$ to maximize

$$\mu_0 U_0 + (1-\mu_0) U_0^\star,$$

subject to (D.3)–(D.4). Let $\lambda_{i,t}$ be the Lagrange multiplier for the respective constraint i. The Lagrangian is

$$\Omega = \mu_0 U_0 + (1-\mu_0) U_0^\star + \lambda_{1,t}(A_t - X_t - X_t^\star) + \lambda_{2,t}(A_t^\star - Y_t - Y_t^\star) + \dots.$$

The optimality condition for good X_t for all t is

$$\mu_0 \left(\prod_{j=1}^{t} W_{2,j} \right) W_{1,t} C_{x,t} = \lambda_{1,t} = C_{x^\star,t}^\star W_{1,t}^\star \left(\prod_{j=1}^{t} W_{2,j}^\star \right) \mu_0^\star, \tag{D.7}$$

where $\mu_0^\star = (1-\mu_0)$. Let μ_t be the time-t Pareto weight for the home country:

$$\mu_t = \mu_0 \left(\prod_{j=1}^{t} W_{2,j} \right) W_{1,t} C_t = \mu_{t-1} W_{2,t} \cdot \frac{W_{1,t}}{W_{1,t-1}} \cdot \frac{C_t}{C_{t-1}} = \mu_{t-1} M_t \cdot \frac{C_t}{C_{t-1}}.$$

Thus, equation (D.7) becomes

$$S_t C_{x,t} \cdot \frac{1}{C_t} = C^\star_{x^\star,t} \cdot \frac{1}{C^\star_t},$$

where $S_t = \mu_t/\mu^\star_t$. Similarly,

$$S_t C_{y,t} \cdot \frac{1}{C_t} = C^\star_{y^\star,t} \cdot \frac{1}{C^\star_t}.$$

Real exchange rate and real interest rates Under complete markets, the real exchange rate satisfies the Backus-Smith condition:

$$\frac{Q_t}{Q_{t-1}} = \frac{M^\star_t}{M_t},$$

where the home stochastic discount factor, M_t, is reported in equation (D.6), and the foreign stochastic discount factor has an analogous structure.

The home and foreign risk-free rates are

$$\mathbb{E}_t[M_{t+1}]R_t = 1; \qquad \mathbb{E}_t[M^\star_{t+1}]R^\star_t = 1.$$

Solution We solve the model around a symmetric steady state.

Log-linear equilibrium system We characterize the log-linear equilibrium system for the complete-markets models as follows.[33] The equilibrium system has 11 endogenous variables $(c_t, c^\star_t, x_t, y_t, x^\star_t, y^\star_t, r_t, r^\star_t, q_t, m_t, m^\star_t)$ with 11 equilibrium conditions:

$$x_t = c_t + \frac{\sigma(1-\varsigma)}{2\varsigma - 1} q_t; \tag{D.8}$$

$$y_t = c_t - \frac{\sigma\varsigma}{2\varsigma - 1} q_t; \tag{D.9}$$

$$x^\star_t = c^\star_t + \frac{\sigma\varsigma}{2\varsigma - 1} q_t; \tag{D.10}$$

$$y^\star_t = c^\star_t - \frac{\sigma(1-\varsigma)}{2\varsigma - 1} q_t; \tag{D.11}$$

$$a_t = \varsigma x_t + (1-\varsigma) x^\star_t; \tag{D.12}$$

$$a^\star_t = (1-\varsigma) y_t + \varsigma y^\star_t; \tag{D.13}$$

$$\mathbb{E}_t[m_{t+1}] = -r_t; \tag{D.14}$$

$$\mathbb{E}_t[m^\star_{t+1}] = -r^\star_t; \tag{D.15}$$

$$q_t - q_{t-1} = m^\star_t - m_t; \tag{D.16}$$

$$m_t = -\frac{1}{\psi}\Delta(c_t) - \left(\gamma - \frac{1}{\psi}\right) \sum_{j=0}^{\infty} \beta^j \left(\mathbb{E}_t - \mathbb{E}_{t-1}\right)[\Delta c_{t+j}], \tag{D.17}$$

[33] We note, however, that in our numerical demonstrations in Section 5 we solve the model with Epstein-Zin preferences with a third-order approximation.

$$m_t^\star = -\frac{1}{\psi}\Delta\left(c_t^\star\right) - \left(\gamma - \frac{1}{\psi}\right)\sum_{j=0}^{\infty}\beta^j\left(\mathbb{E}_t - \mathbb{E}_{t-1}\right)\left[\Delta c_{t+j}^\star\right]. \tag{D.18}$$

Three-equation system To a first-order approximation, the complete-markets model can be expressed as the system of three equations:

$$q_t = -\frac{(2\varsigma-1)^2}{4(1-\varsigma)\sigma\varsigma}(c_t - c_t^\star) + \frac{(2\varsigma-1)}{4(1-\varsigma)\sigma\varsigma}(a_t - a_t^\star); \tag{D.19a}$$

$$\Delta q_t = \frac{1}{\psi}\Delta\left(c_t - c_t^\star\right) + \left(\gamma - \frac{1}{\psi}\right)\mathcal{N}_t; \tag{D.19b}$$

$$\mathcal{N}_t = \sum_{j=0}^{\infty}\beta^j\left(\mathbb{E}_t - \mathbb{E}_{t-1}\right)\left[\Delta c_{t+j} - \Delta c_{t+j}^\star\right]. \tag{D.19c}$$

Equation (D.19a) combines the goods market clearing conditions (D.12)–(D.13) with the goods demand functions (D.8)–(D.11). Equations (D.19b)–(D.19c) combine equations (D.16)–(D.18).

Proof of Proposition 1. Define home–foreign differential variables by, e.g. $\tilde{a}_t \equiv a_t - a_t^\star$. In addition to Θ, defined below (19), let

$$\Theta_\gamma \equiv (2\varsigma-1)^2 + 4\varsigma\sigma(1-\varsigma)\gamma > 0$$

denote the denominator in (21). Because $\sum_{j=0}^{\infty}\beta^j\Delta\tilde{a}_{t+j} = (1-\beta)\sum_{j=0}^{\infty}\beta^j\tilde{a}_{t+j} - \tilde{a}_{t-1}$, equation (21) can be written in levels as

$$\mathcal{N}_t = \frac{(2\varsigma-1)(1-\beta)}{\Theta_\gamma}\sum_{j=0}^{\infty}\beta^j\left(\mathbb{E}_t - \mathbb{E}_{t-1}\right)\left[\tilde{a}_{t+j}\right].$$

Substitute this into (19)–(20) and collect all terms dated $t-1$ or earlier, including date-$(t-1)$ expectations, into Ξ_{t-1}^q and Ξ_{t-1}^c, to find

$$q_t = \Xi_{t-1}^q + \omega_{q,0}^a\tilde{a}_t + (1-\beta)\mathbb{E}_t\left[\sum_{j=1}^{\infty}\beta^j\left(\omega_{q,1}^a\tilde{a}_{t+j}\right)\right], \tag{D.20}$$

$$\tilde{c}_t = \Xi_{t-1}^c + \omega_{c,0}^a\tilde{a}_t + (1-\beta)\mathbb{E}_t\left[\sum_{j=1}^{\infty}\beta^j\left(\omega_{c,1}^a\tilde{a}_{t+j}\right)\right]. \tag{D.21}$$

We can now compute the derivatives in Clause 1 of the proposition. For example,

$$\frac{\partial\tilde{c}_t}{\partial\mathbb{E}_t[a_{t+j}]} = (1-\beta)\beta^j\left(\omega_{c,1}^a\right),$$

where

$$\omega^a_{c,1} = -\frac{4\varsigma\sigma(1-\varsigma)(2\varsigma-1)}{\Theta\,\Theta_\gamma}\left(\gamma - \frac{1}{\psi}\right).$$

Since $(1-\beta)\beta^j > 0$, $\varsigma \in (1/2, 1)$, and $\Theta, \Theta_\gamma > 0$, this derivative is negative when $\gamma > 1/\psi$ and zero when $\gamma = 1/\psi$. Moreover, future productivity does not enter the market-clearing condition (18), so

$$\omega^a_{q,1} = -\frac{(2\varsigma-1)^2}{4\varsigma\sigma(1-\varsigma)}\,\omega^a_{c,1} = \frac{(2\varsigma-1)^3}{\Theta\,\Theta_\gamma}\left(\gamma - \frac{1}{\psi}\right),$$

so that $\partial q_t / \partial \mathbb{E}_t[a_{t+j}] = (1-\beta)\beta^j \omega^a_{q,1}$ is positive when $\gamma > 1/\psi$ and zero when $\gamma = 1/\psi$. This establishes Clause 1 of the proposition.

For Clause 2 of the proposition, note that

$$\begin{aligned}\omega^a_{c,0} &= \frac{2\varsigma-1}{\Theta} + (1-\beta)\,\omega^a_{c,1} \\ &= \frac{2\varsigma-1}{\Theta\,\Theta_\gamma}\left[(2\varsigma-1)^2 + \beta\,4\varsigma\sigma(1-\varsigma)\gamma + (1-\beta)\,\frac{4\varsigma\sigma(1-\varsigma)}{\psi}\right] > 0,\end{aligned}$$

where the second equality uses $\Theta_\gamma - (1-\beta)\,4\varsigma\sigma(1-\varsigma)\left(\gamma - \frac{1}{\psi}\right) = (2\varsigma-1)^2 + \beta\,4\varsigma\sigma(1-\varsigma)\gamma + (1-\beta)\,\frac{4\varsigma\sigma(1-\varsigma)}{\psi}$. Similarly,

$$\begin{aligned}\omega^a_{q,0} &= \frac{2\varsigma-1}{\psi\Theta} + (1-\beta)\,\omega^a_{q,1} \\ &= \frac{2\varsigma-1}{\Theta\,\Theta_\gamma}\left[\beta\,\frac{(2\varsigma-1)^2}{\psi} + \frac{4\varsigma\sigma(1-\varsigma)\gamma}{\psi} + (1-\beta)(2\varsigma-1)^2\gamma\right] > 0,\end{aligned}$$

where the second equality uses $\frac{\Theta_\gamma}{\psi} + (1-\beta)(2\varsigma-1)^2\left(\gamma - \frac{1}{\psi}\right) = \beta\,\frac{(2\varsigma-1)^2}{\psi} + \frac{4\varsigma\sigma(1-\varsigma)\gamma}{\psi} + (1-\beta)(2\varsigma-1)^2\gamma$. Every term in both brackets is positive, so both inequalities hold for any $\gamma, \psi > 0$. This establishes Clause 2 of the proposition. ■

D.1.2 Incomplete markets

Under incomplete markets agents only trade one bond internationally.

Households Let C_t denote the consumption of the home country. A representative home household maximizes the discounted expected utility over consumption under CRRA preferences. The utility function is a special case of equation (D.1) when $\psi = 1/\gamma$:

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\left(\frac{C_t^{1-\gamma}}{1-\gamma}\right)\right], \tag{D.22}$$

where γ represents the coefficient of relative risk aversion.

Home households allocate their within-period consumption expenditure P_tC_t between home and foreign goods, defined by a constant elasticity of substitution (CES) aggregator

featuring home bias in equation (D.2). The households minimize expenditure, resulting in the conventional constant elasticity demand schedules:

$$X_t = \varsigma \left(\frac{p_{X,t}}{P_t}\right)^{-\sigma} C_t, \qquad Y_t = (1-\varsigma)\left(\frac{p_{Y,t}}{P_t}\right)^{-\sigma} C_t.$$

The consumer price level is given by

$$P_t = \left[\varsigma p_{X,t}^{1-\sigma} + (1-\varsigma)\, p_{Y,t}^{1-\sigma}\right]^{\frac{1}{1-\sigma}},$$

where $p_{X,t}$ and $p_{Y,t}$ represent the home-currency prices of the home and foreign goods in the home market and P_t represents the home consumer price level.

The expenditure allocation of the foreign households is characterized by a symmetric demand system. In particular, the demand for home and foreign goods by foreign households is given by

$$Y_t^\star = C_t^\star \varsigma \left(\frac{p_{Y,t}^\star}{P_t^\star}\right)^{-\sigma}, \qquad X_t^\star = C_t^\star (1-\varsigma)\left(\frac{p_{X,t}^\star}{P_t^\star}\right)^{-\sigma},$$

and

$$P_t^\star = \left[\varsigma {p_{Y,t}^\star}^{1-\sigma} + (1-\varsigma)\, {p_{X,t}^\star}^{1-\sigma}\right]^{\frac{1}{1-\sigma}},$$

where $p_{X,t}^\star$ and $p_{Y,t}^\star$ represent the foreign-currency prices of the home and foreign goods in the foreign market and $P_t^\star$ represents the foreign consumer price level.

The budget constraint for the home economy is

$$\frac{B_{t+1}}{R_t} + \frac{Q_t B_{t+1}^\star}{R_t^\star \Phi\left(Q_t B_{t+1}^\star\right)} = \frac{p_{X,t}}{P_t} A_t - \frac{p_{X,t}}{P_t} X_t - \frac{p_{Y,t}}{P_t} Y_t + B_t + Q_t B_t^\star, \tag{D.23}$$

where B_{t+1} and $B_{t+1}^\star$ denote bonds denominated in the home and foreign consumption bundles, respectively; $\frac{p_{X,t}}{P_t}$ and $\frac{p_{Y,t}}{P_t}$ are the prices of the domestic and imported bundles, both relative to the overall consumption bundle. The function $\Phi(\cdot)$ represents an additional cost from international borrowing. As is common in the literature, we assume that higher foreign borrowing, that is, lower $B_{t+1}^\star$, implies higher cost from international borrowing, i.e. $\Phi'(\cdot) < 0$. Households take the overall cost of international borrowing as given. We further assume a zero steady-state intermediation cost by setting $\Phi(B^\star) = 1$. Foreign households only trade foreign-denominated bonds, while home-denominated bonds are in zero net supply. As a result, defining the intermediation costs over the foreign bonds only is sufficient to pin down the overall steady-state net foreign asset position.

The budget constraint for the foreign economy is

$$-\frac{B_{t+1}^\star}{R_t^\star} = \frac{p_{Y,t}^\star}{P_t^\star} A_t^\star - \frac{p_{X,t}^\star}{P_t^\star} X_t^\star - \frac{p_{Y,t}^\star}{P_t^\star} Y_t^\star - B_t^\star + \Pi_t, \tag{D.24}$$

where Π_t are the intermediation profits, rebated lump-sum to the foreign household. $\frac{p_{Y,t}^\star}{P_t^\star}$ and $\frac{p_{X,t}^\star}{P_t^\star}$ are the prices of the domestic and imported bundles, both relative to the overall consumption bundle.

The home household maximizes utility (D.22) subject to (D.23). The first-order conditions with respect to home and foreign bonds imply

$$\beta\,\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\right]R_t = 1; \tag{D.25}$$

$$\beta\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\frac{Q_{t+1}}{Q_t}\right]R_t^\star\,\Phi\big(Q_tB_{t+1}^\star\big) = 1. \tag{D.26}$$

The foreign households' first-order condition for bonds is

$$\beta\,\mathbb{E}_t\left[\left(\frac{C_{t+1}^\star}{C_t^\star}\right)^{-\gamma}\right]R_t^\star = 1. \tag{D.27}$$

Combining equations (D.26) and (D.27) gives

$$\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\frac{Q_{t+1}}{Q_t}\right]\Phi\left(Q_tB_{t+1}^\star\right) = \mathbb{E}_t\left[\left(\frac{C_{t+1}^\star}{C_t^\star}\right)^{-\gamma}\right].$$

The functional form of the interest rate premium is

$$\Phi\left(Q_tB_{t+1}^\star\right) = \left[1 + Q_t(B_{t+1}^\star - B^\star)\right]^{-\phi},$$

where $\phi > 0$. Under this specification, expected excess currency returns depend on the level of external borrowing, akin to the mechanisms highlighted in models of segmented international asset markets (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021; Kekre and Lenel, 2024). Last, the law of one price holds.

Solution We solve the model by taking a log-linear approximation of the equilibrium conditions around a symmetric steady state, where $A = A^\star$, and a zero steady-state net foreign asset position $B^\star = 0$, and thus $Q = 1$.

Log-linear equilibrium system We characterize the log-linear equilibrium system for the incomplete-markets model as follows. The equilibrium system has 10 endogenous variables $(c_t, c_t^\star, x_t, y_t, x_t^\star, y_t^\star, r_t, r_t^\star, q_t, b_{t+1}^\star)$ with 10 equilibrium conditions:

$$x_t = c_t + \frac{\sigma(1-\varsigma)}{2\varsigma - 1}q_t; \tag{D.28}$$

$$y_t = c_t - \frac{\sigma\varsigma}{2\varsigma - 1}q_t; \tag{D.29}$$

$$x_t^\star = c_t^\star + \frac{\sigma\varsigma}{2\varsigma - 1}q_t; \tag{D.30}$$

$$y_t^\star = c_t^\star - \frac{\sigma(1-\varsigma)}{2\varsigma - 1}q_t; \tag{D.31}$$

$$a_t = \varsigma x_t + (1-\varsigma)x_t^\star; \tag{D.32}$$
$$a_t^\star = (1-\varsigma)y_t + \varsigma y_t^\star; \tag{D.33}$$
$$\mathbb{E}_t[c_{t+1}] = c_t + \frac{1}{\gamma} r_t; \tag{D.34}$$
$$\mathbb{E}_t[c_{t+1}^\star] = c_t^\star + \frac{1}{\gamma} r_t^\star; \tag{D.35}$$
$$\mathbb{E}_t[q_{t+1}] - q_t = r_t - r_t^\star + \phi b_{t+1}^\star; \tag{D.36}$$
$$\beta b_{t+1}^\star - b_t^\star = a_t - \varsigma x_t - (1-\varsigma)y_t - \frac{1-\varsigma}{2\varsigma - 1} q_t. \tag{D.37}$$

Three-equation system The incomplete-markets model can be expressed as the system of three equations:

$$q_t = -\frac{(2\varsigma-1)^2}{4(1-\varsigma)\sigma\varsigma}(c_t - c_t^\star) + \frac{(2\varsigma-1)}{4(1-\varsigma)\sigma\varsigma}(a_t - a_t^\star), \tag{D.38a}$$
$$\mathbb{E}_t[q_{t+1}] - q_t = \gamma\, \mathbb{E}_t\left[(c_{t+1} - c_{t+1}^\star) - (c_t - c_t^\star)\right] + \phi\, b_{t+1}^\star, \tag{D.38b}$$
$$\beta b_{t+1}^\star - b_t^\star = (1-\varsigma)\left[-(c_t - c_t^\star) + \frac{(2\sigma\varsigma - 1)}{(2\varsigma - 1)} q_t\right]. \tag{D.38c}$$

Equation (D.38a) combines the goods market clearing conditions (D.32)–(D.33) with the goods demand functions (D.28)–(D.31). Equation (D.38b) combines the modified UIP condition (D.36) with the Euler equations (D.34)–(D.35). Finally, equation (D.38c) links the country's budget constraint (D.37) with the goods market clearing condition (D.32) and the demand functions (D.29) and (D.30).

Equation (D.38b) can be solved forward. Because $b_{t+1}^\star$ is chosen at date t, $b_{t+1}^\star = \mathbb{E}_t[b_{t+1}^\star]$, and by the law of iterated expectations, iterating (D.38b) forward T periods gives

$$q_t - \gamma(c_t - c_t^\star) = \mathbb{E}_t\left[q_{t+T} - \gamma(c_{t+T} - c_{t+T}^\star)\right] - \phi \sum_{j=1}^{T} \mathbb{E}_t[b_{t+j}^\star].$$

So long as the two countries' relative productivity is stationary (i.e. a_t and $a_t^\star$ are cointegrated, as we assume), the debt-elastic premium makes the equilibrium stationary and the first term on the right-hand side vanishes as $T \to \infty$, yielding equation (22) in the main text.

Going forward, we assume that the inverse of the elasticity of intertemporal substitution—the CRRA coefficient—and the trade elasticity are jointly sufficiently high. Specifically, we assume $\gamma(2\sigma\varsigma - 1) > 2\varsigma - 1$. This is the condition of Proposition 2. Since $\gamma > 0$ and $2\varsigma - 1 > 0$, it implies $2\sigma\varsigma > 1$, which we use below. The proposition is stated in terms of the expectations response function defined by Chahrour and Jurado (2025). An advantage of this approach is that it isolates the effects of changes in expectations regardless of their source: the conclusions do not depend on any particular assumptions about the exogenous process for productivity or assumptions about the information that the agents hold.

Proof of Proposition 2. Define home–foreign differential variables by, e.g. $\tilde{a}_t \equiv a_t - a_t^\star$. Com-

bine equations (D.38a)–(D.38c) to deliver a second-order difference equation in $b_t^\star$:

$$\mathbb{E}_t\left[(\beta^{-1}L^2+\omega_{b,0}^b L+1)b_{t+2}^\star\right]=\mathbb{E}_t\left[\omega_{b,0}^a(\tilde{a}_{t+1}-\tilde{a}_t)\right], \tag{D.39}$$

where L represents the lag operator. The coefficient $\omega_{b,0}^b$ on the left-hand side of (D.39) is given by

$$\omega_{b,0}^b=-\beta^{-1}\left[1+\beta+\frac{(1-\varsigma)(2\sigma\varsigma-2\varsigma+1)\,\phi}{\Theta_\gamma}\right], \tag{D.40}$$

where $\Theta_\gamma>0$ is defined below (21). Moreover, the coefficient on the right-hand side of (D.39) is

$$\omega_{b,0}^a=\frac{1-\varsigma}{\beta}\frac{\chi_0}{\Theta_\gamma}, \tag{D.41}$$

where $\chi_0\equiv\gamma(2\sigma\varsigma-1)+(1-2\varsigma)>0$ by the condition of Proposition 2.

Factoring, rewrite the left-hand side of (D.39) as

$$\mathbb{E}_t\left[(1-r_1^{-1}L)(1-r_2^{-1}L)b_{t+2}^\star\right]=\mathbb{E}_t\left[\omega_{b,0}^a(\tilde{a}_{t+1}-\tilde{a}_t)\right], \tag{D.42}$$

where $r_1\in(0,1)$ is the smaller root of the quadratic equation $\beta^{-1}r^2+\omega_{b,0}^b r+1=0$, $r_2>1$ is the larger root, and $\beta=r_1r_2$. The unique stable solution to (D.42) is given by

$$b_{t+1}^\star=r_2^{-1}b_t^\star+r_1\omega_{b,0}^a\tilde{a}_t-(1-r_1)\mathbb{E}_t\left[\sum_{j=1}^{\infty}r_1^j\omega_{b,0}^a\tilde{a}_{t+j}\right]. \tag{D.43}$$

To derive implications for q_t and relative consumption $\tilde{c}_t$, combine (D.38a) and (D.38c) to express both in terms of the bond position:

$$q_t=\frac{(2\varsigma-1)\,\tilde{a}_t+\frac{(2\varsigma-1)^2}{1-\varsigma}\,(\beta b_{t+1}^\star-b_t^\star)}{2\sigma\varsigma-2\varsigma+1}, \tag{D.44}$$

$$\tilde{c}_t=\frac{(2\sigma\varsigma-1)\,\tilde{a}_t-4\sigma\varsigma\,(\beta b_{t+1}^\star-b_t^\star)}{2\sigma\varsigma-2\varsigma+1}. \tag{D.45}$$

Substituting the solution (D.43) for $b_{t+1}^\star$ into (D.44) and (D.45) gives

$$q_t=\omega_q^b b_t^\star+\omega_{q,0}^a\tilde{a}_t+(1-r_1)\mathbb{E}_t\left[\sum_{j=1}^{\infty}r_1^j\omega_{q,1}^a\tilde{a}_{t+j}\right], \tag{D.46}$$

$$\tilde{c}_t=\omega_c^b b_t^\star+\omega_{c,0}^a\tilde{a}_t+(1-r_1)\mathbb{E}_t\left[\sum_{j=1}^{\infty}r_1^j\omega_{c,1}^a\tilde{a}_{t+j}\right]. \tag{D.47}$$

We can now compute the derivatives in Clause 1 of the proposition. For example,

$$\frac{\partial \tilde{c}_t}{\partial \mathbb{E}_t[a_{t+j}]} = (1 - r_1) r_1^j \omega^a_{c,1}.$$

Moreover, $(1 - r_1) > 0$ and $r_1^j > 0$, so that a sufficient condition for the whole term to be positive is that $\omega^a_{c,1} > 0$.

After somewhat lengthy algebra, we have that

$$\omega^a_{c,1} = \frac{4\varsigma\sigma(1-\varsigma)}{2\sigma\varsigma - 2\varsigma + 1} \frac{\chi_0}{\Theta_\gamma} > 0.$$

The inequality follows from observing that $(1-\varsigma) > 0$ since $\varsigma \in (1/2, 1)$ and $(2\sigma\varsigma - 2\varsigma + 1) > 2(1-\varsigma) > 0$ since $2\sigma\varsigma > 1$. Moreover, future productivity does not enter the market-clearing condition (D.38a), so, as in the proof of Proposition 1,

$$\omega^a_{q,1} = -\frac{(2\varsigma - 1)^2}{4\varsigma\sigma(1-\varsigma)} \omega^a_{c,1} < 0.$$

Hence $\tilde{c}_t$ and q_t move in opposite directions in response to anticipated changes in productivity. This establishes Clause 1 of the proposition.

For Clause 2 of the proposition, note that $\frac{\partial \tilde{c}_t}{\partial a_t} = \omega^a_{c,0}$ and $\frac{\partial q_t}{\partial a_t} = \omega^a_{q,0}$. After similar algebra,

$$\begin{aligned}
\omega^a_{c,0} &= (1 - r_1) \frac{2\sigma\varsigma - 1}{2\sigma\varsigma - 2\varsigma + 1} + r_1 \frac{2\varsigma - 1}{\Theta_\gamma}, \\
\omega^a_{q,0} &= (1 - r_1) \frac{2\varsigma - 1}{2\sigma\varsigma - 2\varsigma + 1} + r_1 \frac{\gamma(2\varsigma - 1)}{\Theta_\gamma}.
\end{aligned}$$

Since $r_1 \in (0, 1)$, $2\sigma\varsigma - 1 > 0$, $2\sigma\varsigma - 2\varsigma + 1 > 0$, $\varsigma > 1/2$, and $\Theta_\gamma > 0$, both are positive. This establishes Clause 2 of the proposition, completing the proof. ■

D.2 Blanchard et al. (2013) process

With the Blanchard et al. (2013) process, log productivity, a_t, is the sum of a permanent component μ_t (with autocorrelated growth rates) and a transitory component θ_t, but agents only observe the realizations of productivity a_t and not the two components separately. In addition, agents observe a noisy signal of the current value of the permanent component, $s_t = \mu_t + \xi_t$, which helps them forecast future productivity growth. Specifically, all stochastic processes $\{a_t\}$ and $\{s_t\}$ evolve according to the following system:

$$\begin{aligned}
a_t &= \mu_t + \theta_t, \\
s_t &= \mu_t + \xi_t, \\
\Delta\mu_t &= \rho\Delta\mu_{t-1} + \varepsilon^\mu_t, \\
\theta_t &= \rho\theta_{t-1} + \varepsilon^\theta_t,
\end{aligned} \qquad \begin{bmatrix} \varepsilon^\mu_t \\ \varepsilon^\theta_t \\ \xi_t \end{bmatrix} \overset{\text{iid}}{\sim} \mathcal{N}\left(0, \begin{bmatrix} \sigma^2_\mu & 0 & 0 \\ 0 & \sigma^2_\theta & 0 \\ 0 & 0 & \sigma^2_\xi \end{bmatrix}\right). \tag{D.48}$$

Lastly, it is common to impose the parameter restriction $\rho\sigma_\mu^2 = (1-\rho)^2\sigma_\theta^2$, which ensures that the univariate Wold representation of productivity is a random walk, as is true of TFP in the data.[34]

To properly isolate the independent contributions of beliefs (or the independent effects of technological and noise disturbances), we follow Chahrour and Jurado (2018) to construct a noise representation that is observationally equivalent to representation (D.48).

Lemma 1. *The representation of fundamentals and beliefs in system* (D.48) *is observationally equivalent to the noise representation:*

$$\begin{aligned}
\Delta a_t &= \frac{1-\alpha L}{1-\rho L}\varepsilon_t^a, \\
\Delta \tilde{s}_t &= \sum_{j=0}^{\infty} \alpha^j \varepsilon_{t+j}^a + \tilde{v}_t, \\
\tilde{v}_t &= \frac{\sigma_\theta^2}{\alpha\sigma_\mu^2}\frac{(1-L)(1-\bar{\delta}_1 L)(1-\bar{\delta}_2 L)}{(1-\alpha L)}\varepsilon_t^v, \\
\begin{bmatrix}\varepsilon_t^a \\ \varepsilon_t^v\end{bmatrix} &\overset{\text{iid}}{\sim} \mathcal{N}\left(0, \begin{bmatrix}\sigma_\theta^2/\alpha & 0 \\ 0 & \frac{\alpha\rho\sigma_\xi^2}{\bar{\delta}_1\bar{\delta}_2}\end{bmatrix}\right),
\end{aligned}$$

where α is the stable root of $L^2 - (2+\frac{\sigma_\mu^2}{\sigma_\theta^2})L + 1 = 0$, and $\bar{\delta}_1, \bar{\delta}_2$ are the two stable roots of the polynomial $D(L)$ (excluding the unit roots):

$$D(L) = (1-\alpha L)(1-\alpha L^{-1})\sigma_\mu^2 + (1-\rho L)(1-\rho L^{-1})(1-\alpha L)(1-\alpha L^{-1})(1-L)(1-L^{-1})\sigma_\xi^2 - \alpha\frac{\sigma_\mu^4}{\sigma_\theta^2}.$$

Proof. First, we derive the univariate Wold representation of Δa_t. The process can be written as

$$\Delta a_t = \frac{1}{1-\rho L}\left(\varepsilon_t^\mu + (1-L)\varepsilon_t^\theta\right).$$

The autocovariance generating function of Δa is

$$\Gamma_{\Delta a}(L) = \frac{\sigma_\mu^2 + \sigma_\theta^2(2 - L - L^{-1})}{(1-\rho L)(1-\rho L^{-1})}.$$

After factorizing the numerator, we obtain

$$\Gamma_{\Delta a}(L) = \frac{\sigma_\theta^2}{\alpha}\cdot\frac{(1-\alpha L)(1-\alpha L^{-1})}{(1-\rho L)(1-\rho L^{-1})},$$

where $\alpha \in (0,1)$ is the stable root of the quadratic equation $L^2 - (\frac{\sigma_\mu^2}{\sigma_\theta^2}+2)L + 1 = 0$. This

[34] Blanchard et al. (2013) refer to ε_t^μ as a permanent productivity shock, ε_t^θ as a transitory productivity shock, and ξ_t as an information noise shock.

implies the Wold representation,

$$\Delta a_t = \frac{1-\alpha L}{1-\rho L}\varepsilon_t^a, \quad \varepsilon_t^a \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\theta^2/\alpha).$$

The signal, written as a lag polynomial, is given by

$$\Delta s_t = \frac{1}{1-\rho L}\varepsilon_t^\mu + (1-L)\xi_t.$$

Projecting Δs_t onto the full past, present, and future Δa_t, we can write

$$\Delta s_t = P(L)\Delta a_t + v_t,$$

where

$$P(L) = \frac{\Gamma_{\Delta a, \Delta s}(L)}{\Gamma_{\Delta a}(L)} = \frac{\sigma_\mu^2}{(1-\rho L)(1-\rho L^{-1})}\frac{1}{\Gamma_{\Delta a}(L)} = \frac{\alpha}{(1-\alpha L)(1-\alpha L^{-1})}\frac{\sigma_\mu^2}{\sigma_\theta^2},$$

and v_t is a noise process, characterized below, that is orthogonal to all leads and lags of Δa_t. Substituting the Wold form of Δa_t, we find

$$P(L)\Delta a_t = \frac{\alpha}{(1-\alpha L^{-1})(1-\rho L)}\frac{\sigma_\mu^2}{\sigma_\theta^2}\varepsilon_t^a.$$

Therefore,

$$\Delta s_t = \frac{\alpha}{(1-\alpha L^{-1})(1-\rho L)}\frac{\sigma_\mu^2}{\sigma_\theta^2}\varepsilon_t^a + v_t. \tag{D.49}$$

To simplify without changing the information content of Δs_t, we scale both sides of (D.49) by the invertible polynomial $F(L) \equiv \frac{\sigma_\theta^2(1-\rho L)}{\alpha\sigma_\mu^2}$ to find

$$\Delta\tilde{s}_t \equiv \frac{\sigma_\theta^2(1-\rho L)}{\alpha\sigma_\mu^2}\Delta s_t = \frac{1}{(1-\alpha L^{-1})}\varepsilon_t^a + \tilde{v}_t = \sum_{j=0}^{\infty}\alpha^j\varepsilon_{t+j}^a + \tilde{v}_t.$$

Finally, to characterize $\tilde{v}_t$, compute its autocovariance generating function

$$\begin{aligned}\Gamma_{\tilde{v}}(L) &= F(L)F(L^{-1})\left[\Gamma_{\Delta s}(L) - P(L)\Gamma_{\Delta a}(L)P(L^{-1})\right] \\ &= F(L)F(L^{-1})\left[\frac{D(L)}{(1-\rho L)(1-\rho L^{-1})(1-\alpha L)(1-\alpha L^{-1})}\right],\end{aligned}$$

where the polynomial $D(L)$ is

$$D(L) \equiv (1-\alpha L)(1-\alpha L^{-1})\sigma_\mu^2 + (1-\rho L)(1-\rho L^{-1})(1-\alpha L)(1-\alpha L^{-1})(1-L)(1-L^{-1})\sigma_\xi^2 - \alpha\frac{\sigma_\mu^4}{\sigma_\theta^2}.$$

Table D.1: Calibration of Blanchard et al. (2013) process

Parameter	Description	Value
ρ	Persistence parameter	0.89
σ_θ	Std. dev. "transitory shock"	0.58
σ_ξ	Std. dev. "noise shock"	5×10^{-5}
σ_μ	Std. dev. "permanent shock"	$\rho\sigma_\mu^2 = (1-\rho)^2\sigma_\theta^2$
τ	Cointegration parameter	0.025

Notes: Models are calibrated to quarterly frequency. The rest of the parameters are calibrated according to Table 4.

Then $\Gamma_{\tilde{v}}(L)$ becomes

$$\Gamma_{\tilde{v}}(L) = \left(\frac{\sigma_\theta^2}{\alpha\sigma_\mu^2}\right)^2 \left[\frac{D(L)}{(1-\alpha L)(1-\alpha L^{-1})}\right].$$

Since $\Gamma_{\tilde{v}}(L)$ is a spectral density, $D(L)$ is symmetric and non-negative on the unit circle, ensuring a symmetric factorization exists. Furthermore, because α is defined such that $\alpha^2 - (\frac{\sigma_\mu^2}{\sigma_\theta^2} + 2)\alpha + 1 = 0$, we have $D(1) = 0$. This implies that $(1-L)(1-L^{-1})$ is a factor of $D(L)$. Factoring out the unit root, the square root of $\Gamma_{\tilde{v}}(L)$ gives the following Wold representation:

$$\tilde{v}_t = \frac{\sigma_\theta^2}{\alpha\sigma_\mu^2}\frac{(1-L)(1-\bar{\delta}_1 L)(1-\bar{\delta}_2 L)}{(1-\alpha L)}\varepsilon_t^v, \quad \varepsilon_t^v \overset{\text{iid}}{\sim} \mathcal{N}(0, \frac{\alpha\rho\sigma_\xi^2}{\bar{\delta}_1\bar{\delta}_2}), \tag{D.50}$$

where $\bar{\delta}_1$ and $\bar{\delta}_2$ are the remaining stable roots of $D(L)$. ■

Calibration We assume foreign productivity is cointegrated with home productivity, that is, $\Delta a_t^\star = \tau(a_{t-1} - a_{t-1}^\star)$, where the parameter τ captures cross-country convergence in productivity. Table D.1 summarizes the calibration of this process that we use in Section 5.

D.3 Long-run risk process

In this appendix, we consider a long-run risk process, of the type studied in international finance and macroeconomics by Colacito and Croce (2011; 2013) and Colacito et al. (2018). Letting Δa_t denote the growth rate of productivity (in deviations from its mean), the process $\{\Delta a_t\}$ is assumed to follow the law of motion:

$$\begin{aligned} \Delta a_t &= z_{t-1} + \iota_t, \\ z_t &= \rho z_{t-1} + \varepsilon_t^z, \end{aligned} \qquad \begin{bmatrix} \varepsilon_t^z \\ \iota_t \end{bmatrix} \overset{\text{iid}}{\sim} \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_z^2 & 0 \\ 0 & \sigma_\iota^2 \end{bmatrix}\right), \tag{D.51}$$

where $0 < \rho < 1$. Colacito and Croce (2013) refer to ε_t^z as a long-run shock (or news shock), and ι_t as a short-run shock.

To properly isolate the independent contributions of beliefs (or the independent effects of technological and noise disturbances), we follow Chahrour and Jurado (2018) and construct a noise representation that is observationally equivalent to representation (D.51). The following lemma presents one such noise representation.

Lemma 2. *The representation of fundamentals and beliefs in system* (D.51) *is observationally equivalent to the noise representation*

$$\begin{aligned}
\Delta a_t &= \rho \Delta a_{t-1} + \kappa \varepsilon_t^a - \kappa\alpha\varepsilon_{t-1}^a, \\
\tilde{z}_t &= \sum_{j=0}^{\infty} \alpha^j \varepsilon_{t+1+j}^a + \tilde{v}_t, \\
\tilde{v}_t &= \alpha \tilde{v}_{t-1} + \frac{\sigma_\iota}{\sigma_z}\varepsilon_t^v - \rho\frac{\sigma_\iota}{\sigma_z}\varepsilon_{t-1}^v, \\
\begin{bmatrix} \varepsilon_t^a \\ \varepsilon_t^v \end{bmatrix} &\overset{\text{iid}}{\sim} \mathcal{N}\left(0, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right),
\end{aligned}$$

where α *is the stable root of* $L^2 - \frac{1}{\rho}(1 + \rho^2 + \frac{\sigma_z^2}{\sigma_\iota^2})L + 1 = 0$ *and* $\kappa \equiv \sqrt{\frac{\rho}{\alpha}}\sigma_\iota$.

Proof. The proof follows the same logic as in Lemma 1. First, we derive the univariate Wold representation of Δa_t. The autocovariance generating function (ACGF) of Δa is

$$\Gamma_{\Delta a}(L) = \frac{\sigma_z^2}{(1-\rho L)(1-\rho L^{-1})} + \sigma_\iota^2 = \frac{\sigma_z^2 + \sigma_\iota^2(1-\rho L)(1-\rho L^{-1})}{(1-\rho L)(1-\rho L^{-1})}.$$

We can factor the numerator as a symmetric second-order polynomial:

$$\Gamma_{\Delta a}(L) = \frac{\rho\sigma_\iota^2}{\alpha}\frac{(1-\alpha L)(1-\alpha L^{-1})}{(1-\rho L)(1-\rho L^{-1})}, \tag{D.52}$$

where α is the stable root of the quadratic equation $L^2 - \frac{1}{\rho}(1 + \rho^2 + \frac{\sigma_z^2}{\sigma_\iota^2})L + 1 = 0$. Taking the square root yields the Wold representation:

$$\Delta a_t = \frac{1-\alpha L}{1-\rho L}\sqrt{\frac{\rho}{\alpha}}\sigma_\iota \varepsilon_t^a, \tag{D.53}$$

where $\varepsilon_t^a \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ is the Wold innovation.

Second, we represent the signal process as a function of ε_t^a and noise. Define the linear projection:

$$z_t = P(L)\Delta a_t + v_t, \tag{D.54}$$

where $v_t \perp \Delta a_{t+j}\ \forall j \in \mathbb{Z}$, i.e. the noise component is orthogonal to all leads and lags of

technology. We can write

$$P(L) = \frac{\Gamma_{z,\Delta a}(L)}{\Gamma_{\Delta a}(L)} = \frac{\sigma_z^2 L^{-1}}{\sigma_z^2 + \sigma_\iota^2(1-\rho L)(1-\rho L^{-1})} = \frac{\sigma_z^2 \alpha L^{-1}}{\rho\sigma_\iota^2(1-\alpha L)(1-\alpha L^{-1})}.$$

The noise process v_t has autocovariance generating function

$$\begin{aligned}
\Gamma_v(L) &= \Gamma_z(L) - P(L)\Gamma_{\Delta a}(L)P(L^{-1}) \\
&= \frac{\sigma_z^2}{(1-\rho L)(1-\rho L^{-1})} - \Gamma_{z,\Delta a}(L)\frac{\sigma_z^2 \alpha L}{\rho\sigma_\iota^2(1-\alpha L)(1-\alpha L^{-1})} \\
&= \frac{\sigma_z^2\left((1-\alpha L)(1-\alpha L^{-1}) - \frac{\sigma_z^2\alpha}{\rho\sigma_\iota^2}\right)}{(1-\alpha L)(1-\alpha L^{-1})(1-\rho L)(1-\rho L^{-1})} \\
&= \frac{\sigma_z^2\alpha}{\rho}\frac{1}{(1-\alpha L)(1-\alpha L^{-1})},
\end{aligned}$$

where the last equality uses the factorization in equation (D.52). This gives the Wold representation:

$$v_t = \frac{1}{1-\alpha L}\sqrt{\frac{\alpha}{\rho}}\sigma_z \varepsilon_t^v, \tag{D.55}$$

with $\varepsilon_t^v \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

Plug equations (D.55) and (D.53) into (D.54) and rearrange as

$$\underbrace{\frac{\kappa}{\sigma_z^2}(1-\rho L)z_t}_{\tilde{z}_t} = \frac{L^{-1}}{1-\alpha L^{-1}}\varepsilon_t^a + \underbrace{\frac{\kappa}{\sigma_z^2}(1-\rho L)v_t}_{\tilde{v}_t}, \tag{D.56}$$

where $\kappa \equiv \sqrt{\frac{\rho}{\alpha}}\sigma_\iota$. Thus, the information structure can be written as

$$\begin{aligned}
\Delta a_t &= \rho\Delta a_{t-1} + \kappa\varepsilon_t^a - \kappa\alpha\varepsilon_{t-1}^a, \\
\tilde{z}_t &= \sum_{j=0}^{\infty}\alpha^j\varepsilon_{t+1+j}^a + \tilde{v}_t, \\
\tilde{v}_t &= \alpha\tilde{v}_{t-1} + \frac{\sigma_\iota}{\sigma_z}\varepsilon_t^v - \rho\frac{\sigma_\iota}{\sigma_z}\varepsilon_{t-1}^v.
\end{aligned}$$

■

Calibration We assume the foreign productivity process follows an analogous law of motion and is cointegrated with home productivity:

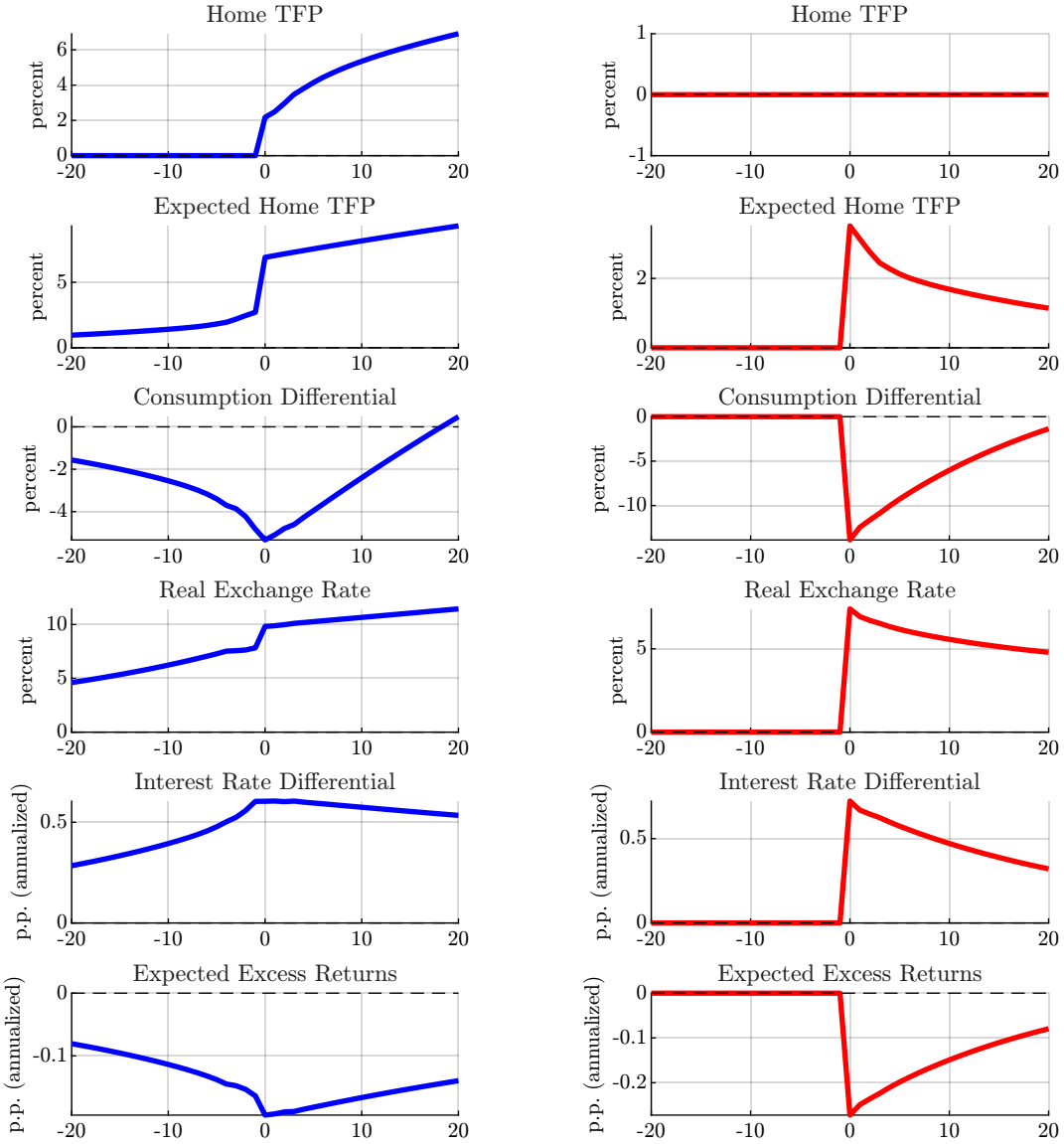
$$\begin{aligned}
\Delta a_t^\star &= z_{t-1}^\star + \tau(a_{t-1} - a_{t-1}^\star) + \iota_t^\star, \\
z_t^\star &= \rho z_{t-1}^\star + \varepsilon_t^{z\star},
\end{aligned} \qquad \begin{bmatrix}\varepsilon_t^{z\star} \\ \iota_t^\star\end{bmatrix} \overset{\text{iid}}{\sim} \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}\sigma_z^2 & 0 \\ 0 & \sigma_\iota^2\end{bmatrix}\right). \tag{D.57}$$

Table D.2: Calibration of long-run risk process

Parameter	Description	Value
ρ	Pers. long-run component	0.995
σ_ι	Std. dev. "short-run shock"	0.01925
σ_z	Std. dev. "long-run shock"	$0.12\sigma_\iota$
τ	Cointegration parameter	0.000175

Notes: Models are calibrated to quarterly frequency, adapting the benchmark annual calibration of Colacito et al. (2018). The rest of the parameters are calibrated according to Table 4.

Table D.2 summarizes the calibration of this process. Figure D.1 characterizes the impulse responses of the complete-markets model with Epstein-Zin preferences under the long-run risk process.



(a) Technological disturbances (b) Noise disturbances

Figure D.1: Impulse responses to technological and noise disturbances (long-run risk process)

Notes: The model features complete markets with Epstein-Zin preferences. Table D.2 reports the parameter values of equation (D.51). Expected excess returns refer to those of holding foreign bonds, as defined in (5). The model is calibrated to quarterly frequency. Each period is a quarter.

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